

Efficient estimation of binary treatment effects under local differential privacy

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Abstract

We characterize the exact regular local asymptotic minimax squared-error bound for estimating the average treatment effect in an independently sampled randomized trial with binary outcomes, known interior assignment probability, interior arm means, and a fixed positive local privacy budget. Each subject releases one privatized assignment–outcome record, and release rules may adapt to earlier outputs on arbitrary measurable spaces. The optimal stationary contrast variance is the reciprocal of a finite, fourteen-pattern staircase saddle that profiles out the common-mean nuisance direction, and it lower-bounds local asymptotic minimax squared-error risk for every sequential private procedure. The attainment results assume a fixed measurable strong-saddle selector satisfying the stated conditions. Fixing any such selector, a private-pilot affine estimator attains the bound within the regular class and supplies consistent variance estimation and pointwise asymptotically valid Wald intervals, including at optimal-support changes. An attaining stationary release always exists with at most five active outputs; certified open neighborhoods have exact minimum cardinalities of five and three. Balanced assignment and complementary arm means yield an optimal binary randomized-response release. A comparison with the published known-assignment custom-Laplace sample mean translates the variance improvement at a specified balanced design point into prospective interval-length and recruitment calculations.

1 Introduction

This paper identifies the exact regular local asymptotic minimax squared-error constant for estimating a treatment effect when both treatment assignment and outcome are locally private. The experiment consists of independently sampled subjects in a randomized binary trial, with a known assignment probability strictly between zero and one, interior outcome probabilities in both arms, and a fixed positive privacy budget. Each subject contributes one private assignment–outcome record. Release rules may depend on preceding private outputs and may take values in arbitrary measurable spaces. Within this model, the stationary optimal contrast variance is the reciprocal of a finite nuisance-profiled staircase optimization. Fixing any measurable strong-saddle selector satisfying the stated conditions on the interior parameter domain, a private-pilot estimator attains the constant and supports pointwise Wald inference.

Randomization identifies the average treatment effect as the difference between the treatment-arm and control-arm outcome probabilities. Joint local privatization determines how much information about that difference reaches the analyst. A release must satisfy the

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privacy inequality across all four assignment–outcome pairs, including pairs that differ in both coordinates. The resulting design problem concerns the scalar treatment-effect contrast within a two-parameter model: both arm means vary, and their common shift is a nuisance direction. Efficient release selection therefore depends on how the channel preserves information about the contrast after accounting for that nuisance variation.

Our first result connects this statistical objective to finite channel geometry. Under the stated interiority and fixed-privacy conditions, [Theorem 1](#) characterizes the smallest stationary contrast variance over all measurable output spaces. Its reciprocal is obtained by maximizing nuisance-profiled information over weights on fourteen nonconstant staircase patterns. For each fixed profiling direction, information is linear in these weights; minimization over the common-mean direction produces the saddle objective. Equivalently, the optimized information is the minimum of a finite upper envelope of convex quadratics. This representation gives an exact finite optimization at each specified pair of arm means, assignment probability, and privacy budget.

The stationary optimum also governs the sequential experiment. Conditional on a fixed measurable selector satisfying the strong-saddle conditions in [Algorithm 1](#), [Theorem 2](#) establishes a local asymptotic squared-error lower bound for every sequence of sequential private protocols and transcript-measurable estimators. The criterion takes the worst risk over bounded admissible local perturbations, then the lower asymptotic limit, and finally the supremum over perturbation radii. The same constant is attained within the regular class by a private-pilot construction with pilot size equal to the integer part of the square root of the total sample size. Thus history adaptation and arbitrary measurable releases are incorporated into the exact efficiency comparison.

The construction first estimates the arm means using four-category randomized response and then selects a staircase channel for the remaining subjects. Its treatment-effect estimate combines the pilot contrast with an affine correction computed from the main-sample releases. Under a fixed measurable strong-saddle selector, [Theorem 3](#) establishes exact conditional centering given the private pilot history, regular asymptotic normality, consistent empirical variance estimation, and pointwise asymptotic Wald coverage. These conclusions hold for every pilot schedule satisfying [Assumptions 10](#) and [11](#). They apply throughout the interior parameter domain, including points where the optimal channel support changes.

The optimal output alphabet has a corresponding statistical structure. [Theorem 4](#) guarantees an attaining stationary staircase release with at most five active outputs at every interior design point. At assignment probability $3/10$ and privacy budget $\log 3$, [Theorem 7](#) identifies open neighborhoods of arm means $(1/2, 1/2)$ and $(1/10, 1/10)$ in which the minimum attaining cardinalities among all stationary channels are five and three, respectively. Under balanced assignment and complementary arm means, [Theorem 8](#) instead yields an optimal two-output randomized-response release. These results establish specific optimal release cardinalities under contrast-specific efficiency at selected outcome prevalences and designs.

The analysis connects two established strands of private inference. [Steinberger \[2024, Theorems 3.3 and 3.5, Remark 3.6, and Section 4\]](#) develops sequential asymptotic efficiency theory under his stated measurable-space and information-limit conditions, and identifies scalar functionals of inverse information as objectives in multiparameter models. Here the four-record trial model permits an exact nuisance-profiled contrast calculation over arbitrary measurable sequential output spaces. The staircase geometry follows the extremal-mechanism framework of [Kairouz et al. \[2016, Theorems 2 and 4\]](#); nuisance profiling gives the present optimization

its concave weight objective and associated support structure. The exact fixed-privacy constant complements the statistical-rate analyses of [Duchi et al. \[2018\]](#), [Rohde and Steinberger \[2020\]](#), [Duchi and Ruan \[2024\]](#).

For a direct causal benchmark, we evaluate the known-assignment custom-Laplace release and sample-mean estimator of [Ohnishi and Awan \[2025, Sections 3.1 and 3.3, Theorem 7, and Appendix A.5\]](#). At assignment probability $1/2$, both arm means $1/2$, and privacy budget one, [Theorem 9](#) establishes a strict variance improvement for the efficient procedure. The implied asymptotic half-length ratio is approximately 2.69, with the custom-Laplace sample mean in the numerator; the corresponding sample-size ratio at equal precision is approximately 7.26. We illustrate these quantities through a prospective binary endpoint defined by meeting the 80% attendance threshold of the Colombian cash-transfer program studied by [Barrera-Orsorio et al. \[2011\]](#).

The paper next reviews related work, defines the private trial model, derives the efficiency bound and finite optimization, develops private-pilot inference, studies optimal release cardinality, and gives the interval comparison and prospective design calculation. Discussion is followed by the proof exposition in [Section 9](#) and the appended proofs.

2 Related work

The closest asymptotic efficiency framework is that of [Steinberger \[2024, Theorems 3.3 and 3.5, Remark 3.6, and Section 4\]](#). For parametric models differentiable in quadratic mean, his sequential theory establishes local asymptotic mixed normality along subsequences with convergent conditional information, under countably generated input and output sigma-fields. The convolution theorem additionally imposes positive-definiteness conditions on the conditional information matrices and their limiting accumulation points. His scalar optimization maximizes output Fisher information, and the private two-step procedure uses a preliminary estimate to select an asymptotically optimal mechanism under the stated regularity conditions. Remark 3.6 explicitly identifies scalar functionals of inverse information as objectives for multiparameter models. Our analysis treats a treatment-effect contrast in a two-parameter randomized binary trial with binary outcomes and known interior assignment probability, as specified in [Definition 1](#). At interior arm means and a fixed positive privacy budget, [Theorems 1 and 2](#) connect finite channel optimization with the exact efficiency bound for the stated class of regular procedures using arbitrary measurable sequential output spaces. The scalar target therefore retains a nuisance parameter throughout the optimization.

The finite reduction is closely related to the staircase mechanisms of [Kairouz et al. \[2016, Theorems 2 and 4\]](#). Their utility class consists of sums over output columns of a sublinear function, meaning a positively homogeneous, subadditive function of the column. For this class, they establish an optimal staircase mechanism with output alphabet no larger than the input alphabet and express the optimum as a finite linear program. The present fourteen-pattern staircase program, characterized in [Definitions 4 and 5](#) and [Theorem 1](#), combines this finite channel geometry with nuisance profiling. It also parallels contrast-specific optimal-design geometry: [Elfving \[1952, Sections 2–4\]](#) represents optimal designs through a convex set generated by signed regression vectors, and [Pukelsheim \[2006, Chapter 2\]](#) develops the corresponding information-matrix framework. For a fixed parameter direction, directional Fisher information here is additive over staircase outputs and linear in their weights. Contrast information minimizes that directional information over directions normalized to have unit treatment-effect derivative. This minimiza-

tion makes the objective concave in the staircase weights and gives the optimization its saddle structure. The relevant support geometry is consequently governed jointly by channel normalization and profiling. Under the conditions stated in the support results, [Theorem 4](#) supplies a five-output upper bound, while [Theorem 5](#) identifies five-output optimality through its certificate and score-separation conditions. Comparison by randomization provides the broader experiment-ordering context [[Blackwell, 1953](#), pp. 265–272]; the results characterize contrast-specific efficiency within this framework.

The nuisance projection and local risk formulation follow the local-asymptotic tradition of [Le Cam and Yang \[2000, Chapter 6\]](#) and [van der Vaart \[1998, Chapters 7–8 and 25\]](#). The sequential lower bound uses the Bayesian information-inequality lineage of [Van Trees \[2001\]](#) and [Gill and Levit \[1995, pp. 59–79\]](#). These references provide intellectual antecedents; the formal verification contract records no external theorem as a Lean proof dependency.

The direct causal benchmark is [Ohnishi and Awan \[2025, Sections 3.1 and 3.3, Lemma 4, Theorem 7, and Appendix A.5\]](#). Their randomized-experiment model permits potential outcomes in the unit interval. Lemma 4 gives a worst-case squared-error lower bound for privacy budgets between zero and one. In the known-assignment custom scenario, each subject forms the inverse-probability-weighted observed outcome and adds independent Laplace noise calibrated to the sum of the reciprocal treatment and control assignment probabilities. The sample mean of these releases is unbiased and consistent; Theorem 7 supplies asymptotic normality, a variance estimate, confidence intervals, and the privacy-dependent convergence rate. These statements provide both a rate benchmark and a concrete inferential procedure. For binary outcomes, our comparison uses precisely this release and sample-mean estimator, specified in [Definition 14](#). The comparison in [Theorem 9](#) evaluates its asymptotic variance against the optimized contrast bound at the stated trial parameters and privacy budget.

These exact variance comparisons complement the literature on statistical rates under local privacy. In the bounded scalar-mean problem, [Duchi et al. \[2018, Section 3.2.1\]](#) obtain squared-error order $\min\{1, (n\varepsilon^2)^{-1}\}$, up to numerical constants, in the high-privacy regime $0 < \varepsilon \leq 1$. [Ohnishi and Awan \[2025, Sections 3.1 and 3.3, Lemma 4\]](#) obtain the same benchmark order for their bounded-outcome randomized-experiment model. This rate describes the joint sample-size/privacy scaling; the fixed-privacy efficiency constant derived here resolves the model- and design-specific numerical factor that rate notation suppresses. [Rohde and Steinberger \[2020\]](#) characterize the difficulty of functional estimation through total-variation moduli, including rate-optimal binary releases for linear functionals over convex models. Instance-specific local minimax complexity is developed further by [Duchi and Ruan \[2024\]](#). Information-based approaches also include the Fisher-information contraction inequalities of [Barnes et al. \[2020\]](#) and the communication-based estimation lower bounds of [Duchi and Rogers \[2019\]](#). Here the finite binary-trial experiment permits an exact optimization of the asymptotic contrast variance at a fixed privacy budget.

Exact scalar mechanisms provide a further point of comparison. For a Gaussian mean with known unit variance, [Kalinin and Steinberger \[2025\]](#) identify randomized response applied to the centered sign as an information-maximizing mechanism in their high-privacy regime, and attain efficiency through a private preliminary estimate. The private-pilot construction in [Algorithm 1](#) and [Theorem 3](#) uses the same broad principle of learning the parameter before selecting the main-sample release. Its channel selection solves the nuisance-profiled contrast problem, with pilot size diverging and occupying a vanishing fraction of the sample under [Assumptions 10 and 11](#).

Recent work further develops extremal-channel optimization. [Amorino and Gloter \[2025\]](#) study factorization through extremal mechanisms and Fisher-information maximization in discrete and continuous settings, while [Amorino and Gloter \[2026\]](#) analyze symmetry and asymmetry in staircase information across privacy regimes. For binary hypothesis testing between two known finite-alphabet distributions, [Ghazi et al. \[2026\]](#) optimize output f -divergences through likelihood-ratio ordering, contiguous partitions, and randomized response. These objectives illuminate the role of staircase structure. Our contrast objective additionally accounts for variation in both arm means, and its support certificates characterize the resulting release cardinality.

Finally, causal efficiency connects the analysis to [Hahn \[1998\]](#), [Hirano et al. \[2003\]](#), whose work studies efficient average treatment-effect estimation under unconfoundedness. In the private setting, [Kormos and van der Vaart \[2026\]](#) construct suitable identifying mechanisms that transfer rate-double-robust and semiparametric inferential properties to noisy observations, including for causal targets. The present binary randomized-trial model makes the release mechanism itself an object of efficiency optimization: the channel, nuisance adjustment, and attainable treatment-effect variance are characterized together under the stated sequential privacy conditions.

3 Trial model and private releases

Throughout, n denotes the positive sample size, subjects are indexed by $i = 1, \dots, n$, record categories by $j \in \{0, 1, 2, 3\}$, and treatment arms by $a \in \{0, 1\}$. Thus subject indices are one-based while the four-record alphabet is zero-based. Vectors are columns, $^\top$ denotes transpose, and $\|\cdot\|$ is the Euclidean norm. Generic positive constants are denoted by C ; O_p and o_p denote stochastic order, and $\Rightarrow$ denotes convergence in distribution.

3.1 Data generation

For subject i , let $Y_i(0)$ be the control potential outcome and $Y_i(1)$ the treatment potential outcome. The binary assignment W_i indicates treatment, and Y_i denotes the observed outcome. The subject's private record is the ordered assignment–outcome pair X_i , written $X_i = (W_i, Y_i)$.

Write p for the treatment-assignment probability and q for the control-assignment probability, with $q = 1 - p$. For arm a , the population mean μ_a is $\mu_a = E[Y_i(a)]$; thus μ_0 and μ_1 are the control-arm and treatment-arm means. Collect these means in the parameter vector θ , with $\theta = (\mu_0, \mu_1)^\top$. The average treatment effect τ and its contrast vector c satisfy

$$c = (-1, 1)^\top, \quad \tau = E[Y_i(1) - Y_i(0)] = \mu_1 - \mu_0 = c^\top \theta.$$

The following data-generation conditions connect this causal target to the distribution of the observed records.

Assumption 1. `[ass:iid-subjects]` (Independent identically distributed subjects) *The subject sequence $\{(Y_i(0), Y_i(1), W_i) : i \geq 1\}$, comprising control potential outcomes, treatment potential outcomes, and treatment assignments, is independent and identically distributed. Equivalently, for every $n \geq 1$, the first n subject triples are jointly independent and have a common law. Zero-based indexing $i \in \mathbb{N}$ corresponds to subject indexing by the relabeling $i \leftrightarrow i + 1$.*

Independent superpopulation sampling makes each subject a draw from the same population and gives the sample a product structure [[Ohnishi and Awan, 2025](#)]. Within each subject, the

joint distribution of the two potential outcomes may accommodate dependence. Treatment assignment is governed by the next two conditions.

Assumption 2. `[ass:random-assignment]` (Independence of treatment assignment) *For every subject $i \geq 1$, the treatment assignment W_i is independent of the potential outcomes $Y_i(0)$ and $Y_i(1)$:*

$$W_i \perp (Y_i(0), Y_i(1)).$$

Random assignment makes the potential-outcome distribution in each assigned arm representative of the corresponding population distribution [Ohnishi and Awan, 2025]. Its probability is a known feature of the trial design.

Assumption 3. `[ass:known-assignment]` (Known treatment assignment probability) *For every subject $i \geq 1$, the treatment assignment W_i has known treatment probability p :*

$$P(W_i = 1) = p.$$

The known Bernoulli assignment probability condition treats p as a design quantity when estimating the arm means [Ohnishi and Awan, 2025]. To specify the record that enters the privacy mechanism, we next connect assignment to the observed outcome.

Assumption 4. `[ass:consistency]` (Consistency of observed outcomes) *For every subject $i \geq 1$, the observed outcome Y_i is determined by the treatment assignment W_i and the treatment and control potential outcomes $Y_i(1)$ and $Y_i(0)$:*

$$Y_i = W_i Y_i(1) + (1 - W_i) Y_i(0).$$

Potential-outcome consistency identifies the observed response with the potential outcome under the assigned treatment [Ohnishi and Awan, 2025]. The outcome restriction then gives the record a finite alphabet.

Assumption 5. `[ass:binary-outcomes]` (Binary potential outcome support) *For every subject $i \geq 1$, the control and treatment potential outcomes $Y_i(0)$ and $Y_i(1)$ satisfy*

$$(Y_i(0), Y_i(1)) \in \{0, 1\}^2$$

almost surely.

The binary potential outcomes condition describes trials with a dichotomous response, such as whether a specified event occurs [Ohnishi and Awan, 2025]. Under Assumptions 4 and 5, the private record takes values in the ordered four-point alphabet $\mathcal{X}$:

$$\mathcal{X} = \{x_0, x_1, x_2, x_3\}, \quad x_0 = (0, 0), \quad x_1 = (0, 1), \quad x_2 = (1, 0), \quad x_3 = (1, 1).$$

Let the record law P_θ have probability vector π_θ , whose coordinates are $\pi_{\theta,j} = P_\theta(X_i = x_j)$. Random assignment and consistency give

$$\pi_\theta = \begin{pmatrix} q(1 - \mu_0) \\ q\mu_0 \\ p(1 - \mu_1) \\ p\mu_1 \end{pmatrix}.$$

Thus the distribution of the observed record is determined by the two arm means and the known assignment probability.

3.2 Interiority

The analysis uses an interior parameter domain. The next two conditions ensure positive assignment probabilities and positive record probabilities throughout that domain.

Assumption 6. `[ass:interior-assignment]` (Interior treatment assignment probability) *The treatment assignment probability p satisfies*

$$0 < p < 1.$$

Strict treatment overlap assigns positive probability to each arm, so both arm means enter the observed-record distribution [Ohnishi and Awan, 2025]. We also impose interiority on those means.

Assumption 7. `[ass:interior-means]` (Interior arm means) *The arm-mean parameter θ satisfies*

$$\theta \in (0, 1)^2.$$

The interior Bernoulli parameter condition gives each outcome category positive probability within its arm and provides an open parameter domain for information calculations [Steinberger, 2024]. Together with strict overlap, it makes all four coordinates of π_θ positive. We collect the resulting product experiments in the following model.

Assumption 8. `[ass:fixed-privacy]` (Fixed positive privacy budget) *The privacy budget is a fixed real number $\varepsilon > 0$, and the privacy parameter at every sample size $n \in \mathbb{N}$ equals ε .*

In Definition 1, the assignment probability remains fixed as the arm means vary. This separates the known trial design from the two-dimensional outcome parameter whose contrast is the treatment effect.

3.3 Private releases

A stationary locally private channel, denoted by Q , maps each private record to a randomized output on a measurable space $(\mathcal{Z}, \mathcal{G})$. Stationarity means that the same release rule is used for each subject, with fresh randomization. For every measurable event A and every pair of records $x, x' \in \mathcal{X}$, its local privacy requirement is

$$Q(A \mid x) \leq e^\varepsilon Q(A \mid x').$$

Here ε is the privacy budget. Because the inputs are assignment–outcome pairs, the comparison protects assignment and outcome jointly: x and x' may differ in either or both coordinates.

The privacy budget is held fixed across the sequence of experiments.

Assumption 9. `[ass:sequential-ldp]` (Sequential local privacy) *The release rules Q_i satisfy*

$$Q_i(A \mid x, h) \leq \exp(\varepsilon) Q_i(A \mid x', h)$$

for every stage i , every measurable output event $A \in \mathcal{G}_i$, every pair of private records $x, x' \in \mathcal{X}$, and every history tuple $h \in \prod_{j < i} \mathcal{Z}_j$, including null or otherwise unattainable histories in the full history product.

The fixed local privacy budget condition keeps the per-subject protection level constant as sample size grows [Steinberger, 2024]. It therefore supplies a common privacy constraint for comparing release rules.

Sequential releases allow the rule for a new subject to depend on earlier private outputs. Let Z_i denote subject i 's released observation, taking values in the measurable output space $(\mathcal{Z}_i, \mathcal{G}_i)$. The preceding-release history H_{i-1} is

$$H_{i-1} = (Z_1, \dots, Z_{i-1}),$$

with the empty history at $i = 1$. Conditional on the current record and this history, the release rule Q_i generates Z_i according to $Q_i(\cdot \mid X_i, H_{i-1})$. Each subject contributes one release; adaptation proceeds through the outputs of preceding subjects. The privacy requirement applies uniformly over the full history space.

Definition 1. [def:binary-trial-model] (Binary-trial model $\mathcal{M}$) *The model is*

$$\mathcal{M} = \left\{ (P_\theta^{\otimes n})_{n \geq 1} : \theta = (\mu_0, \mu_1)^\top \in (0, 1)^2, P_\theta(X_i = x_j) = \pi_{\theta,j} \right\},$$

under Assumptions 6 and 7. Here μ_0 and μ_1 are the control-arm and treatment-arm means, X_i is subject i 's private record, x_j is an enumerated record value, and $\pi_{\theta,j}$ is its probability under the record law P_θ . The law $P_\theta^{\otimes n}$ is the product law of n subject records.

Sequentially interactive local differential privacy permits history-dependent release rules while retaining the same recordwise protection at each stage [Steinberger, 2024]. The quantification over every history makes the privacy guarantee a property of the release rule throughout its domain, including histories with zero probability under a particular parameter. The protocol class $\mathfrak{Q}_{n,\varepsilon}^{\text{seq}}$ combines this guarantee with measurability and nonanticipation.

Definition 2. [def:sequential-channels] (Sequential private protocols $\mathfrak{Q}_{n,\varepsilon}^{\text{seq}}$) *The class $\mathfrak{Q}_{n,\varepsilon}^{\text{seq}}$ consists of n -stage protocols*

$$Q^{(n)} = (Q_i)_{i=1}^n$$

on arbitrary measurable output spaces $(\mathcal{Z}_i, \mathcal{G}_i)$, subject to Assumptions 8 and 9, with the following properties:

- **(Kernels.)** Each $Q_i(\cdot \mid x, h)$ is a Markov kernel from $\mathcal{X} \times \prod_{j < i} \mathcal{Z}_j$ to $(\mathcal{Z}_i, \mathcal{G}_i)$.
- **(Nonanticipation.)** Each release rule is nonanticipating.
- **(Privacy.)** For every stage i , measurable set $A \in \mathcal{G}_i$, inputs $x, x' \in \mathcal{X}$, and history $h \in \prod_{j < i} \mathcal{Z}_j$,

$$Q_i(A \mid x, h) \leq e^\varepsilon Q_i(A \mid x', h).$$

Nonanticipation in Definition 2 restricts each rule to the current subject's record and preceding releases. A stationary channel yields a member of this class by taking $Q_i(\cdot \mid x, h) = Q(\cdot \mid x)$. More generally, earlier private releases may determine the channel used for later subjects.

3.4 Output information and cardinality

Efficiency comparisons use the output Fisher-information matrix $I_\theta(Q)$ of a stationary channel and the active-output cardinality $\kappa(Q)$. The following definition fixes these quantities for measurable output spaces and specifies how singular information is treated when evaluating the treatment-effect contrast.

For a stationary channel Q on $(\mathcal{Z}, \mathcal{G})$, let

$$\nu = \sum_{j=0}^3 Q(\cdot | x_j), \quad k_j = \frac{dQ(\cdot | x_j)}{d\nu}, \quad f_\theta = \sum_{j=0}^3 \pi_{\theta,j} k_j.$$

Holding Q and p fixed, define

$$I_\theta(Q) = \int_{\{f_\theta > 0\}} \frac{(\nabla_\theta f_\theta)(\nabla_\theta f_\theta)^\top}{f_\theta} d\nu.$$

The contrast variance is $c^\top I_\theta(Q) c$ when $c \in \text{range}(I_\theta(Q))$, and $+\infty$ otherwise. Let $P_\theta Q$ be the output law and $(P_\theta Q)^*$ its induced outer measure. For every finite underlying output space, write

$$\kappa(Q) = \# \{z \in \mathcal{Z} : (P_\theta Q)^*({z}) > 0\};$$

for every infinite underlying output space, set $\kappa(Q) = +\infty$. On a finite power-set alphabet each singleton is measurable, so $(P_\theta Q)^*({z}) = (P_\theta Q)({z})$, recovering the ordinary count of positive-probability output labels. The outer-measure form also gives the stated convention when a finite carrier has a coarser sigma-field.

This density formulation expresses information directly in terms of the released observation. The range condition accommodates channels that retain information about the treatment-effect contrast while their information matrix is singular in the full arm-mean parameter. Its extended-value convention makes estimability part of the channel comparison.

For finite power-set alphabets, $\kappa(Q)$ counts output labels with positive probability. Since every input record has positive probability under [Definition 1](#), an output label is active exactly when at least one of the four conditional release probabilities is positive. The count is therefore independent of the interior arm means and measures the number of active labels used by a fixed finite release rule.

4 Exact efficiency bound and finite optimization

The contrast-variance convention from the preceding section provides a common criterion for stationary private channels, including channels with singular information matrices that identify the treatment effect. We now characterize its optimum by a finite staircase program and connect that optimum to local squared-error risk for the sequential protocols in [Definition 2](#). Throughout this section, assignment is interior, the two arm means are interior, and privacy is fixed as specified in [Assumptions 6 to 8](#).

4.1 Staircase channels and nuisance projection

A staircase release associates each output with a nonconstant pattern on the four-record alphabet. Its conditional release probabilities take two values whose ratio is e^ε . The following

enumeration fixes the fourteen patterns S_j used to index channel weights and, subsequently, active supports.

Definition 3. [synth_8] (Ordered staircase patterns) *For the ordered private-record alphabet*

$$\mathcal{X} = \{x_0, x_1, x_2, x_3\} = \{(0, 0), (0, 1), (1, 0), (1, 1)\},$$

let $\mathcal{S} = \{S : \emptyset \neq S \subsetneq \{0, 1, 2, 3\}\}$. Define its fixed enumeration by

$$(S_1, \dots, S_{14}) = (\{0\}, \{1\}, \{0, 1\}, \{2\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}, \\ \{3\}, \{0, 3\}, \{1, 3\}, \{0, 1, 3\}, \{2, 3\}, \{0, 2, 3\}, \{1, 2, 3\}).$$

Thus S_j consists of the positions of the ones in the four-bit binary expansion of j , with position 0 the least significant bit. A pattern S corresponds to the record subset $\{x_k : k \in S\}$. This order indexes all staircase-weight coordinates and certified active sets.

The channel is determined by a nonnegative weight vector α . Row normalization restricts these weights to a polytope, while the record probabilities in Definition 1 determine the information carried by each pattern. The next definition gathers the normalization constraints and the directional information objective.

The privacy increment d determines the two entries of each pattern vector b_S . The feasible set $\mathcal{A}_\varepsilon$ enforces all four channel row sums simultaneously. For each pattern, the denominator $h_S(\theta)$ is its probability factor, and the vector g_S is the derivative of that factor with respect to the arm means. These quantities give a direct construction of the staircase channel Q_α and its projected output score $\rho_S(\theta, t)$.

For $\alpha \in \mathcal{A}_\varepsilon$, the stationary staircase channel on $\mathcal{S}$ is

$$Q_\alpha(S \mid x_k) = \alpha_S b_S(k), \quad \Pr_\theta(S) = \alpha_S h_S(\theta).$$

For an active pattern S , its output score and projected score are

$$s_S^{\text{out}}(\theta) = \frac{g_S}{h_S(\theta)}, \quad \rho_S(\theta, t) = \frac{g_S^\top v(t)}{h_S(\theta)}.$$

Thus $I_\theta(Q_\alpha) = I_\theta(\alpha)$ and

$$\kappa(Q_\alpha) = \#\{S \in \mathcal{S} : \alpha_S > 0\}.$$

Write $\text{vert}(\mathcal{A}_\varepsilon)$ for the vertices of the feasible weight polytope.

Feasibility in Definition 4 makes these release probabilities sum to one for every record. Within an active output, their largest-to-smallest ratio is e^ε . The output score $s_S^{\text{out}}(\theta)$ therefore yields the staircase information matrix $I_\theta(\alpha)$ directly from the released pattern.

Nuisance projection enters through the direction $v(t)$. Since $c^\top v(t) = 1$, these directions parameterize all arm-mean perturbations that change the treatment effect by one unit. Varying t adds a common shift to both arm means, which leaves their difference unchanged. Minimizing the directional information $F_\theta(\alpha, t)$ over t thus profiles out that common-mean nuisance direction. When the contrast is estimable, the resulting minimum is the reciprocal of the contrast variance under the preceding convention.

Optimizing this profiled value over feasible releases defines the information benchmark $J^*(\theta, p, \varepsilon)$ and its reciprocal variance benchmark $V^*(\theta, p, \varepsilon)$.

Definition 4. [synth_4] (Directional information objective) Fix $\theta = (\mu_0, \mu_1)^\top \in (0, 1)^2$, $p \in (0, 1)$, and $\varepsilon > 0$. Write $q = 1 - p$, $d = e^\varepsilon - 1$, and

$$\pi_\theta = (q(1 - \mu_0), q\mu_0, p(1 - \mu_1), p\mu_1)^\top.$$

For $S \in \mathcal{S} = \{S : \emptyset \neq S \subsetneq \{0, 1, 2, 3\}\}$, define

$$b_S = \mathbf{1} + d\mathbf{1}_S, \quad h_S(\theta) = \pi_\theta^\top b_S, \quad g_S = d \begin{pmatrix} q\{\mathbf{1}_S(1) - \mathbf{1}_S(0)\} \\ p\{\mathbf{1}_S(3) - \mathbf{1}_S(2)\} \end{pmatrix}.$$

For feasible weights

$$\alpha \in \mathcal{A}_\varepsilon = \left\{ \alpha \in \mathbb{R}_+^{14} : \sum_{S \in \mathcal{S}} \alpha_S b_S = \mathbf{1} \right\}$$

and $t \in \mathbb{R}$, set $v(t) = (t, t + 1)^\top$. The directional information objective is

$$F_\theta(\alpha, t) = \sum_{S \in \mathcal{S}} \alpha_S \frac{(g_S^\top v(t))^2}{h_S(\theta)} = v(t)^\top I_\theta(\alpha) v(t), \quad I_\theta(\alpha) = \sum_{S \in \mathcal{S}} \alpha_S \frac{g_S g_S^\top}{h_S(\theta)}.$$

Its nuisance-profiled value at α is $\min_{t \in \mathbb{R}} F_\theta(\alpha, t)$; the dependence on p and ε enters through π_θ , g_S , b_S , and $\mathcal{A}_\varepsilon$.

The stationary comparison ranges over every measurable output space, with contrast estimability incorporated into its extended-value criterion. The following result identifies its optimum with the finite staircase benchmark and gives an equivalent optimization over the vertices of the weight polytope.

Definition 5. [def:staircase-saddle] (Profiled information J^* and variance V^*) The optimized profiled information and its reciprocal variance are

$$J^*(\theta, p, \varepsilon) = \max_{\alpha \in \mathcal{A}_\varepsilon} \min_{t \in \mathbb{R}} \sum_{S \in \mathcal{S}} \alpha_S \frac{(g_S^\top v(t))^2}{h_S(\theta)}, \quad V^*(\theta, p, \varepsilon) = \frac{1}{J^*(\theta, p, \varepsilon)}.$$

The finite reduction reflects two features of the objective: it is linear in the channel weights for a fixed profiling direction, and it is quadratic in the profiling coordinate for fixed weights. Accordingly, [Theorem 1](#) expresses the optimized information as the minimum of a finite upper envelope of convex quadratics. This gives a one-dimensional representation of the efficiency surface at each specified pair of arm means, assignment probability, and privacy level. The contrast-variance convention also permits a singular channel to attain the optimum when its information range contains the contrast.

The staircase refinement underlying the comparison with arbitrary stationary output spaces is presented in [Section 9](#). For the sequential lower-bound construction, we also record the smooth scalar localization density used there.

Theorem 1. [thm:finite-oracle] (Finite staircase oracle) Let $\theta = (\mu_0, \mu_1)^\top$, let p be the treatment-assignment probability, and let ε be the privacy parameter. Suppose:

- (Assignment.) p satisfies [Assumption 6](#).
- (Means.) θ satisfies [Assumption 7](#).

- **(Fixed privacy.)** The privacy sequence and ε satisfy [Assumption 8](#).

Use the staircase objective and optimized values $J^*(\theta, p, \varepsilon)$ and $V^*(\theta, p, \varepsilon) = J^*(\theta, p, \varepsilon)^{-1}$ from [Definition 5](#). For a stationary channel Q , let $I_\theta(Q)$ be the output Fisher-information matrix, obtained by integrating the products of output-density derivatives divided by the output density against the channel's dominating measure. Evaluate its contrast variance as

$$c^\top I_\theta(Q)^+ c = \begin{cases} \inf_{\{v: I_\theta(Q)v=c\}} c^\top v, & c \in \text{range}(I_\theta(Q)), \\ +\infty, & c \notin \text{range}(I_\theta(Q)). \end{cases}$$

For every measurable output space and every Markov channel Q on the four-record alphabet satisfying

$$Q(A \mid x) \leq e^\varepsilon Q(A \mid x') \quad \text{for every measurable } A \text{ and every } x, x',$$

the oracle bound is

$$V^*(\theta, p, \varepsilon) \leq c^\top I_\theta(Q)^+ c.$$

There exists a vector α of fourteen nonnegative staircase-pattern weights, with all four channel row sums equal to one, such that its staircase channel Q_α satisfies the same privacy inequalities and

$$c^\top I_\theta(Q_\alpha)^+ c = c^\top I_\theta(\alpha)^+ c = V^*(\theta, p, \varepsilon), \quad I_\theta(\alpha) = \sum_{S \in \mathcal{S}} \alpha_S \frac{g_S g_S^\top}{h_S(\theta)}.$$

Here the pattern vectors and denominators are those entering the staircase objective in [Definition 5](#). Consequently,

$$\inf_Q c^\top I_\theta(Q)^+ c = V^*(\theta, p, \varepsilon),$$

where the infimum ranges over all such stationary private channels and measurable output spaces. Moreover,

$$J^*(\theta, p, \varepsilon) = \inf_{t \in \mathbb{R}} \max_{\alpha \in \mathcal{A}_\varepsilon} F_\theta(\alpha, t) = \inf_{t \in \mathbb{R}} \max_{\alpha \in \text{vert}(\mathcal{A}_\varepsilon)} F_\theta(\alpha, t),$$

and there exists $t^* \in \mathbb{R}$ such that

$$\max_{\alpha \in \mathcal{A}_\varepsilon} F_\theta(\alpha, t^*) = J^*(\theta, p, \varepsilon).$$

The density in [Definition 15](#) combines compact localization with vanishing density and derivative at the endpoints. Its prior-information contribution is explicit and decreases as the localization radius grows. The directional construction and sequential lower-bound apparatus using this density are deferred to [Section 9](#).

4.2 Local alternatives and regular procedures

We first define the local perturbations used to compare procedures near an interior baseline. A perturbation of order $n^{-1/2}$ changes the treatment-effect contrast on the same scale as the estimator's limiting error.

Definition 6. [synth_2] (Local parameter alternative) The local alternative $\theta_{n,h}$ is defined coordinatewise by

$$\theta_{n,h}(k) = \theta_0(k) + \frac{h(k)}{\sqrt{n}}, \quad k \in \{0, 1\},$$

for a baseline vector $\theta_0 \in \mathbb{R}^2$, a perturbation vector $h \in \mathbb{R}^2$, and a nonnegative integer n . At $n = 0$, the quotient is defined to be zero, so $\theta_{0,h} = \theta_0$.

The corresponding scaled error centers the estimate at the contrast under this local parameter, so fixed perturbations can share a common limiting error law.

Definition 7. [synth_3] (Scaled local estimation error) *The scaled local estimation error $U_{n,h}^{\mathcal{P}}$ is defined by*

$$U_{n,h}^{\mathcal{P}}(z) = \sqrt{n} \left(\hat{\tau}_n^{\mathcal{P}}(z) - \left[\theta_0(1) + \frac{h(1)}{\sqrt{n}} - \theta_0(0) - \frac{h(0)}{\sqrt{n}} \right] \right).$$

Here $\mathcal{P}$ is a sequence of sequential ε -locally private protocols, their transcript laws factorizing through the release channels, and measurable real-valued estimators; $\hat{\tau}_n^{\mathcal{P}}(z)$ denotes its estimate from transcript z . The parameters are $\varepsilon \in \mathbb{R}$, $\theta_0, h \in \mathbb{R}^2$, and $n \in \{0, 1, \dots\}$, with z any element of the product of the n measurable release spaces. At $n = 0$, each quotient $h(k)/\sqrt{n}$ is defined to be zero.

Regularity combines a common local limiting law with squared-error tail control on bounded local neighborhoods. The next definition packages the admissible perturbations, procedure sequence, scaled error, and regular class used by the minimax theorem.

Definition 8. [synth_6] (Local alternatives and regular sequential private procedures) *Fix $p \in (0, 1)$, $\varepsilon > 0$, and $\theta_0 \in (0, 1)^2$. For $n \geq 1$ and $h \in \mathbb{R}^2$, set*

$$\theta_{n,h} = \theta_0 + \frac{h}{\sqrt{n}}.$$

For $0 < H < \infty$, the admissible local perturbations are

$$\mathcal{H}_{n,H}(\theta_0) = \left\{ h \in \mathbb{R}^2 : \|h\|_2 \leq H, \ 0 < \theta_{0,k} + \frac{h_k}{\sqrt{n}} < 1 \text{ for } k = 0, 1 \right\}.$$

A sequential private procedure sequence is

$$\mathcal{P} = ((Q^{(n)}, \hat{\tau}_n))_{n \geq 1},$$

where $Q^{(n)} \in \mathfrak{Q}_{n,\varepsilon}^{\text{seq}}$ for every n , and $\hat{\tau}_n$ is $\sigma(Z_1, \dots, Z_n)$ -measurable. With $c = (-1, 1)^\top$, its scaled local error is

$$U_{n,h}^{\mathcal{P}} = \sqrt{n} \{ \hat{\tau}_n - c^\top \theta_{n,h} \}.$$

We say that $\mathcal{P}$ is regular at θ_0 when there exists a Borel probability law $L_{\mathcal{P}}$ on $\mathbb{R}$ satisfying

$$\int u^2 dL_{\mathcal{P}}(u) < \infty$$

such that, for every fixed $h \in \mathbb{R}^2$,

$$U_{n,h}^{\mathcal{P}} \rightsquigarrow L_{\mathcal{P}}$$

under the experiment at $\theta_{n,h}$, with the same law $L_{\mathcal{P}}$ for every fixed h , and the local squared errors are asymptotically uniformly integrable: for every $0 < H < \infty$,

$$\lim_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} \sup_{h \in \mathcal{H}_{n,H}(\theta_0)} \mathbb{E}_{\theta_{n,h}}^{\mathcal{P}} [(U_{n,h}^{\mathcal{P}})^2 \mathbf{1}\{|U_{n,h}^{\mathcal{P}}| > M\}] = 0.$$

For fixed h , the local alternative is interior for all sufficiently large n . The class $\mathfrak{P}_{\varepsilon}^{\text{reg}}(\theta_0)$ consists of all complete procedure sequences satisfying these requirements.

The common limit makes local error behavior comparable across fixed perturbations, while uniform integrability connects convergence in distribution to squared-error risk. We now turn to the pilot allocation used for attainment. Its first requirement supplies an increasing number of private observations for learning the arm means [Kalinin and Steinberger, 2025].

Assumption 10. [ass:pilot-diverges] (Diverging pilot size) *The pilot sample size m_n satisfies*

$$m_n \rightarrow \infty.$$

The second pilot requirement appears next. Together, the two conditions let the pilot learn the arm means while reserving an asymptotically full fraction of subjects for the selected channel.

4.3 Private pilot, channel selection, and sequential efficiency

The pilot fraction vanishes, subject to the finite-sample restrictions that give both stages observations once $n \geq 2$.

Assumption 11. [ass:pilot-sublinear] (Sublinear pilot sample size) *The pilot sample sizes m_n are nonnegative integers satisfying*

$$\frac{m_n}{n} \rightarrow 0 \quad \text{as } n \rightarrow \infty, \quad 1 \leq m_n < n \quad \text{for every integer } n \geq 2.$$

The preliminary release is four-category randomized response on the zero-based record alphabet. Its probabilities are affine in the record probabilities, so their inverse estimates the two success cells used for the arm means.

Definition 9. [synth_10] (Four-category randomized-response pilot) *For $\varepsilon > 0$ and the ordered private-record categories*

$$x_0 = (0, 0), \quad x_1 = (0, 1), \quad x_2 = (1, 0), \quad x_3 = (1, 1),$$

four-category ε -randomized response is the release kernel on $\{0, 1, 2, 3\}$ defined by

$$Q_{\text{RR}}(k \mid x_j) = \begin{cases} \frac{\exp(\varepsilon)}{\exp(\varepsilon) + 3}, & k = j, \\ \frac{1}{\exp(\varepsilon) + 3}, & k \neq j, \end{cases} \quad j, k \in \{0, 1, 2, 3\}.$$

The pilot releases $R_1, \dots, R_{m_n}$ are generated by applying this kernel to the first m_n private records with independent randomization. Consequently, for each reported category k ,

$$\Pr_{\theta}(R_i = k) = \frac{1 + (\exp(\varepsilon) - 1)\pi_{\theta,k}}{\exp(\varepsilon) + 3}.$$

Inverting these probabilities and clipping the result supplies the interior pilot estimate $\tilde{\theta}$. Under interior assignment and positive privacy, a measurable strong-saddle selector exists; the construction fixes any such selector. It jointly returns feasible channel weights, a minimizing profiling coordinate, and the optimized information value. Conditional on the pilot, the selected channel and projected-score correction are fixed.

Algorithm 1 (Private pilot estimator)

The private pilot estimator is the sequence of sequential ε -locally private protocols and measurable estimators defined below, for an assignment probability p satisfying [Assumption 6](#), a privacy parameter $\varepsilon > 0$, and an arbitrary pilot-size schedule $m : \mathbb{N} \rightarrow \mathbb{N}$, with $m_n = m(n)$.

1. Fix a measurable selector defined on all real parameter pairs,

$$\vartheta \mapsto (\alpha_\vartheta, t_\vartheta, J_\vartheta),$$

whose weight component has fourteen real coordinates. For every interior parameter ϑ , its values satisfy

$$\begin{aligned} \alpha_\vartheta &\in \mathcal{A}_\varepsilon, \\ t_\vartheta &\in \arg \min_{t \in \mathbb{R}} F_\vartheta(\alpha_\vartheta, t), \\ J_\vartheta &= F_\vartheta(\alpha_\vartheta, t_\vartheta) = J^*(\vartheta, p, \varepsilon), \\ F_\vartheta(\alpha, t_\vartheta) &\leq J_\vartheta \quad \text{for every } \alpha \in \mathcal{A}_\varepsilon. \end{aligned}$$

The staircase quantities are as in [Definition 5](#).

2. At sample size n , release each available record with index $i \leq m_n$ through four-category ε -randomized response, obtaining pilot releases $R_i \in \{0, 1, 2, 3\}$, with the category indexing of [Definition 9](#). Define

$$\hat{u}_k = \frac{1}{m_n} \sum_{i=1}^{\min\{m_n, n\}} \mathbf{1}\{R_i = k\}, \quad \hat{\pi}_k = \frac{(\exp(\varepsilon) + 3)\hat{u}_k - 1}{\exp(\varepsilon) - 1}, \quad k \in \{0, 1, 2, 3\}.$$

Reciprocals of zero are interpreted as zero, and empty sums as zero.

3. Define the clipping margin and pilot parameter estimate by

$$\begin{aligned} \delta_n &= (m_n + 2)^{-1}, \\ \tilde{\theta}_0 &= \max \left\{ \delta_n, \min \left\{ 1 - \delta_n, \frac{\hat{\pi}_1}{1 - p} \right\} \right\}, \\ \tilde{\theta}_1 &= \max \left\{ \delta_n, \min \left\{ 1 - \delta_n, \frac{\hat{\pi}_3}{p} \right\} \right\}, \\ \tilde{\theta} &= (\tilde{\theta}_0, \tilde{\theta}_1)^\top. \end{aligned}$$

Here $\hat{\pi}_1$ and $\hat{\pi}_3$ use the zero-based control-success and treatment-success categories, respectively. Evaluate the selector at this estimate:

$$(\tilde{\alpha}, \tilde{t}, \tilde{J}) = (\alpha_{\tilde{\theta}}, t_{\tilde{\theta}}, J_{\tilde{\theta}}).$$

4. Release each remaining record through the selected staircase channel:

$$S_i \sim Q_{\tilde{\alpha}}(\cdot \mid X_i), \quad m_n < i \leq n,$$

conditionally on the pilot releases and the remaining records. The transcript law is the law generated by the successive release kernels. Output

$$\hat{\tau}^* = c^\top \tilde{\theta} + (\max\{n - m_n, 0\})^{-1} \sum_{\substack{1 \leq i \leq n \\ m_n < i}} \frac{g_{S_i}^\top v(\tilde{t})}{\tilde{J} h_{S_i}(\tilde{\theta})}.$$

Selection depends only on released pilot observations, so the construction belongs to the history-dependent protocol class in [Definition 2](#). The theorem below uses distinct quantifiers for its two conclusions: the lower bound applies to every sequential private procedure sequence with measurable estimators, while attainment and the infimum equality are stated within the regular class. Both conclusions allow arbitrary measurable release spaces and history-dependent channels, and attainment retains the selector condition in [Algorithm 1](#).

Theorem 2. [thm:sequential-minimax] (Sequential local minimax efficiency) *Fix the treatment-assignment probability p , the privacy level ε , and the arm-mean parameter θ_0 . Suppose:*

- *(Interior assignment.) p satisfies [Assumption 6](#).*
- *(Interior means.) θ_0 satisfies [Assumption 7](#).*
- *(Fixed privacy.) The privacy sequence and ε satisfy [Assumption 8](#).*
- *(Strong-saddle selection.) Fix any selector $\vartheta \mapsto (\alpha_\vartheta, t_\vartheta, J_\vartheta)$ satisfying the strong-saddle selector conditions in [Algorithm 1](#).*

Write $V^*(\theta_0, p, \varepsilon) = J^*(\theta_0, p, \varepsilon)^{-1}$ for the reciprocal optimized profiled information in [Definition 5](#). For an arm-mean parameter $\theta = (\mu_0, \mu_1)^\top$, the treatment-effect target is

$$\tau(\theta) = \mu_1 - \mu_0.$$

For $n \geq 1$ and $H > 0$, define the local alternatives and admissible perturbations by

$$\theta_{n,h} = \theta_0 + \frac{h}{\sqrt{n}}, \quad \mathcal{H}_{n,H}(\theta_0) = \left\{ h \in \mathbb{R}^2 : \|h\|_2 \leq H, \quad \theta_0 + \frac{h}{\sqrt{n}} \in (0, 1)^2 \right\}.$$

For a procedure sequence $\mathcal{P}$, let its scaled local estimation error be

$$U_{n,h}^{\mathcal{P}} = \sqrt{n} \left(\hat{\tau}_n^{\mathcal{P}} - \tau(\theta_{n,h}) \right),$$

where $\hat{\tau}_n^{\mathcal{P}}$ is its transcript-measurable estimator.

For every sequence $\mathcal{P}$ of sequential ε -locally private protocols and measurable estimators, with arbitrary measurable output spaces indexed by sample size and release stage and with transcript laws factorizing according to the history-dependent kernels in [Definition 2](#),

$$V^*(\theta_0, p, \varepsilon) \leq \sup_{H>0} \liminf_{n \rightarrow \infty} \sup_{h \in \mathcal{H}_{n,H}(\theta_0)} \mathbb{E}_{\theta_{n,h}}^{\mathcal{P}} [(U_{n,h}^{\mathcal{P}})^2].$$

Let $\mathfrak{P}_\varepsilon^{\text{reg}}(\theta_0)$ denote the class of these procedure sequences whose scaled local errors have a common local weak limit law $L_{\mathcal{P}}$ with finite second moment,

$$\int_{\mathbb{R}} u^2 L_{\mathcal{P}}(du) < \infty,$$

and whose squared-error tails satisfy, for every $H > 0$,

$$\lim_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} \sup_{h \in \mathcal{H}_{n,H}(\theta_0)} \mathbb{E}_{\theta_{n,h}}^{\mathcal{P}} [(U_{n,h}^{\mathcal{P}})^2 \mathbf{1}\{|U_{n,h}^{\mathcal{P}}| > M\}] = 0.$$

There exists a pilot schedule, specifically

$$m_n = \lfloor \sqrt{n} \rfloor,$$

satisfying [Assumptions 10](#) and [11](#), for which the procedure in [Algorithm 1](#), using the fixed selector above, belongs to $\mathfrak{P}_\varepsilon^{\text{reg}}(\theta_0)$ and attains equality in the lower bound. Moreover,

$$\inf_{\mathcal{P} \in \mathfrak{P}_\varepsilon^{\text{reg}}(\theta_0)} \sup_{H > 0} \liminf_{n \rightarrow \infty} \sup_{h \in \mathcal{H}_{n,H}(\theta_0)} \mathbb{E}_{\theta_{n,h}}^{\mathcal{P}} [(U_{n,h}^{\mathcal{P}})^2] = V^*(\theta_0, p, \varepsilon).$$

All risks and the infimum are interpreted in the extended nonnegative reals, with the real variance constant embedded therein.

The equality in [Theorem 2](#) connects the stationary channel problem to the full sequential experiment: the same variance constant governs the local risk lower bound and is attained by the regular private-pilot construction. Intuitively, the diverging pilot learns the design point, while its vanishing sample fraction leaves the first-order allocation to the selected staircase release. With the fixed measurable strong-saddle selector, the schedule $m_n = \lfloor \sqrt{n} \rfloor$ implements this allocation and attains the benchmark.

The risk criterion first takes the worst squared error over admissible perturbations at a fixed radius, then its lower asymptotic limit, and finally the supremum over radii. Under these quantifiers, [Theorem 2](#) identifies $V^*(\theta_0, p, \varepsilon)$ as the exact regular local asymptotic squared-error constant at the specified interior design point.

5 Private-pilot estimation and Wald inference

The efficiency benchmark in [Theorem 2](#) is attained by a procedure that uses private pilot releases to select the main-sample channel. The construction combines the pilot treatment-effect estimate with an affine correction computed from the remaining releases. Under interior assignment and positive privacy, a measurable strong-saddle selector exists. The procedure fixes any such selector satisfying the interface in [Algorithm 1](#): at every interior parameter, it returns feasible staircase weights, a minimizing profiling coordinate, and the optimal information value, with the selected weights also maximizing the objective at that coordinate.

Two restrictions govern the pilot allocation:

$$m_n \longrightarrow \infty, \quad \frac{m_n}{n} \longrightarrow 0,$$

as specified in [Assumptions 10](#) and [11](#). The first supplies an increasing number of private observations for learning the arm means; the second allocates an asymptotically full fraction of subjects to the selected main-sample channel. The attainment result below applies to an arbitrary schedule satisfying these restrictions. The schedule $m_n = \lfloor \sqrt{n} \rfloor$ exhibited for the minimax equality in [Theorem 2](#) is one such choice.

To describe the estimator, we first specify the selected affine correction $\tilde{\phi}(S)$. It combines the selected profiling direction with the pattern-specific score and normalizes by the selected information value.

Definition 10. [[synth_9](#)] (Pilot-selected affine correction) *Let $p \in (0, 1)$, $\varepsilon > 0$, and let $\tilde{\theta} \in (0, 1)^2$ be the private pilot estimate. Apply the measurable strong-saddle selector of [Algorithm 1](#) to obtain*

$$(\tilde{\alpha}, \tilde{t}, \tilde{J}) = (\alpha_{\tilde{\theta}}, t_{\tilde{\theta}}, J_{\tilde{\theta}}), \quad \tilde{J} = F_{\tilde{\theta}}(\tilde{\alpha}, \tilde{t}) = J^*(\tilde{\theta}, p, \varepsilon) > 0.$$

For each $S \in \mathcal{S}$, define the selected affine correction by

$$\tilde{\phi}(S) = \frac{g_S^\top v(\tilde{t})}{\tilde{J} h_S(\tilde{\theta})}.$$

Here $v(t) = (t, t+1)^\top$, $h_S(\vartheta) = \pi_\vartheta^\top b_S$, and $g_S = \nabla_\vartheta h_S(\vartheta)$. The selected weights determine the main-sample channel

$$Q_{\tilde{\alpha}}(S \mid x_j) = \tilde{\alpha}_S b_S(j)$$

and the information value

$$\tilde{J} = \sum_{T \in \mathcal{S}} \tilde{\alpha}_T \frac{(g_T^\top v(\tilde{t}))^2}{h_T(\tilde{\theta})}.$$

For a transcript entry that is a pilot release, set its correction to zero; for a main-sample pattern release S , use $\tilde{\phi}(S)$.

The pilot determines both the release probabilities and the numerical correction attached to each main-sample pattern. Conditional on the private pilot history $H_{m_n} = (R_1, \dots, R_{m_n})$, these choices are fixed. The remaining subjects then supply releases under the selected channel. The following boxed algorithm collects the implementation in [Algorithm 1](#), displaying the steps for sample sizes with $1 \leq m_n < n$, as eventually implied by the two pilot-size restrictions.

Private-pilot estimation algorithm

Inputs. The known assignment probability $p \in (0, 1)$, fixed privacy level $\varepsilon > 0$, a pilot schedule satisfying

$$m_n \rightarrow \infty, \quad m_n/n \rightarrow 0,$$

and a fixed measurable strong-saddle selector

$$\vartheta \mapsto (\alpha_\vartheta, t_\vartheta, J_\vartheta)$$

defined on all real parameter pairs and satisfying all the interior-parameter selection conditions in [Algorithm 1](#), including maximization over feasible weights at the selected profiling coordinate. The attainment results assume a fixed measurable strong-saddle selector satisfying these conditions. Numerical evaluation of this input requires a separately specified solver and tie-breaking rule.

1. **Release and estimate the pilot.** For the first m_n subjects, use the four-category ε -randomized-response rule in [Definition 9](#). From the category frequencies, compute

$$\hat{u}_k = \frac{1}{m_n} \sum_{i=1}^{m_n} \mathbf{1}\{R_i = k\}, \quad \hat{\pi}_k = \frac{(e^\varepsilon + 3)\hat{u}_k - 1}{e^\varepsilon - 1}, \quad k \in \{0, 1, 2, 3\}.$$

With $\delta_n = (m_n + 2)^{-1}$, set

$$\begin{aligned} \tilde{\theta}_0 &= \max \left\{ \delta_n, \min \left\{ 1 - \delta_n, \frac{\hat{\pi}_1}{1-p} \right\} \right\}, \\ \tilde{\theta}_1 &= \max \left\{ \delta_n, \min \left\{ 1 - \delta_n, \frac{\hat{\pi}_3}{p} \right\} \right\}. \end{aligned}$$

The zero-based categories 1 and 3 represent control success and treatment success, respectively.

2. **Select the main-sample channel.** Evaluate the input selector at $\tilde{\theta} = (\tilde{\theta}_0, \tilde{\theta}_1)^\top$ to obtain $(\tilde{\alpha}, \tilde{t}, \tilde{J})$.
3. **Release the remaining records.** For $m_n < i \leq n$, generate the pattern release

$$S_i \sim Q_{\tilde{\alpha}}(\cdot \mid X_i)$$

using the selected channel in [Definition 10](#).

4. **Estimate the treatment effect.** Report the pilot contrast plus the average selected correction:

$$\hat{\tau}^* = \tilde{\theta}_1 - \tilde{\theta}_0 + \frac{1}{n - m_n} \sum_{i=m_n+1}^n \tilde{\phi}(S_i).$$

The complete procedure, including its conventions for every sample size, is defined in [Algorithm 1](#).

Clipping places the pilot estimate in the interior parameter space where the selector condi-

tions apply. The main-sample correction then targets the discrepancy between the pilot contrast and the true treatment effect. The conditional unbiasedness clause in [Theorem 3](#) gives this centering an exact interpretation: the estimation error has mean zero conditional on the private pilot history. Consequently, the theorem accommodates every pilot schedule satisfying the two stated limits.

For studentization, use the empirical variance estimator $\hat{V}^*$. It centers the main-sample corrections at their empirical mean, so the pilot contrast, which is fixed conditional on the pilot history, contributes no term to this empirical variance.

Definition 11. `[synth_1]` (Empirical variance of selected corrections) *The variance estimator $\hat{V}^*$ is defined for any $p, \varepsilon \in \mathbb{R}$, pilot-size schedule $m : \mathbb{N} \rightarrow \mathbb{N}$, sample size $n \in \mathbb{N}$, transcript z of n pilot-or-pattern releases, and selector mapping each real parameter pair to a staircase-weight vector and two real numbers. Write $m_n = m(n)$, and let $\tilde{\theta}$ be the pilot parameter estimate computed from p, ε, m, z . For each transcript entry, $\tilde{\phi}(z_i)$ denotes zero when the entry is a pilot release and the selected affine correction $\tilde{\phi}(S_i)$, evaluated at $\tilde{\theta}, p, \varepsilon$ using the supplied selector, when the entry is a pattern release. Then*

$$\hat{V}^* = \frac{1}{\max\{n - m_n, 0\}} \sum_{\substack{0 \leq i < n \\ m_n \leq i}} \left(\tilde{\phi}(z_i) - \frac{1}{\max\{n - m_n, 0\}} \sum_{\substack{0 \leq j < n \\ m_n \leq j}} \tilde{\phi}(z_j) \right)^2.$$

Here reciprocals are interpreted as zero at zero, so the definition also gives $\hat{V}^* = 0$ when $m_n \geq n$.

For a nonempty main sample, this is the empirical second central moment with divisor $n - m_n$. Its zero-based transcript indexing corresponds to the one-based main-sample indices in the algorithm. Under the attainment conditions, it consistently estimates the asymptotic variance constant $V^*(\theta, p, \varepsilon)$ defined in [Definition 5](#). The resulting standard error is $\sqrt{\hat{V}^*/n}$, reflecting the theorem's normalization by the total sample size.

Fix the Wald error probability γ in $(0, 1)$, and let Φ denote the standard normal distribution function. The normal critical value $z_{1-\gamma/2}$ uses the generalized-inverse convention

$$z_{1-\gamma/2} = \inf\{z \in \mathbb{R} : 1 - \gamma/2 \leq \Phi(z)\}.$$

Together, the selected estimator and empirical variance yield the interval

$$\left[\hat{\tau}^* - z_{1-\gamma/2} \sqrt{\hat{V}^*/n}, \quad \hat{\tau}^* + z_{1-\gamma/2} \sqrt{\hat{V}^*/n} \right].$$

The next result establishes privacy, conditional centering, regular attainment, variance consistency, and pointwise coverage for this construction.

Theorem 3. `[thm:pilot-attainment]` (Private pilot attainment) *Let $\theta = (\mu_0, \mu_1)^\top$ be the arm-mean parameter, p the treatment-assignment probability, and ε the fixed privacy level. Suppose:*

- *(Assignment.)* p satisfies [Assumption 6](#).
- *(Means.)* θ satisfies [Assumption 7](#).
- *(Privacy.)* The privacy sequence and ε satisfy [Assumption 8](#).

- **(Selection.)** The selector in [Algorithm 1](#) satisfies the saddle-selection conditions there and, at every interior parameter ϑ , its selected weights maximize the objective at its selected profiling coordinate:

$$F_{\vartheta}(\alpha, t_{\vartheta}) \leq F_{\vartheta}(\alpha_{\vartheta}, t_{\vartheta}) \quad \text{for every } \alpha \in \mathcal{A}_{\varepsilon}.$$

- **(Pilot schedule.)** The pilot sizes m_n satisfy [Assumptions 10](#) and [11](#).
- **(Coverage level.)** The Wald error probability γ satisfies $0 < \gamma < 1$.

Consider the complete procedure sequence $\mathcal{P}$ constructed in [Algorithm 1](#), with estimator $\hat{\tau}^*$, and write $\tau = \mu_1 - \mu_0$ for the treatment-effect contrast. For every sample size, its channels satisfy sequential ε -local differential privacy as specified in [Assumption 9](#).

For $n \geq 2$, let $H_{m_n} = (R_1, \dots, R_{m_n})$ be the private pilot history. For every bounded measurable function f of the transcript whose value depends only on H_{m_n} , the product $(\hat{\tau}^* - \tau)f$ is integrable and

$$\mathbb{E}_{\theta}[(\hat{\tau}^* - \tau)f] = 0.$$

Under every fixed local perturbation h , the scaled local error $U_{n,h}^{\mathcal{P}}$, centered at the contrast of the local parameter $\theta_{n,h}$, satisfies

$$U_{n,h}^{\mathcal{P}} = \sqrt{n}(\hat{\tau}^* - ((\theta_{n,h})_1 - (\theta_{n,h})_0)) \rightsquigarrow N(0, V^*(\theta, p, \varepsilon)) \quad \text{under the transcript law at } \theta_{n,h}.$$

Here $V^*(\theta, p, \varepsilon)$ is the optimized contrast variance in [Definition 5](#). In particular,

$$\sqrt{n}(\hat{\tau}^* - \tau) \rightsquigarrow N(0, V^*(\theta, p, \varepsilon)).$$

The variance estimator $\hat{V}^*$ satisfies

$$\Pr_{\theta} \left(\left| \hat{V}^* - V^*(\theta, p, \varepsilon) \right| > \delta \right) \longrightarrow 0 \quad \text{for every } \delta > 0.$$

The procedure is regular at θ , with common local limit law

$$L_{\mathcal{P}} = N(0, V^*(\theta, p, \varepsilon)).$$

In addition to the local convergence above, its squared scaled errors satisfy asymptotic uniform integrability over every bounded admissible local-perturbation set: writing $\mathcal{H}_{n,H}(\theta)$ for that set at radius H , for every $H > 0$,

$$\lim_{\substack{M \rightarrow \infty \\ M \in \mathbb{N}}} \limsup_{n \rightarrow \infty} \sup_{h \in \mathcal{H}_{n,H}(\theta)} \mathbb{E}_{\theta_{n,h}} \left[(U_{n,h}^{\mathcal{P}})^2 \mathbf{1}\{|U_{n,h}^{\mathcal{P}}| > M\} \right] = 0.$$

Finally, let Φ be the standard normal distribution function and define its critical value by

$$z_{1-\gamma/2} = \inf \{z \in \mathbb{R} : 1 - \gamma/2 \leq \Phi(z)\}.$$

Then

$$\Pr_{\theta} \left(|\hat{\tau}^* - \tau| \leq z_{1-\gamma/2} \sqrt{\hat{V}^*/n} \right) \longrightarrow 1 - \gamma.$$

The construction separates learning the release rule from centering the treatment-effect estimate. The increasing pilot informs channel selection, while the affine correction exactly centers the estimator conditional on that selection. A vanishing pilot fraction then leaves the main sample to determine the first-order variance. Under the measurable strong-saddle conditions, these features deliver the optimized variance for every admissible pilot schedule, including the exhibited schedule used in [Theorem 2](#).

Regularity strengthens the fixed-parameter normal approximation by giving the same limiting error law under every fixed local perturbation, with centering at the corresponding treatment effect. The additional uniform-integrability clause controls squared-error tails over bounded admissible perturbation sets. Together, these conclusions connect the estimator's distributional behavior to the regular squared-error benchmark in [Theorem 2](#).

The variance-consistency and coverage conclusions apply pointwise at every interior arm-mean parameter satisfying the stated conditions, including parameter points at which the optimal channel support changes. At such points, the measurable strong-saddle selector remains the governing selection rule, and the empirical variance estimates the same optimized variance constant appearing in the normal limit. Thus the procedure supplies a selected private release rule, a treatment-effect estimate, and an asymptotically valid Wald interval throughout this interior domain.

6 Optimal release cardinality

Output cardinality is a property of the stationary release design that attains the optimized treatment-effect information. Although the optimization built from the fourteen patterns in [Definitions 3](#) and [4](#) uses all candidate rays, an efficient design can always be chosen with at most five active outputs. The following result establishes this bound throughout the interior trial model.

Theorem 4. [\[thm:five-output-upper-bound\]](#) (Five-output optimal staircase channel) *Let $\theta = (\mu_0, \mu_1)^\top$, p , and ε satisfy:*

- *(Interior means.)* $0 < \mu_0 < 1$ and $0 < \mu_1 < 1$.
- *(Assignment probability.)* $0 < p < 1$.
- *(Privacy parameter.)* $\varepsilon > 0$.

With the feasible staircase weights and profiled objective of [Definition 5](#), there exists $\alpha \in \mathcal{A}_\varepsilon$ such that

$$J^*(\theta, p, \varepsilon) = \inf_{t \in \mathbb{R}} F_\theta(\alpha, t), \quad |\{S \in \mathcal{S} : \alpha_S \neq 0\}| \leq 5, \quad \kappa(Q_\alpha) \leq 5.$$

Here the active-output cardinality of the associated fourteen-symbol staircase channel is

$$\kappa(Q_\alpha) = |\{S \in \mathcal{S} : (P_\theta Q_\alpha)(\{S\}) \neq 0\}|.$$

The fourteen patterns provide a common representation for the optimization, while the active weights determine the alphabet used by a particular release. [Theorem 4](#) therefore gives a uniform bound on the number of output symbols needed to attain the optimized profiled information. Its practical content is an existence guarantee: at each admissible parameter point, the efficient stationary experiment admits a release with at most five active symbols.

To identify where particular supports attain the optimum, we use certificates that combine channel normalization, minimization over the profiling coordinate, and dual optimality for the weights. The fixed pattern enumeration in [Definition 3](#) gives the support indices a common meaning across parameter points.

Definition 12. [\[synth_5\]](#) (Staircase support certificates) *Fix $\theta \in (0, 1)^2$, $p \in (0, 1)$, and $\varepsilon > 0$, and use the staircase rays, masses, and derivative vectors defined above. For a proposed active set $A \subseteq \mathcal{S}$, a support certificate consists of weights $\alpha \in \mathbb{R}_+^{14}$, a profiling coordinate $t \in \mathbb{R}$, and a dual vector $\eta \in \mathbb{R}^4$, with $\alpha_S > 0$ for $S \in A$ and $\alpha_S = 0$ for $S \notin A$. Write*

$$f_S(t) = \frac{(g_S^\top v(t))^2}{h_S(\theta)}.$$

We say that primal normalization holds when

$$B\alpha = \mathbf{1}, \quad \text{equivalently} \quad \sum_{S \in \mathcal{S}} \alpha_S b_S = \mathbf{1},$$

where $B = (b_{S_1}, \dots, b_{S_{14}})$.

We say that profiling stationarity holds at (α, t) when

$$\frac{\partial}{\partial t} F_\theta(\alpha, t) = \sum_{S \in \mathcal{S}} \alpha_S f'_S(t) = 2 \sum_{S \in \mathcal{S}} \alpha_S \frac{(g_S^\top v(t))(g_S^\top (1, 1)^\top)}{h_S(\theta)} = 0.$$

For feasible α , the function $F_\theta(\alpha, \cdot)$ is a differentiable convex quadratic, so this condition is equivalent to $t \in \arg \min_{u \in \mathbb{R}} F_\theta(\alpha, u)$.

The dual vector η corresponds to the normalization constraint in the linear program

$$\max_{\alpha \geq 0: B\alpha = \mathbf{1}} \sum_{S \in \mathcal{S}} \alpha_S f_S(t);$$

its dual feasibility conditions are $b_S^\top \eta \geq f_S(t)$ for every $S \in \mathcal{S}$. We say that active dual equalities hold when

$$b_S^\top \eta = f_S(t) \quad (S \in A),$$

and that strict inactive dual inequalities hold when

$$b_S^\top \eta > f_S(t) \quad (S \in \mathcal{S} \setminus A).$$

The active sets for the three-output and five-output certificates are respectively

$$A = \{S_1, S_6, S_8\}, \quad A = \{S_3, S_4, S_6, S_8, S_9\},$$

using the fixed pattern enumeration. The five-output certificate additionally requires pairwise distinct values of

$$\rho_S(\theta, t) = \frac{g_S^\top v(t)}{h_S(\theta)} \quad (S \in A).$$

Normalization makes the weights a feasible channel, and profiling stationarity selects a minimizing coordinate for its directional information. The dual vector η bounds each pattern's directional contribution $f_S(t)$; equality on active patterns identifies where the bound is attained. Strict inequalities on inactive patterns distinguish the proposed support within this optimization. The additional separation of projected scores in the five-output certificate supports the cardinality conclusion for arbitrary attaining stationary channels. These conditions define the following parameter regions.

Definition 13. [def:support-regions] (Support regions $\mathcal{R}_3$ and $\mathcal{R}_5$) *The region $\mathcal{R}_3$ is the parameter set for which, using the fixed pattern enumeration $S_1, \dots, S_{14}$, the following conditions hold with active set $\{S_1, S_6, S_8\}$:*

- *(Normalization.) Primal normalization holds.*
- *(Stationarity.) Profiling stationarity holds.*
- *(Active patterns.) The active dual equalities hold.*
- *(Inactive patterns.) The inactive dual inequalities are strict.*

The region $\mathcal{R}_5$ is the analogous parameter set with active set $\{S_3, S_4, S_6, S_8, S_9\}$, additionally satisfying

$$\rho_S(\theta, t) \neq \rho_T(\theta, t) \quad \text{for all distinct } S, T \in \{S_3, S_4, S_6, S_8, S_9\}.$$

The regions $\mathcal{R}_3$ and $\mathcal{R}_5$, defined in Definition 13, describe parameters admitting the respective support certificates. Their coordinates are ordered as treatment-assignment probability, control-arm mean, treatment-arm mean, and privacy parameter. The five-output region makes the global existence bound sharp on an open parameter set and gives explicit examples at equal arm means.

Theorem 5. [thm:five-output-region] (The five-output region) *The five-pattern certificate and score-separation region $\mathcal{R}_5$, associated with the support $\{S_3, S_4, S_6, S_8, S_9\}$, is open in $\mathbb{R}^4$ and nonempty. Write $J^*(\theta, p, \varepsilon)$ and $V^*(\theta, p, \varepsilon)$ for the optimized profiled information and its reciprocal from Definition 5.*

For every $0 < p < 1/2$,

$$(p, 1/2, 1/2, \log 3) \in \mathcal{R}_5.$$

At $\theta = (1/2, 1/2)^\top$ and $\varepsilon = \log 3$, there is a unique feasible staircase-weight vector α attaining the optimized profiled information:

$$J^*(\theta, p, \log 3) = \inf_{t \in \mathbb{R}} F_\theta(\alpha, t), \quad \{S \in \mathcal{S} : \alpha_S \neq 0\} = \{S_3, S_4, S_6, S_8, S_9\}.$$

Here feasibility means that the weights are nonnegative and their weighted staircase rays sum to one at each of the four input records. Uniqueness means that every feasible weight vector β satisfying

$$J^*(\theta, p, \log 3) = \inf_{t \in \mathbb{R}} F_\theta(\beta, t)$$

equals α .

For every $1/2 < p < 1$,

$$(1 - p, 1/2, 1/2, \log 3) \in \mathcal{R}_5.$$

At $\theta = (1/2, 1/2)^\top$ and the original assignment probability p , there is likewise a unique feasible staircase-weight vector α satisfying

$$J^*(\theta, p, \log 3) = \inf_{t \in \mathbb{R}} F_\theta(\alpha, t).$$

Its support is the image of $\{S_3, S_4, S_6, S_8, S_9\}$ under exchange of the treatment labels.

In each of these two cases, every stationary $(\log 3)$ -LDP channel Q from the four-record alphabet to an arbitrary measurable output space that attains contrast variance $V^*(\theta, p, \log 3)$ satisfies $\kappa(Q) \geq 5$. More generally, for every

$$(p, \mu_0, \mu_1, \varepsilon) \in \mathcal{R}_5, \quad \theta = (\mu_0, \mu_1)^\top,$$

every stationary ε -LDP channel Q attaining contrast variance $V^*(\theta, p, \varepsilon)$ satisfies

$$\kappa(Q) \geq 5.$$

A stationary ε -LDP channel is a Markov kernel satisfying

$$Q(x, A) \leq e^\varepsilon Q(x', A)$$

for every measurable output event A and every pair of input records x, x' . Attainment means that the extended nonnegative contrast variance of the output Fisher-information matrix $I_\theta(Q)$ equals the nonnegative part of $V^*(\theta, p, \varepsilon)$. Explicitly, this contrast variance is

$$\begin{cases} \max \left\{ \inf_{v: I_\theta(Q)v=c} c^\top v, 0 \right\}, & c \in \text{range}(I_\theta(Q)), \\ +\infty, & c \notin \text{range}(I_\theta(Q)), \end{cases}$$

where c is the treatment-effect contrast vector. The output-cardinality convention is

$$\kappa(Q) = \begin{cases} \#\{z \in \mathcal{Z} : (P_\theta Q)(\{z\}) \neq 0\}, & \mathcal{Z} \text{ finite}, \\ +\infty, & \mathcal{Z} \text{ infinite}, \end{cases}$$

where $P_\theta Q$ is the channel's output law.

Five outputs here reflect a requirement of efficient stationary release design. Throughout the certified five-output region, the lower bound applies to every attaining stationary channel under the stated output-cardinality convention. Together with [Theorem 4](#), it makes five the sharp cardinality bound on this region. At equal arm means and privacy parameter $\log 3$, the theorem also identifies a unique optimizing staircase-weight vector for each assignment probability below one half; exchanging treatment labels gives the corresponding characterization above one half. The uniqueness conclusion concerns weights in the fixed staircase representation, while the cardinality conclusion concerns the full class of attaining stationary channels.

A smaller optimal support occurs at the same privacy parameter when both arm means are low. The next result provides an exact three-pattern optimizer and its information value.

Theorem 6. [\[thm:three-output-region\]](#) (Three-output region and optimizer) *The three-pattern certificate region $\mathcal{R}_3$ is open in $\mathbb{R}^4$ and contains*

$$(p, \mu_0, \mu_1, \varepsilon) = \left(\frac{3}{10}, \frac{1}{10}, \frac{1}{10}, \log 3 \right).$$

At this point, with $\theta = (1/10, 1/10)^\top$, there exists a feasible staircase-weight vector α , in the optimization of [Definition 5](#), whose support and weights satisfy

$$\{S \in \mathcal{S} : \alpha_S \neq 0\} = \{S_1, S_6, S_8\}, \quad \alpha_{S_1} = \alpha_{S_6} = \alpha_{S_8} = \frac{1}{5}.$$

This vector attains the optimized profiled information:

$$J^*\left(\theta, \frac{3}{10}, \log 3\right) = \inf_{t \in \mathbb{R}} F_\theta(\alpha, t),$$

where the profiled objective is evaluated at $p = 3/10$ and $\varepsilon = \log 3$. Moreover, every feasible staircase-weight vector β at $\varepsilon = \log 3$ satisfying

$$J^*\left(\theta, \frac{3}{10}, \log 3\right) = \inf_{t \in \mathbb{R}} F_\theta(\beta, t)$$

equals α . The optimized information and its reciprocal variance, as defined in [Definition 5](#), are

$$J^*\left(\theta, \frac{3}{10}, \log 3\right) = \frac{441}{3994}, \quad V^*\left(\theta, \frac{3}{10}, \log 3\right) = \frac{3994}{441}.$$

At this parameter point, three active patterns suffice to achieve the optimized information, and their equal weights give an explicit benchmark for the finite optimization. The rational information and variance values permit a direct comparison with numerical solutions. Openness of the certificate region places this example within a neighborhood of certified parameters. The following result connects the local staircase supports to exact minimum output cardinalities among arbitrary attaining stationary channels.

Theorem 7. `[thm:support-transition]` (Local support and cardinality transition) *Fix the treatment-assignment probability $p = 3/10$ and privacy parameter $\varepsilon = \log 3$. There exist disjoint open sets $U_5, U_3 \subseteq \mathbb{R}^2$ containing $(1/2, 1/2)$ and $(1/10, 1/10)$, respectively, with the following properties:*

- For every $\theta = (\mu_0, \mu_1)^\top \in U_5$, there exists a unique feasible staircase-weight vector $\alpha \in \mathcal{A}_\varepsilon$ satisfying

$$J^*(\theta, p, \varepsilon) = \inf_{t \in \mathbb{R}} F_\theta(\alpha, t).$$

Its active support is $\{S_3, S_4, S_6, S_8, S_9\}$. Moreover, a feasible staircase channel attains contrast variance $\max\{V^(\theta, p, \varepsilon), 0\}$ with exactly five active outputs, and every stationary ε -LDP channel on any measurable output space attaining this contrast variance has at least five active outputs.*

- For every $\theta = (\mu_0, \mu_1)^\top \in U_3$, there exists a unique feasible staircase-weight vector $\alpha \in \mathcal{A}_\varepsilon$ satisfying

$$J^*(\theta, p, \varepsilon) = \inf_{t \in \mathbb{R}} F_\theta(\alpha, t).$$

Its active support is $\{S_1, S_6, S_8\}$. Moreover, a feasible staircase channel attains contrast variance $\max\{V^(\theta, p, \varepsilon), 0\}$ with exactly three active outputs, and every stationary ε -LDP channel on any measurable output space attaining this contrast variance has at least three active outputs.*

In each case, uniqueness is among all feasible staircase-weight vectors satisfying the displayed equality.

Holding assignment and privacy fixed, the minimum cardinality changes from five near arm means $(1/2, 1/2)$ to three near $(1/10, 1/10)$. Each conclusion persists on an open neighborhood, so the comparison captures a local dependence of efficient release design on outcome prevalence. In both neighborhoods, uniqueness selects the optimizer within the staircase family, and the matching universal lower bound establishes the minimum cardinality for all attaining stationary channels. The certificate algebra and the projected-score arguments establishing these cardinality bounds are presented in [Section 9](#).

Balanced assignment and complementary arm means give a further interpretable special case of the general optimization. Under this symmetry, binary randomized response applied to the treatment–outcome sign attains the optimized contrast variance.

Theorem 8. [\[thm:balanced-reduction\]](#) (Balanced randomized response reduction) *Let $\theta = (\mu_0, \mu_1)^\top$ be the arm-mean parameter and $\tau = \mu_1 - \mu_0$ the treatment-effect contrast. Suppose:*

- *(Interior means.) θ satisfies [Assumption 7](#).*
- *(Balance.) $p = 1/2$ and $\mu_0 + \mu_1 = 1$.*
- *(Privacy parameter.) $\varepsilon > 0$.*

Let Q be binary randomized response applied to $(2W_i - 1)(2Y_i - 1)$: it releases the input sign with probability

$$\frac{e^\varepsilon}{e^\varepsilon + 1}$$

and the opposite sign with probability

$$\frac{1}{e^\varepsilon + 1}.$$

Write $V^(\theta, p, \varepsilon) = J^*(\theta, p, \varepsilon)^{-1}$ for the reciprocal optimized nuisance-profiled information. Then*

$$V^*(\theta, 1/2, \varepsilon) = \left(\frac{\cosh(\varepsilon/2)}{\sinh(\varepsilon/2)} \right)^2 - \tau^2.$$

The channel Q is a stationary Markov channel satisfying, for every measurable output event and every pair of input records,

$$Q(x_j, A) \leq e^\varepsilon Q(x_{j'}, A).$$

Moreover, let $I_\theta(Q)$ be the Fisher-information matrix of its output experiment at $p = 1/2$, and let $c = (-1, 1)^\top$. With contrast variance defined as $c^\top I_\theta(Q)^+ c$ when $c \in \text{range}(I_\theta(Q))$, and as $+\infty$ otherwise, this channel attains

$$c^\top I_\theta(Q)^+ c = V^*(\theta, 1/2, \varepsilon), \quad c \in \text{range}(I_\theta(Q)).$$

The input sign records whether the treatment and outcome signs agree, aligning the binary release with the treatment-effect contrast. Under balanced assignment and complementary arm means, this release retains the information needed for efficient contrast estimation: the range condition makes estimability explicit, and its contrast variance equals the optimized bound. Both release probabilities are positive, giving exactly two active outputs. The closed-form variance therefore supplies a simple benchmark for the general optimization under the stated symmetry, alongside the three- and five-output designs characterized above.

7 Interval comparison and prospective trial use

The efficiency bound translates into a comparison of asymptotic Wald interval lengths for a specified release and estimator. We use the known-assignment custom-Laplace construction of [Ohnishi and Awan \[2025\]](#), applied to the binary trial model in [Definition 1](#). Its release $\tilde{A}_i$ adds noise to an inverse-probability-weighted outcome contribution, and its estimator $\hat{\tau}_{\text{OA}}$ averages these releases. The following definition fixes both components of the benchmark.

Definition 14. [def:oa-release] (Custom-Laplace release $\tilde{A}_i$) *The custom-Laplace release and its sample-mean estimator are*

$$\tilde{A}_i = \frac{W_i Y_i}{p} - \frac{(1 - W_i) Y_i}{q} + L_i, \quad \hat{\tau}_{\text{OA}} = \frac{1}{n} \sum_{i=1}^n \tilde{A}_i.$$

The noise variables L_i are independent and identically distributed, independent of all subject records, with

$$L_i \sim \text{Lap}(0, \Delta_A/\varepsilon), \quad \Delta_A = p^{-1} + q^{-1} = \frac{1}{pq}.$$

The weighted outcome contribution has expectation equal to the treatment effect under randomized assignment, following the inverse-probability-weighting principle of [Horvitz and Thompson \[1952\]](#). The independent, centered noise L_i preserves this expectation. Its scale is determined by the sensitivity constant Δ_A , the distance between the largest and smallest weighted contributions on the four-record alphabet. Thus the benchmark combines variation from assignment and outcomes with the variance of the added Laplace noise.

Write $V_{\text{OA}}(\theta, p, \varepsilon)$ for the asymptotic variance constant of this sample mean. Comparing it with $V^*(\theta, p, \varepsilon)$, defined in [Definition 5](#), gives an interval-length comparison at a common sample size and coverage level. The next result supplies the benchmark variance, its consistent estimator, and the corresponding comparison with the selected private-pilot procedure.

Theorem 9. [thm:laplace-comparison] (Laplace variance and interval comparison) *Let $\theta = (\mu_0, \mu_1)^\top$ be the pair of arm means, p the treatment-assignment probability, ε the privacy parameter, and γ the Wald error probability. Suppose the following conditions hold, under the usual measurability conditions.*

- **(Binary randomized trial.)** *The subjects $\{(Y_i(0), Y_i(1), W_i) : i \geq 1\}$ are independent and identically distributed. The potential outcomes are binary, treatment assignment is independent of the potential outcomes, and*

$$\Pr(W_i = 1) = p, \quad Y_i = (1 - W_i)Y_i(0) + W_i Y_i(1), \quad \mu_0 = \mathbb{E}[Y_i(0)], \quad \mu_1 = \mathbb{E}[Y_i(1)],$$

with $0 < p < 1$ and $0 < \mu_0, \mu_1 < 1$. Set $q = 1 - p$ and $\tau = \mu_1 - \mu_0$.

- **(Public experiment and representation.)** *The specified public experiment is $P_\theta^{\otimes n}$ for every positive sample size n , and the private records $X_i = (W_i, Y_i)$, ordered as $(0, 0), (0, 1), (1, 0), (1, 1)$, have this joint law. Their one-subject probabilities are*

$$(q(1 - \mu_0), q\mu_0, p(1 - \mu_1), p\mu_1).$$

- **(Comparator noise.)** The pairs (X_i, L_i) are jointly independent across subjects, each L_i is independent of X_i , and

$$L_i \sim \text{Lap}(0, \Delta_A/\varepsilon), \quad \Delta_A = \frac{1}{p} + \frac{1}{q}.$$

- **(Privacy and critical value.)** We have $\varepsilon > 0$, $0 < \gamma < 1$, and

$$z_{1-\gamma/2} = \inf\{z \in \mathbb{R} : \Phi(z) \geq 1 - \gamma/2\} > 0,$$

where Φ is the standard-normal distribution function.

- **(Strong-saddle selection.)** Fix any measurable strong-saddle selector $\vartheta \mapsto (\alpha_\vartheta, t_\vartheta, J_\vartheta)$ as in [Algorithm 1](#). In particular, at every interior ϑ , its selected weights maximize the profiled objective at its selected minimizing coordinate:

$$F_\vartheta(\alpha, t_\vartheta) \leq F_\vartheta(\alpha_\vartheta, t_\vartheta) \quad \text{for every } \alpha \in \mathcal{A}_\varepsilon.$$

Define the comparator releases, their sample mean, and their empirical variance by

$$\tilde{A}_i = \frac{W_i Y_i}{p} - \frac{(1 - W_i) Y_i}{q} + L_i, \quad \hat{\tau}_{\text{OA}} = \frac{1}{n} \sum_{i=1}^n \tilde{A}_i, \quad \hat{V}_{\text{OA}} = \frac{1}{n} \sum_{i=1}^n \tilde{A}_i^2 - \hat{\tau}_{\text{OA}}^2,$$

and set

$$V_{\text{OA}}(\theta, p, \varepsilon) = \frac{\mu_1}{p} + \frac{\mu_0}{q} - \tau^2 + \frac{2\Delta_A^2}{\varepsilon^2}.$$

The specified public experiment and its private-record representation hold jointly with the following conclusions:

$$\begin{aligned} \sqrt{n}(\hat{\tau}_{\text{OA}} - \tau) &\rightsquigarrow N(0, V_{\text{OA}}(\theta, p, \varepsilon)), \\ \frac{z_{1-\gamma/2} \sqrt{V_{\text{OA}}(\theta, p, \varepsilon)/n}}{z_{1-\gamma/2} \sqrt{V^*(\theta, p, \varepsilon)/n}} &\longrightarrow \sqrt{\frac{V_{\text{OA}}(\theta, p, \varepsilon)}{V^*(\theta, p, \varepsilon)}}. \end{aligned}$$

Moreover, for every $\delta > 0$,

$$\Pr\left(\left|\hat{V}_{\text{OA}} - V_{\text{OA}}(\theta, p, \varepsilon)\right| > \delta\right) \longrightarrow 0.$$

There exists a pilot schedule satisfying [Assumptions 10](#) and [11](#), witnessed by

$$m_n = \lfloor \sqrt{n} \rfloor.$$

For this schedule, use the selected pilot procedure in [Algorithm 1](#) and its variance statistic $\hat{V}^*$. For every sequence of couplings whose first marginal is the trial probability law and whose second marginal is the transcript law of that pilot procedure at (θ, p) , and for every $\delta > 0$, the coupled probabilities satisfy

$$\Pr\left(\left|\sqrt{\frac{\hat{V}_{\text{OA}}}{\hat{V}^*}} - \sqrt{\frac{V_{\text{OA}}(\theta, p, \varepsilon)}{V^*(\theta, p, \varepsilon)}}\right| > \delta\right) \longrightarrow 0.$$

$$\text{At } (p, \mu_0, \mu_1, \varepsilon) = (1/2, 1/2, 1/2, 1),$$

$$V_{\text{OA}} = 34, \quad V^* = \left(\frac{\cosh(1/2)}{\sinh(1/2)}\right)^2 < V_{\text{OA}}.$$

The variance decomposition explains the comparison. The first three terms of V_{OA} describe the variability of the weighted binary outcome; the final term is the independent noise variance. At the stated balanced point, these contributions are 2 and 32, respectively. The optimized variance instead reflects the release chosen for efficient estimation of the treatment-effect contrast. A common normal critical value $z_{1-\gamma/2}$ and sample size cancel from the half-length ratio, leaving the square root of the variance ratio. The strict inequality at the displayed point therefore establishes shorter asymptotic intervals for the selected efficient procedure relative to the custom-Laplace sample mean.

The statistical comparison is specific to the release and sample-mean estimator in [Definition 14](#). Consistency of $\hat{V}_{\text{OA}}$, together with the selected pilot variance statistic, makes the population ratio operational through estimated Wald half-lengths. The coupling conclusion permits any dependence between the two procedures consistent with their stated marginal laws. Its pilot schedule supplies both an increasing number of pilot observations and a vanishing pilot fraction, as required by [Assumptions 10](#) and [11](#).

7.1 Prospective attendance-threshold design

The Colombian cash-transfer program studied by [Barrera-Osorio et al. \[2011\]](#) motivates a prospective binary endpoint based on meeting its 80% attendance threshold. For a prospective randomized evaluation, let $D_i(a)$ denote subject i 's fraction of scheduled school days attended under assignment a , measured over a prespecified observation window. Define the proposed potential outcome by

$$Y_i(a) = \mathbf{1}\{D_i(a) \geq 0.80\}, \quad a \in \{0, 1\}.$$

The observation window, denominator of scheduled days, and rules for recording attendance would be fixed before assignment. Under consistency, the observed indicator is $Y_i = Y_i(W_i)$, and the treatment effect is the difference between the two probabilities of meeting the threshold. This construction uses the program's threshold to define an endpoint for a prospective trial.

The design inputs are the known assignment probability p , the fixed privacy budget ε , the planned sample size n , the Wald error probability γ , and interior planning values for the threshold probabilities μ_0 and μ_1 . For independently sampled subjects with randomized assignment and the record representation in [Definition 1](#), these inputs determine the two variance constants in [Theorem 9](#). Implementation of the efficient procedure additionally uses the measurable strong-saddle selector in [Algorithm 1](#) and the stated pilot schedule. The planning arm means specify scenarios for evaluating expected precision; the private pilot supplies the parameter estimates used to select the release during the trial.

For illustration, take the hypothetical planning values $(p, \mu_0, \mu_1, \varepsilon) = (1/2, 1/2, 1/2, 1)$. The exact witness in [Theorem 9](#) then gives the following prospective design calculations:

$$\text{Asymptotic half-length ratio at equal sample size} = \sqrt{34} \tanh(1/2) \approx 2.69,$$

$$\text{Asymptotic sample-size ratio at equal half-length} = 34 \tanh^2(1/2) \approx 7.26.$$

Both ratios place the custom-Laplace sample mean in the numerator and the selected efficient procedure in the denominator. The second calculation follows by equating their asymptotic Wald half-lengths at a common coverage level. These are prospective calculations for the specified binary endpoint and planning values, translating the variance comparison into precision and recruitment quantities for trial design.

8 Discussion, extensions, and limitations

The efficiency analysis links three aspects of private trial design: optimization for the treatment-effect contrast, adaptation to unknown arm means, and the number of outputs required by an attaining stationary release. Under the interior binary-trial model, known assignment probability, and fixed positive privacy level, [Theorem 1](#) characterizes the stationary optimum through a finite staircase program. After fixing any measurable strong-saddle selector satisfying the conditions in [Algorithm 1](#), [Theorem 2](#) identifies the optimized variance as the exact regular local asymptotic squared-error constant for sequential private procedures. Its solution determines the attainable first-order precision for estimating the treatment effect.

Contrast-specific optimization allocates information according to the scalar estimand while accounting for the unknown arm means through nuisance profiling. The balanced case in [Theorem 8](#) makes this principle especially transparent. With balanced assignment and complementary arm means, binary randomized response applied to the treatment–outcome sign attains the optimized contrast variance. More generally, [Theorem 4](#) guarantees an attaining staircase release with at most five active outputs at every admissible interior design point.

The private pilot makes parameter-dependent optimization operational. Under [Assumptions 10](#) and [11](#), its size m_n grows while its fraction m_n/n vanishes. The increasing pilot learns the arm means through randomized-response releases, and the remaining subjects use the channel selected at the pilot estimate. Conditional on the pilot history, that channel and its affine correction are fixed. The exact conditional centering in [Theorem 3](#) allows the pilot to guide release selection while the main sample determines the first-order variance. With the stated strong-saddle selection conditions, every pilot schedule satisfying these restrictions yields the optimized normal limit and consistent studentization.

Support changes describe how the attaining stationary experiment varies with outcome prevalence. At assignment probability $p = 3/10$ and privacy level $\varepsilon = \log 3$, [Theorem 7](#) establishes open neighborhoods in which the minimum attaining cardinality is respectively five and three. These conclusions concern stationary channels at specified parameter points; the pilot construction adapts by selecting a channel from private observations. Its inference guarantee applies pointwise throughout the interior domain, including points at which an optimal support changes. The measurable strong-saddle conditions in [Theorem 3](#) govern selection at these points, and the empirical variance consistently estimates the optimized variance appearing in the limiting distribution. Thus support changes are compatible with pointwise asymptotic Wald coverage.

For prospective trial design, these results connect endpoint prevalence, privacy, and recruitment requirements through the optimized variance. The attendance-threshold endpoint provides one prospective binary outcome to which this analysis could apply, with the observation window and recording rules fixed before assignment. Planning arm means determine precision scenarios, while the private pilot supplies the estimates used during the trial. The comparison in [Theorem 9](#) translates the variance difference between the efficient procedure and the specified custom-Laplace sample mean into asymptotic interval-length and sample-size quantities.

8.1 Limitations and future work

A complete classification of optimal supports remains open. The five-output upper bound, the certified three- and five-output neighborhoods, and the balanced binary reduction establish specific features of the efficiency surface. Further work could characterize all attaining supports,

ties between optimizers, and the boundaries separating support regions. The pointwise coverage result also leaves uniform coverage over parameter sequences approaching such boundaries as a separate question.

The attainment result applies to any measurable strong-saddle selector satisfying the stated conditions. Constructing a numerical reference solver with a deterministic tie-breaking rule and an independently checkable numerical certificate for a computed selector is a separate computational task.

The asymptotic analysis assumes interior arm means, a fixed assignment probability strictly between zero and one, and a fixed positive privacy level. Boundary arm means require an analysis of the corresponding local experiments. Assignment probabilities or privacy budgets that change with sample size may also alter pilot informativeness, the appropriate normalization, and the attainable variance. These regimes require conditions tailored to their rates of change.

Broader outcome models would require new record representations and information calculations, together with an appropriate private learning procedure for release selection. Repeated-user interaction introduces an additional distinction: the sequential protocols here adapt across independently sampled subjects, each supplying a private record. Multiple releases from the same individual require a privacy accounting and statistical model that accommodate dependence within that individual’s transcript.

Finally, [Theorem 9](#) compares the efficient procedure with the sample mean of the specified custom-Laplace releases. Comparing that sample mean with full-likelihood estimation under the same release would clarify the contribution of estimator choice within the comparator experiment. Such an analysis would complement the present comparison of release-and-estimator procedures.

9 Proofs and verification note

The auxiliary statements in this section connect measurable private experiments to the finite information problem and supply the localization used for the sequential risk bound. We first present the refinement underlying the stationary comparison, then organize its role in the oracle and cardinality results. The private-pilot construction, sequential lower bound, and comparator calculation follow in that order.

9.1 Measurable refinement and information equality

A staircase refinement represents a stationary private release as a post-processing of a finite private experiment. For cardinality comparisons, the relevant equality condition concerns information in the profiling direction: patterns with distinct projected scores must remain distinguishable whenever post-processing preserves that directional information. The following statement supplies both properties for arbitrary measurable output spaces. Within that environment only, x_1, x_2, x_3, x_4 and $\pi_{\theta,1}, \dots, \pi_{\theta,4}$ denote the standing zero-based records x_0, x_1, x_2, x_3 and their probabilities in the same order; this local one-based relabeling leaves subject indices unchanged.

Lemma 1. `[lem:staircase-refinement]` (Staircase refinement and score equality) *Let p be the treatment-assignment probability, let $\theta = (\mu_0, \mu_1)^\top$ be the arm-mean parameter, and let ε be the privacy level. Suppose that:*

- *(Interior assignment.) p satisfies [Assumption 6](#).*

- **(Interior means.)** θ satisfies [Assumption 7](#).
- **(Fixed privacy.)** The privacy sequence and ε satisfy [Assumption 8](#).
- **(Stationary privacy.)** Q is a probability kernel from the four-record alphabet $\mathcal{X}$ to an arbitrary measurable space Z , and, for every measurable $A \subseteq Z$ and every $x, x' \in \mathcal{X}$,

$$Q(A \mid x) \leq e^\varepsilon Q(A \mid x').$$

Index the fourteen nonempty proper subsets of $\mathcal{X}$ by $\mathcal{S}$. The feasible staircase weights and their channel are given by

$$\mathcal{A}_\varepsilon = \left\{ \alpha : \alpha_S \geq 0 \text{ for every } S \in \mathcal{S}, \quad \sum_{S \in \mathcal{S}} \alpha_S \begin{cases} e^\varepsilon, & x \in S, \\ 1, & x \notin S \end{cases} = 1 \text{ for every } x \in \mathcal{X} \right\}, \quad Q_\alpha(S \mid x) = \alpha_S \begin{cases} e^\varepsilon, & x \in S, \\ 1, & x \notin S \end{cases}$$

Writing $\pi_{\theta,j} = P_\theta(X = x_j)$ for the record probabilities, define the pattern masses, gradients, profiling directions, and projected scores by

$$h_S(\theta) = \sum_{j=1}^4 \pi_{\theta,j} \begin{cases} e^\varepsilon, & x_j \in S, \\ 1, & x_j \notin S, \end{cases} \quad g_S = \nabla_\theta h_S(\theta), \quad v(t) = \begin{pmatrix} t \\ t+1 \end{pmatrix}, \quad \rho_S(\theta, t) = \frac{g_S^\top v(t)}{h_S(\theta)},$$

with p and ε fixed in the gradient. Let $I_\theta(Q)$ denote the output Fisher-information matrix, obtained by integrating the outer product of the output score against the output law, and set

$$I_\theta(\alpha) = \sum_{S \in \mathcal{S}} \frac{\alpha_S g_S g_S^\top}{h_S(\theta)}.$$

Then there exist $\alpha \in \mathcal{A}_\varepsilon$ and a probability kernel K from $\mathcal{S}$ to Z such that

$$Q = K \circ Q_\alpha, \quad I_\theta(\alpha) - I_\theta(Q) \succeq 0.$$

For every $t \in \mathbb{R}$, if

$$v(t)^\top I_\theta(\alpha) v(t) = v(t)^\top I_\theta(Q) v(t),$$

then every pair $S, T \in \mathcal{S}$ with $\alpha_S \neq 0$, $\alpha_T \neq 0$, and conditional output measures $K(S, \cdot)$ and $K(T, \cdot)$ that are not mutually singular satisfies

$$\rho_S(\theta, t) = \rho_T(\theta, t).$$

The proof is deferred to [Section A](#).

Definition 15. [synth_7] (Smooth prior density w_R) For $R > 0$, define the Lebesgue density w_R by

$$w_R(a) = \begin{cases} \frac{15}{16R} \left(1 - \frac{a^2}{R^2}\right)^2, & |a| < R, \\ 0, & |a| \geq R. \end{cases}$$

This density is nonnegative, is positive exactly on $(-R, R)$, has closed support $[-R, R]$, and satisfies

$$\int_{\mathbb{R}} w_R(a) da = \int_{-R}^R w_R(a) da = 1.$$

It is continuously differentiable on $\mathbb{R}$, with

$$w'_R(a) = \begin{cases} -\frac{15a}{4R^3} \left(1 - \frac{a^2}{R^2}\right), & |a| < R, \\ 0, & |a| \geq R, \end{cases}$$

so both the density and its derivative vanish at the endpoints. Its squared score is integrable under the prior, with

$$\int_{-R}^R \frac{\{w'_R(a)\}^2}{w_R(a)} da = \frac{10}{R^2},$$

where the integrand is taken to be zero at points where $w_R(a) = 0$.

The information ordering in [Lemma 1](#) makes the finite experiment a benchmark for every stationary release satisfying the stated privacy inequalities, including releases with continuous outputs. Its equality clause concerns the selected direction rather than equality of the entire information matrix. In that direction, overlapping conditional output measures can preserve information when their active patterns have the same projected score. Pairwise distinct projected scores therefore require mutually singular conditional output measures under directional equality.

9.2 Finite optimization and optimal supports

For [Theorem 1](#), the refinement supplies the comparison between arbitrary stationary experiments and the staircase family. The finite objective is defined in [Definition 4](#), with pattern coordinates fixed by [Definition 3](#). Linearity in the weights and convexity in the profiling coordinate explain the theorem's representation as a minimum of the upper envelope of finitely many vertex objectives. The contrast-estimability convention in [Theorem 1](#) retains singular experiments whose information range contains the treatment-effect contrast.

The support arguments use the normalization, profiling stationarity, and dual conditions collected in [Definition 12](#). These conditions identify a feasible release and its information value at a saddle; the strict inactive inequalities distinguish the certified active patterns. The existence conclusion in [Theorem 4](#) gives an attaining staircase experiment with at most five active outputs throughout the interior domain.

The equality clause of [Lemma 1](#) supplies the additional distinction needed for universal cardinality comparisons. The five-pattern region in [Definition 13](#) explicitly includes projected-score separation, and [Theorem 5](#) establishes the matching lower cardinality bound for every attaining stationary channel in that region. The three-pattern optimizer and its exact information value are given in [Theorem 6](#). At fixed assignment probability $3/10$ and privacy level $\log 3$, [Theorem 7](#) supplies the separate neighborhoods where three and five are the respective minimum attaining cardinalities. Thus uniqueness of staircase weights and minimum cardinality among arbitrary stationary releases remain distinct conclusions with their stated domains.

Under balanced assignment and complementary arm means, [Theorem 8](#) identifies a binary experiment aligned with the contrast. Its information-range condition explains how two outputs suffice for efficient contrast estimation under that symmetry.

9.3 Private-pilot attainment and studentization

The pilot argument is governed by the fixed measurable strong-saddle selector in [Algorithm 1](#). At every interior parameter, its selected weights are feasible, its profiling coordinate minimizes

the selected objective, and its weights maximize the objective at that coordinate. These joint conditions connect the correction in [Definition 10](#) to the optimized information value.

The diverging pilot condition in [Assumption 10](#) supports parameter learning through the randomized-response experiment in [Definition 9](#). The sublinear allocation in [Assumption 11](#) preserves an asymptotically full main sample. Conditional centering in [Theorem 3](#) gives the affine correction its exact statistical role: the estimator targets the treatment effect conditional on the released pilot history.

Local regularity is specified in [Definition 8](#). Under the attainment conditions, [Theorem 3](#) supplies both a common normal limit under every fixed local perturbation and uniform integrability of squared scaled errors over bounded admissible perturbation sets. These conclusions connect the distributional approximation to the squared-error criterion in [Theorem 2](#).

The empirical variance in [Definition 11](#) estimates the variance constant of that normal limit. Its consistency and the coverage conclusion in [Theorem 3](#) apply pointwise at every admissible interior parameter, including points where the optimal support changes. The governing condition at such points is the measurable strong-saddle selection rule. Accordingly, studentization remains tied to the optimized variance at the parameter under consideration.

9.4 Sequential localization and the lower bound

The sequential lower-bound construction localizes the experiment along a direction that changes the treatment effect by one unit, following the local-risk framework of [Le Cam and Yang \[2000, Chapter 6\]](#) and [van der Vaart \[1998, Chapters 7–8 and 25\]](#). The smooth scalar prior in [Definition 15](#) supplies a normalized, continuously differentiable density with compact support, vanishing density and derivative at the endpoints, and prior information $10/R^2$. Pairing this prior with the minimizing direction selected by [Theorem 1](#) gives the following handle for the Bayesian information inequality [[Van Trees, 2001](#)], as used statistically by [Gill and Levit \[1995, pp. 59–79\]](#).

Definition 16. `[def:sequential-vt-handle]` (Sequential van Trees handle) *The sequential van Trees handle $\mathfrak{H}_{\text{VT}}$ is defined for parameters $\theta, p, \varepsilon, R$ satisfying:*

- *(Interiority.)* [Assumptions 6 and 7](#).
- *(Privacy.)* $\varepsilon > 0$.
- *(Prior radius.)* $R > 0$.

It is the direction–prior pair

$$\mathfrak{H}_{\text{VT}} = (v(t^*), w_R), \quad v(t^*) = (t^*, t^* + 1)^\top,$$

where t^ is the selected minimizing coordinate supplied by [Theorem 1](#), satisfying*

$$\max_{\alpha \in \mathcal{A}_\varepsilon} F_\theta(\alpha, t^*) = J^*(\theta, p, \varepsilon),$$

with the feasible weights, profiled objective, and optimized value as in [Definition 5](#). The prior density is

$$w_R(u) = \begin{cases} \frac{15}{16R} \left(1 - \left(\frac{u}{R}\right)^2\right)^2, & |u| < R, \\ 0, & |u| \geq R. \end{cases}$$

The sequential van Trees handle $\mathfrak{H}_{\text{VT}}$ fixes the direction at the interior baseline and supplies scalar localization along that direction. Its directional choice links the sequential information comparison to the same optimized value used in the stationary oracle.

In the lower-bound argument for [Theorem 2](#), the conditional-score martingale accounts for information accumulated through history-dependent releases. The preceding release tuple H_{i-1} determines the channel used at the next stage, as specified in [Definition 2](#). Conditional centering of the stage scores is the basis for the martingale information accounting. Uniform domination over compact interior parameter neighborhoods supplies the control required for the score integrals and their localization, while the privacy condition at every history permits the stationary information comparison at each stage.

The lower-bound conclusion in [Theorem 2](#) ranges over complete sequential private procedure sequences with measurable estimators and arbitrary measurable release spaces. Its risk criterion takes the worst risk over admissible perturbations, the lower asymptotic limit, and then the supremum over localization radii, in the displayed order. The regular attainment conclusion uses the fixed selector and the exhibited pilot schedule $m_n = \lfloor \sqrt{n} \rfloor$. The local convergence and squared-error tail control in [Theorem 3](#) supply the corresponding regularity properties.

9.5 Comparator variance and interval lengths

The comparator in [Definition 14](#) combines an inverse-probability-weighted binary outcome with independent centered Laplace noise. The variance formula in [Theorem 9](#) separates these contributions:

$$V_{\text{OA}}(\theta, p, \varepsilon) = \left(\frac{\mu_1}{p} + \frac{\mu_0}{q} - \tau^2 \right) + \frac{2\Delta_A^2}{\varepsilon^2}.$$

At the balanced parameter point specified there, the two contributions are 2 and 32, giving $V_{\text{OA}} = 34$. The efficient variance at that point is the balanced-reduction value in [Theorem 8](#), and the strict comparison is stated in [Theorem 9](#).

A common sample size and normal critical value leave the square root of the variance ratio as the asymptotic Wald half-length ratio. The empirical comparison in [Theorem 9](#) uses consistent variance statistics and applies to every coupling with the specified marginal laws. Its scope is the selected efficient procedure and the custom-Laplace sample mean defined in [Definition 14](#).

9.6 Verification note

The formal statements and their accompanying proofs are machine-checked in Lean 4. Their scope is conditional on the stated trial model, measurable experiment representations, privacy restrictions, and procedure inputs, including the strong-saddle selector and the regularity requirements where invoked. The checked presentation layer contains 28 frozen statement environments and ten synthesized definitions, and the ten theorem or lemma proof receipts were audited as faithful. The formal contract lists no external published theorem as a Lean proof dependency; literature citations motivate and position the results rather than serving as unchecked formal premises.

Appendices

A Proofs of the main results

Proof of Theorem 1. Fix the parameters satisfying the stated assumptions. Throughout the proof, write

$$c = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad v(t) = \begin{pmatrix} t \\ t+1 \end{pmatrix}, \quad F_\theta(\alpha, t) = \sum_{S \in \mathcal{S}} \alpha_S \frac{(g_S^\top v(t))^2}{h_S(\theta)} \quad (\alpha \in \mathbb{R}^{\mathcal{S}}, t \in \mathbb{R}).$$

Thus $c^\top v(t) = 1$. The staircase quantities are those in Lemma 1, and the optimized values are those in Definition 5.

1. We first establish the existence facts needed for the minimax argument. For $S \in \mathcal{S}$ and $x \in \mathcal{X}$, define

$$b_S(x) = \begin{cases} e^\varepsilon, & x \in S, \\ 1, & x \notin S. \end{cases}$$

Throughout this proof, relabel the four records indexed 1, 2, 3, 4 in Lemma 1 by 0, 1, 2, 3, respectively, preserving their order and reindexing their probability coordinates accordingly. The record probabilities are therefore

$$(\pi_{\theta,0}, \pi_{\theta,1}, \pi_{\theta,2}, \pi_{\theta,3}) = ((1-p)(1-\mu_0), (1-p)\mu_0, p(1-\mu_1), p\mu_1).$$

They are positive, and hence

$$h_S(\theta) = \sum_{j=0}^3 \pi_{\theta,j} b_S(x_j) > 0 \quad (S \in \mathcal{S}).$$

Let the complementary-pattern weights be

$$(\alpha_{\text{pair}})_S = \begin{cases} (e^\varepsilon + 1)^{-1}, & S = \{x_0\} \text{ or } S = \mathcal{X} \setminus \{x_0\}, \\ 0, & \text{otherwise.} \end{cases}$$

Each record belongs to exactly one of these two patterns, so

$$\sum_{S \in \mathcal{S}} (\alpha_{\text{pair}})_S b_S(x) = \frac{e^\varepsilon + 1}{e^\varepsilon + 1} = 1 \quad (x \in \mathcal{X}).$$

Thus $\mathcal{A}_\varepsilon$ is nonempty. Its defining inequalities and equalities are closed and convex. Moreover, since $b_S(x) \geq 1$,

$$0 \leq \alpha_S \leq \sum_{T \in \mathcal{S}} \alpha_T \leq \sum_{T \in \mathcal{S}} \alpha_T b_T(x) = 1 \quad (\alpha \in \mathcal{A}_\varepsilon, S \in \mathcal{S}).$$

Consequently, $\mathcal{A}_\varepsilon$ is compact.

For every feasible α , define the quadratic coefficients

$$a_{\text{quad}}(\alpha) = 2 \sum_{S \in \mathcal{S}} \alpha_S \frac{(g_S^\top v(0))(g_{S,0} + g_{S,1})}{h_S(\theta)}, \quad b_{\text{quad}}(\alpha) = \sum_{S \in \mathcal{S}} \alpha_S \frac{(g_{S,0} + g_{S,1})^2}{h_S(\theta)} \geq 0,$$

where $g_S = (g_{S,0}, g_{S,1})^\top$. Expanding the squares gives

$$F_\theta(\alpha, u) = F_\theta(\alpha, 0) + u a_{\text{quad}}(\alpha) + u^2 b_{\text{quad}}(\alpha) \quad (u \in \mathbb{R}).$$

If $b_{\text{quad}}(\alpha) = 0$, nonnegativity of $F_\theta(\alpha, u)$ for every u forces $a_{\text{quad}}(\alpha) = 0$: otherwise,

$$F_\theta\left(\alpha, -\frac{F_\theta(\alpha, 0) + 1}{a_{\text{quad}}(\alpha)}\right) = -1.$$

If $b_{\text{quad}}(\alpha) > 0$, completing the square gives a minimizer. In both cases, a minimizing coordinate is

$$t_\alpha = \begin{cases} 0, & b_{\text{quad}}(\alpha) = 0, \\ -\frac{a_{\text{quad}}(\alpha)}{2b_{\text{quad}}(\alpha)}, & b_{\text{quad}}(\alpha) > 0, \end{cases}$$

and

$$F_\theta(\alpha, u) - F_\theta(\alpha, t_\alpha) = b_{\text{quad}}(\alpha)(u - t_\alpha)^2 \geq 0.$$

In particular, the profile

$$\mathcal{P}(\alpha) = \min_{t \in \mathbb{R}} F_\theta(\alpha, t) \quad (\alpha \in \mathcal{A}_\varepsilon)$$

is finite and nonnegative. For each fixed t , the objective is continuous in α ; its infimum over t is therefore upper semicontinuous. Compactness supplies a feasible $\alpha_{\max}$ such that

$$\mathcal{P}(\alpha) \leq \mathcal{P}(\alpha_{\max}) \quad (\alpha \in \mathcal{A}_\varepsilon), \quad J^*(\theta, p, \varepsilon) = \mathcal{P}(\alpha_{\max}).$$

Define the upper envelope by

$$\mathcal{E}(t) = \max_{\alpha \in \mathcal{A}_\varepsilon} F_\theta(\alpha, t) \quad (t \in \mathbb{R}).$$

The maximum exists by compactness and continuity. Joint continuity of F_θ , together with compactness of the weight set, also makes $\mathcal{E}$ continuous.

For the first singleton pattern,

$$g_{\{x_0\}} = \begin{pmatrix} -(e^\varepsilon - 1)(1 - p) \\ 0 \end{pmatrix}.$$

Define the positive envelope coefficient

$$\eta_{\text{env}} = \frac{(e^\varepsilon - 1)^2(1 - p)^2}{(e^\varepsilon + 1)h_{\{x_0\}}(\theta)} > 0.$$

Nonnegativity of every summand yields

$$\eta_{\text{env}} t^2 \leq F_\theta(\alpha_{\text{pair}}, t) \leq \mathcal{E}(t) \quad (t \in \mathbb{R}).$$

Hence the sublevel set at $\mathcal{E}(0)$ satisfies

$$\{t \in \mathbb{R} : \mathcal{E}(t) \leq \mathcal{E}(0)\} \subseteq \left[-\sqrt{\frac{\mathcal{E}(0)}{\eta_{\text{env}}}}, \sqrt{\frac{\mathcal{E}(0)}{\eta_{\text{env}}}} \right].$$

It is nonempty and closed, and therefore compact. Minimizing on this set gives a coordinate $t_{\text{env}} \in \mathbb{R}$ with

$$\mathcal{E}(t_{\text{env}}) \leq \mathcal{E}(t) \quad (t \in \mathbb{R}).$$

2. We now prove the minimax identity. Define

$$\mathcal{N}(\alpha, t) = -F_\theta(\alpha, t) \quad (\alpha \in \mathcal{A}_\varepsilon, t \in \mathbb{R}).$$

For fixed t , this function is continuous and affine in α , hence quasiconvex. For fixed feasible α , it is continuous and concave in t , since $b_{\text{quad}}(\alpha) \geq 0$, hence quasiconcave. The weight set is nonempty, compact, and convex, and $\mathbb{R}$ is convex. The extrema established above identify the quantities entering Sion's minimax theorem [Sion, 1958]:

$$\sup_{t \in \mathbb{R}} \mathcal{N}(\alpha, t) = -\mathcal{P}(\alpha), \quad \inf_{\alpha \in \mathcal{A}_\varepsilon} \sup_{t \in \mathbb{R}} \mathcal{N}(\alpha, t) = -J^*(\theta, p, \varepsilon),$$

and

$$\inf_{\alpha \in \mathcal{A}_\varepsilon} \mathcal{N}(\alpha, t) = -\mathcal{E}(t), \quad \sup_{t \in \mathbb{R}} \inf_{\alpha \in \mathcal{A}_\varepsilon} \mathcal{N}(\alpha, t) = -\inf_{t \in \mathbb{R}} \mathcal{E}(t).$$

Sion's theorem equates the last quantities in these two displays. Changing signs gives

$$J^*(\theta, p, \varepsilon) = \inf_{t \in \mathbb{R}} \mathcal{E}(t) = \inf_{t \in \mathbb{R}} \max_{\alpha \in \mathcal{A}_\varepsilon} F_\theta(\alpha, t).$$

3. For fixed t , define the vertex envelope

$$\mathcal{E}_{\text{vert}}(t) = \sup_{\alpha \in \text{vert}(\mathcal{A}_\varepsilon)} F_\theta(\alpha, t).$$

The compact convex set $\mathcal{A}_\varepsilon$ has extreme points, and its continuous objective is bounded above there. Since its vertices belong to the feasible set,

$$\mathcal{E}_{\text{vert}}(t) \leq \mathcal{E}(t).$$

Conversely, define the sublevel set

$$\mathcal{C}_t = \{\alpha \in \mathbb{R}^S : F_\theta(\alpha, t) \leq \mathcal{E}_{\text{vert}}(t)\}.$$

It is closed by continuity and convex by affinity of the objective. It contains all extreme points. The closed-convex-hull characterization of a compact convex set therefore gives

$$\mathcal{A}_\varepsilon = \overline{\text{conv}(\text{vert}(\mathcal{A}_\varepsilon))} \subseteq \mathcal{C}_t.$$

Evaluating at a maximizing feasible weight proves

$$\mathcal{E}(t) \leq \mathcal{E}_{\text{vert}}(t), \quad \mathcal{E}(t) = \mathcal{E}_{\text{vert}}(t).$$

The feasible set is a bounded polyhedron in $\mathbb{R}^S$, so its vertex set is finite and the vertex supremum is a maximum. Thus the two envelope value sets coincide, and

$$J^*(\theta, p, \varepsilon) = \inf_{t \in \mathbb{R}} \max_{\alpha \in \text{vert}(\mathcal{A}_\varepsilon)} F_\theta(\alpha, t).$$

4. Let Q be any admissible stationary channel on an arbitrary measurable output space. By Lemma 1, there are feasible weights α_Q and a probability kernel K_Q such that

$$Q = K_Q \circ Q_{\alpha_Q}, \quad I_\theta(\alpha_Q) - I_\theta(Q) \succeq 0.$$

Choose a minimizing coordinate $t_Q \in \mathbb{R}$ for the quadratic objective of α_Q . The profile maximum established above gives

$$F_\theta(\alpha_Q, t_Q) = \mathcal{P}(\alpha_Q) \leq \mathcal{P}(\alpha_{\max}) = J^*(\theta, p, \varepsilon).$$

Expanding the information matrix shows, for every feasible α and every $t \in \mathbb{R}$, that

$$v(t)^\top I_\theta(\alpha) v(t) = \sum_{S \in \mathcal{S}} \alpha_S \frac{(g_S^\top v(t))^2}{h_S(\theta)} = F_\theta(\alpha, t).$$

Applying the positive semidefinite difference to $v(t_Q)$ gives the intermediate bound

$$v(t_Q)^\top I_\theta(Q) v(t_Q) \leq v(t_Q)^\top I_\theta(\alpha_Q) v(t_Q) = F_\theta(\alpha_Q, t_Q) \leq J^*(\theta, p, \varepsilon).$$

We next verify that the quantity on the right is positive. Define the singleton weights

$$(\alpha_{\text{sing}})_S = \begin{cases} (e^\varepsilon + 3)^{-1}, & S = \{x_j\} \text{ for some } j \in \{0, 1, 2, 3\}, \\ 0, & \text{otherwise.} \end{cases}$$

For each record, one singleton contributes e^ε and the other three contribute 1, so

$$\sum_{S \in \mathcal{S}} (\alpha_{\text{sing}})_S b_S(x) = \frac{e^\varepsilon + 3}{e^\varepsilon + 3} = 1.$$

These weights are feasible. Define the positive coefficients

$$\eta_{\text{control}} = \frac{(e^\varepsilon - 1)^2 (1 - p)^2}{(e^\varepsilon + 3) h_{\{x_0\}}(\theta)}, \quad \eta_{\text{treatment}} = \frac{(e^\varepsilon - 1)^2 p^2}{(e^\varepsilon + 3) h_{\{x_3\}}(\theta)}.$$

The first and fourth singleton summands give, respectively,

$$0 \leq \eta_{\text{control}} t^2 \leq F_\theta(\alpha_{\text{sing}}, t), \quad 0 \leq \eta_{\text{treatment}} (t + 1)^2 \leq F_\theta(\alpha_{\text{sing}}, t).$$

At a minimizing coordinate t_{sing} , a zero objective would force both $t_{\text{sing}} = 0$ and $t_{\text{sing}} + 1 = 0$. Therefore

$$0 < F_\theta(\alpha_{\text{sing}}, t_{\text{sing}}) = \mathcal{P}(\alpha_{\text{sing}}) \leq J^*(\theta, p, \varepsilon).$$

For completeness, the output information matrix is positive semidefinite. Define its dominating measure and density quantities by

$$\nu_Q = \sum_{j=0}^3 Q(\cdot \mid x_j), \quad d_{Q,j} = \frac{dQ(\cdot \mid x_j)}{d\nu_Q},$$

$$f_{\theta,Q}(z) = \sum_{j=0}^3 \pi_{\theta,j} d_{Q,j}(z), \quad \dot{f}_{\theta,Q}(z) = \sum_{j=0}^3 (\nabla_\theta \pi_{\theta,j}) d_{Q,j}(z).$$

The information integrand is assigned zero when its denominator vanishes. Symmetry follows from its outer-product form, and, for every $w \in \mathbb{R}^2$,

$$w^\top I_\theta(Q) w = \int \frac{(w^\top \dot{f}_{\theta,Q}(z))^2}{f_{\theta,Q}(z)} d\nu_Q(z) \geq 0.$$

Here $f_{\theta,Q} \geq 0$; the interior parameters and the finite record alphabet justify the finite-sum and integral operations.

We apply the contrast-variance bound directly, including its range case split. If $c \notin \text{range}(I_\theta(Q))$, the variance is $+\infty$, which proves the required inequality. Otherwise choose $w_Q \in \mathbb{R}^2$ with $I_\theta(Q)w_Q = c$, and define

$$\ell_Q = c^\top w_Q = w_Q^\top I_\theta(Q)w_Q.$$

Positive semidefiniteness gives $\ell_Q \geq 0$. Equality would imply $I_\theta(Q)w_Q = 0$, contradicting $c \neq 0$, so $\ell_Q > 0$. Any other solution $\tilde{w}_Q \in \mathbb{R}^2$ satisfies

$$c^\top \tilde{w}_Q = w_Q^\top I_\theta(Q)\tilde{w}_Q = w_Q^\top c = \ell_Q.$$

Thus the infimum in the contrast-variance definition equals ℓ_Q .

Define the vector

$$y_Q = \ell_Q v(t_Q) - w_Q \in \mathbb{R}^2.$$

Using $v(t_Q)^\top c = 1$, expansion yields

$$0 \leq y_Q^\top I_\theta(Q)y_Q = \ell_Q \left(\ell_Q v(t_Q)^\top I_\theta(Q)v(t_Q) - 1 \right).$$

Since $\ell_Q > 0$, the intermediate bound above implies

$$1 \leq \ell_Q v(t_Q)^\top I_\theta(Q)v(t_Q) \leq \ell_Q J^*(\theta, p, \varepsilon).$$

Taking the reciprocal of the positive optimized information gives

$$V^*(\theta, p, \varepsilon) = \frac{1}{J^*(\theta, p, \varepsilon)} \leq \ell_Q = c^\top I_\theta(Q)^+ c.$$

5. Choose an optimal feasible weight $\alpha_{\text{opt}} \in \mathcal{A}_\varepsilon$ and a minimizing coordinate $t_{\text{opt}} \in \mathbb{R}$. Optimality of α_{opt} and minimality of t_{opt} give

$$F_\theta(\alpha_{\text{opt}}, t_{\text{opt}}) = \mathcal{P}(\alpha_{\text{opt}}) = J^*(\theta, p, \varepsilon).$$

Define

$$I_{\text{opt}} = I_\theta(\alpha_{\text{opt}}), \quad j_{\text{opt}} = v(t_{\text{opt}})^\top I_{\text{opt}} v(t_{\text{opt}}) = J^*(\theta, p, \varepsilon) > 0.$$

The matrix is symmetric, and

$$w^\top I_{\text{opt}} w = \sum_{S \in \mathcal{S}} (\alpha_{\text{opt}})_S \frac{(g_S^\top w)^2}{h_S(\theta)} \geq 0 \quad (w \in \mathbb{R}^2).$$

Moreover, the minimizing property becomes

$$v(t_{\text{opt}})^\top I_{\text{opt}} v(t_{\text{opt}}) \leq v(u)^\top I_{\text{opt}} v(u) \quad (u \in \mathbb{R}).$$

To compute its contrast variance, define

$$\mathbf{1}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad a_{\text{shift}} = \mathbf{1}_2^\top I_{\text{opt}} \mathbf{1}_2, \quad b_{\text{shift}} = 2\mathbf{1}_2^\top I_{\text{opt}} v(t_{\text{opt}}).$$

Since $v(t_{\text{opt}} + \zeta) = v(t_{\text{opt}}) + \zeta \mathbf{1}_2$, minimality implies

$$0 \leq \zeta b_{\text{shift}} + \zeta^2 a_{\text{shift}} \quad (\zeta \in \mathbb{R}).$$

This forces $b_{\text{shift}} = 0$. Indeed, if it were nonzero, define

$$D_{\text{shift}} = |a_{\text{shift}}| + 1 > 0, \quad \zeta_{\text{shift}} = -\frac{b_{\text{shift}}}{2D_{\text{shift}}}.$$

Then $a_{\text{shift}} \leq D_{\text{shift}}$ gives

$$\zeta_{\text{shift}} b_{\text{shift}} + \zeta_{\text{shift}}^2 a_{\text{shift}} = -\frac{b_{\text{shift}}^2}{2D_{\text{shift}}} + \frac{a_{\text{shift}} b_{\text{shift}}^2}{4D_{\text{shift}}^2} \leq -\frac{b_{\text{shift}}^2}{4D_{\text{shift}}} < 0,$$

a contradiction.

Thus the two coordinates of $I_{\text{opt}} v(t_{\text{opt}})$ sum to zero. Combining this with the definition of j_{opt} gives

$$I_{\text{opt}} v(t_{\text{opt}}) = \begin{pmatrix} -j_{\text{opt}} \\ j_{\text{opt}} \end{pmatrix} = j_{\text{opt}} c.$$

Consequently, the vector

$$w_{\text{opt}} = \frac{v(t_{\text{opt}})}{j_{\text{opt}}}$$

satisfies

$$I_{\text{opt}} w_{\text{opt}} = c, \quad c^\top w_{\text{opt}} = \frac{1}{j_{\text{opt}}}.$$

For every $w \in \mathbb{R}^2$ satisfying $I_{\text{opt}} w = c$, symmetry gives

$$c^\top w = w_{\text{opt}}^\top I_{\text{opt}} w = w_{\text{opt}}^\top c = \frac{1}{j_{\text{opt}}}.$$

Thus the defining infimum gives

$$c^\top I_\theta(\alpha_{\text{opt}})^+ c = \frac{1}{j_{\text{opt}}} = V^*(\theta, p, \varepsilon).$$

It remains to identify the corresponding channel information. Feasibility makes each row of $Q_{\alpha_{\text{opt}}}$ a probability distribution. For all $S \in \mathcal{S}$ and $x, x' \in \mathcal{X}$,

$$b_S(x) \leq e^\varepsilon b_S(x'), \quad (\alpha_{\text{opt}})_S b_S(x) \leq e^\varepsilon (\alpha_{\text{opt}})_S b_S(x').$$

Summing over any subset of the finite output alphabet gives the required privacy inequality.

For $S \in \mathcal{S}$, define

$$r_S = \sum_{j=0}^3 b_S(x_j) > 0.$$

The dominating measure and conditional densities of this channel satisfy

$$\nu_{Q_{\alpha_{\text{opt}}}}(\{S\}) = (\alpha_{\text{opt}})_S r_S, \quad d_{Q_{\alpha_{\text{opt}}}, j}(S) = \begin{cases} b_S(x_j)/r_S, & (\alpha_{\text{opt}})_S > 0, \\ 0, & (\alpha_{\text{opt}})_S = 0. \end{cases}$$

The zero-weight outputs have zero dominating mass. At every positive-weight output, the output density and its derivative are

$$f_{\theta, Q_{\alpha_{\text{opt}}}}(S) = \frac{h_S(\theta)}{r_S}, \quad \dot{f}_{\theta, Q_{\alpha_{\text{opt}}}}(S) = \frac{g_S}{r_S}.$$

Hence integration over the finite alphabet gives

$$\begin{aligned} I_{\theta}(Q_{\alpha_{\text{opt}}}) &= \sum_{\substack{S \in \mathcal{S} \\ (\alpha_{\text{opt}})_S > 0}} (\alpha_{\text{opt}})_S r_S \frac{g_S g_S^{\top}}{h_S(\theta) r_S} \\ &= \sum_{S \in \mathcal{S}} (\alpha_{\text{opt}})_S \frac{g_S g_S^{\top}}{h_S(\theta)} = I_{\theta}(\alpha_{\text{opt}}). \end{aligned}$$

Together with the preceding variance calculation, this proves

$$c^{\top} I_{\theta}(Q_{\alpha_{\text{opt}}})^+ c = c^{\top} I_{\theta}(\alpha_{\text{opt}})^+ c = V^*(\theta, p, \varepsilon).$$

The universal lower bound and this admissible attaining channel then yield

$$\inf_Q c^{\top} I_{\theta}(Q)^+ c = V^*(\theta, p, \varepsilon),$$

with the infimum taken over the channel class in the statement.

6. Finally, set

$$t^* = t_{\text{env}} \in \mathbb{R}.$$

Since t_{env} minimizes $\mathcal{E}$, the minimax identity gives

$$\max_{\alpha \in \mathcal{A}_{\varepsilon}} F_{\theta}(\alpha, t^*) = \mathcal{E}(t_{\text{env}}) = \inf_{t \in \mathbb{R}} \mathcal{E}(t) = J^*(\theta, p, \varepsilon),$$

which proves the asserted attainment of the upper-envelope value.

□

Proof of Theorem 2. Throughout, nonnegative risks are interpreted in $[0, \infty]$. Set

$$c = (-1, 1)^{\top}, \quad J_0 = J^*(\theta_0, p, \varepsilon), \quad V_0 = V^*(\theta_0, p, \varepsilon), \quad \iota(a) = \max\{a, 0\} \in [0, \infty] \quad (a \in \mathbb{R}).$$

For any procedure sequence $\mathcal{P}$, write

$$\text{Risk}_{n,H}(\mathcal{P}) = \sup_{h \in \mathcal{H}_{n,H}(\theta_0)} \mathbb{E}_{\theta_{n,h}}^{\mathcal{P}} [(U_{n,h}^{\mathcal{P}})^2], \quad \text{Risk}_{\infty}(\mathcal{P}) = \sup_{H > 0} \liminf_{n \rightarrow \infty} \text{Risk}_{n,H}(\mathcal{P}).$$

1. Fix an arbitrary sequentially private procedure sequence $\mathcal{P}$, including its output spaces and estimator. Take the oracle coordinate t^* supplied by Theorem 1, as used in Definition 16, and define

$$v^* = v(t^*) = (t^*, t^* + 1)^{\top}, \quad G(\theta', t) = \max_{\alpha \in \mathcal{A}_{\varepsilon}} F_{\theta'}(\alpha, t) \quad (\theta' \in (0, 1)^2, t \in \mathbb{R}).$$

Then

$$c^{\top} v^* = 1, \quad G(\theta_0, t^*) = J_0.$$

In particular, $v^* \neq 0$. For $R > 0$, put

$$H(R) = R\|v^*\|_2 > 0, \quad \vartheta_n(a) = \theta_0 + \frac{av^*}{\sqrt{n}} \quad (n \geq 1, a \in \mathbb{R}).$$

For every fixed R , the entire path $\vartheta_n([-R, R])$ is interior for sufficiently large n , because

$$\sup_{|a| \leq R} \|\vartheta_n(a) - \theta_0\|_2 \leq \frac{R\|v^*\|_2}{\sqrt{n}} \longrightarrow 0.$$

Use the prior family in [Definition 16](#) at half the path radius:

$$\bar{w}_R(a) = w_{R/2}(a) = \begin{cases} \frac{15}{8R} \left(1 - \frac{4a^2}{R^2}\right)^2, & |a| < R/2, \\ 0, & |a| \geq R/2. \end{cases}$$

Its derivative is

$$\bar{w}'_R(a) = \begin{cases} -\frac{30a}{R^3} \left(1 - \frac{4a^2}{R^2}\right), & |a| < R/2, \\ 0, & |a| \geq R/2. \end{cases}$$

Thus $\bar{w}_R$ is continuously differentiable, is nonnegative, and vanishes at both endpoints of $[-R, R]$. Direct integration gives

$$\int_{-R}^R \bar{w}_R(a) da = 1, \quad \int_{-R}^R \frac{\bar{w}'_R(a)^2}{\bar{w}_R(a)} da = \frac{480}{R^5} \int_{-R/2}^{R/2} a^2 da = \frac{40}{R^2},$$

where the quotient is zero wherever $\bar{w}_R = 0$.

For sufficiently large n , define the prior-averaged information envelope

$$\mathcal{J}_{n,R} = \int_{-R}^R \bar{w}_R(a) G(\vartheta_n(a), t^*) da.$$

The feasible weight set is compact: its row-normalization equations and nonnegative coordinates imply

$$0 \leq \alpha_S \leq 1 \quad (S \in \mathcal{S}, \alpha \in \mathcal{A}_\varepsilon).$$

Moreover, for interior θ' ,

$$1 \leq h_S(\theta') \leq e^\varepsilon.$$

Consequently, $F_{\theta'}(\alpha, t^*)$ is jointly continuous in (θ', α) . Compactness gives continuity of its maximum, with

$$|G(\theta', t^*) - G(\theta_0, t^*)| \leq \max_{\alpha \in \mathcal{A}_\varepsilon} |F_{\theta'}(\alpha, t^*) - F_{\theta_0}(\alpha, t^*)|.$$

It follows that

$$\sup_{|a| \leq R} |G(\vartheta_n(a), t^*) - J_0| \longrightarrow 0, \quad \mathcal{J}_{n,R} \longrightarrow J_0.$$

Equivalently, one may integrate the pointwise convergence using the eventual dominating function

$$|\bar{w}_R(a) G(\vartheta_n(a), t^*)| \leq (|J_0| + 1) \bar{w}_R(a).$$

2. We next obtain the transcript information bound. The product record law in [Definition 1](#), with the category ordering in [Definition 9](#), has probabilities

$$\begin{aligned}\pi_{\theta',0} &= (1-p)(1-\theta'_0), & \pi_{\theta',1} &= (1-p)\theta'_0, \\ \pi_{\theta',2} &= p(1-\theta'_1), & \pi_{\theta',3} &= p\theta'_1\end{aligned}\quad (\theta' \in (0,1)^2).$$

Their directional derivatives along v^* are

$$\dot{\pi}_j^* = (v^*)^\top \nabla_{\theta'} \pi_{\theta',j} \quad (j = 0, 1, 2, 3), \quad \sum_{j=0}^3 \dot{\pi}_j^* = 0.$$

Fix $n \geq 1$. For $0 \leq k \leq n$, let

$$\mathbf{H}_{n,k} = \prod_{i=1}^k \mathcal{Z}_{n,i}.$$

For each input-category string $\mathbf{j} = (j_1, \dots, j_k) \in \{0, 1, 2, 3\}^k$, let $\Lambda_{n,k,\mathbf{j}}$ be the probability measure on $\mathbf{H}_{n,k}$ obtained by successively applying the first k release kernels to that fixed input string. Define

$$\nu_{n,k} = \sum_{\mathbf{j} \in \{0,1,2,3\}^k} \Lambda_{n,k,\mathbf{j}}, \quad \lambda_{n,k,\mathbf{j}} = \frac{d\Lambda_{n,k,\mathbf{j}}}{d\nu_{n,k}}.$$

At $k = 0$, the history space is a singleton and its law is the unit point mass. Factorization of the transcript laws in [Definition 2](#) yields the prefix density

$$f_{n,k}^{\theta'}(y) = \sum_{\mathbf{j} \in \{0,1,2,3\}^k} \left(\prod_{i=1}^k \pi_{\theta',j_i} \right) \lambda_{n,k,\mathbf{j}}(y).$$

Its directional derivative is the finite sum

$$\dot{f}_{n,k}^{\theta'}(y) = \sum_{\mathbf{j} \in \{0,1,2,3\}^k} \left[\sum_{i=1}^k \dot{\pi}_{j_i}^* \prod_{\substack{1 \leq i' \leq k \\ i' \neq i}} \pi_{\theta',j_{i'}} \right] \lambda_{n,k,\mathbf{j}}(y).$$

Let $\mathbb{P}_{n,k}^{\theta'}$ denote this prefix law, and define its score by

$$s_{n,k}^{\theta'}(y) = \begin{cases} \dot{f}_{n,k}^{\theta'}(y) / f_{n,k}^{\theta'}(y), & f_{n,k}^{\theta'}(y) > 0, \\ 0, & f_{n,k}^{\theta'}(y) = 0. \end{cases}$$

All record probabilities are positive. Hence these scores are square integrable; indeed,

$$|s_{n,k}^{\theta'}| \leq k \max_{0 \leq j \leq 3} \frac{|\dot{\pi}_j^*|}{\pi_{\theta',j}} \quad \mathbb{P}_{n,k}^{\theta'}\text{-almost everywhere.}$$

For $k < n$, freeze the prefix law at θ' and form the joint channel

$$\overline{Q}_{n,k}^{\theta'}(dy, dz \mid x) = \mathbb{P}_{n,k}^{\theta'}(dy) Q_{k+1}^{(n)}(dz \mid x, y).$$

For a measurable joint event A , write

$$A_y = \{z : (y, z) \in A\}.$$

The all-history privacy inequalities give

$$\overline{Q}_{n,k}^{\theta'}(A | x) = \int Q_{k+1}^{(n)}(A_y | x, y) \mathbb{P}_{n,k}^{\theta'}(dy) \leq e^\varepsilon \overline{Q}_{n,k}^{\theta'}(A | x').$$

Thus this is a stationary private channel on the joint history–release space. Define its output law and directional score by

$$\rho_{n,k}^{\theta'} = \sum_{j=0}^3 \pi_{\theta',j} \overline{Q}_{n,k}^{\theta'}(\cdot | x_j), \quad \Delta_{n,k}^{\theta'} = \frac{d\left(\sum_{j=0}^3 \dot{\pi}_j^* \overline{Q}_{n,k}^{\theta'}(\cdot | x_j)\right)}{d\rho_{n,k}^{\theta'}}.$$

Choose a measurable version of the latter derivative, with value zero on a null set.

Differentiating the finite-mixture expression for the successor prefix law gives, almost everywhere under $\rho_{n,k}^{\theta'}$,

$$s_{n,k+1}^{\theta'}(y, z) = s_{n,k}^{\theta'}(y) + \Delta_{n,k}^{\theta'}(y, z).$$

Here the first summand differentiates the prefix law and the second differentiates the current-record mixture. For every measurable prefix event B ,

$$\int_{B \times \mathcal{Z}_{n,k+1}} \Delta_{n,k}^{\theta'} d\rho_{n,k}^{\theta'} = \sum_{j=0}^3 \dot{\pi}_j^* \mathbb{P}_{n,k}^{\theta'}(B) = 0.$$

Therefore the current score has conditional mean zero given the prefix, and

$$\int s_{n,k}^{\theta'}(y) \Delta_{n,k}^{\theta'}(y, z) d\rho_{n,k}^{\theta'}(y, z) = 0.$$

Define the prefix score energy

$$\mathcal{I}_{n,k}(\theta') = \int (s_{n,k}^{\theta'})^2 d\mathbb{P}_{n,k}^{\theta'}.$$

Expanding the square in the score identity gives

$$\mathcal{I}_{n,0}(\theta') = 0, \quad \mathcal{I}_{n,k+1}(\theta') = \mathcal{I}_{n,k}(\theta') + \int (\Delta_{n,k}^{\theta'})^2 d\rho_{n,k}^{\theta'}.$$

Apply [Lemma 1](#) to the joint channel. It supplies feasible weights $\beta_{n,k}^{\theta'} \in \mathcal{A}_\varepsilon$ with

$$\begin{aligned} \int (\Delta_{n,k}^{\theta'})^2 d\rho_{n,k}^{\theta'} &= (v^*)^\top I_{\theta'}(\overline{Q}_{n,k}^{\theta'}) v^* \\ &\leq F_{\theta'}(\beta_{n,k}^{\theta'}, t^*) \leq G(\theta', t^*). \end{aligned}$$

Induction on the prefix length now yields

$$\mathcal{I}_{n,k}(\theta') \leq kG(\theta', t^*) \quad (0 \leq k \leq n).$$

For the local scalar path $a \mapsto \vartheta_n(a)$, define

$$s_n^{\text{loc}}(a, z) = \frac{1}{\sqrt{n}} s_{n,n}^{\vartheta_n(a)}(z), \quad I_n^{\text{loc}}(a) = \int (s_n^{\text{loc}}(a, z))^2 \mathbb{P}_{n,n}^{\vartheta_n(a)}(dz).$$

For sufficiently large n , uniformly over $|a| \leq R$,

$$0 \leq I_n^{\text{loc}}(a) = \frac{\mathcal{I}_{n,n}(\vartheta_n(a))}{n} \leq G(\vartheta_n(a), t^*).$$

Multiplying by the nonnegative prior density and integrating gives

$$0 \leq \int_{-R}^R \bar{w}_R(a) I_n^{\text{loc}}(a) da \leq \mathcal{J}_{n,R}.$$

The finite-mixture domination and the displayed score bounds also give the integrability needed for this calculation on arbitrary measurable output spaces.

3. Fix $R > 0$ and take n sufficiently large that the preceding path is interior. Define the prior-transcript probability measure

$$\mathbb{B}_{n,R}(da, dz) = \bar{w}_R(a) da \mathbb{P}_{n,n}^{\vartheta_n(a)}(dz) \quad (a \in [-R, R]),$$

the estimation error and its integrated square by

$$e_n(a, z) = \hat{\tau}_n^{\mathcal{P}}(z) - \tau(\vartheta_n(a)), \quad \mathcal{E}_{n,R} = \int e_n(a, z)^2 \mathbb{B}_{n,R}(da, dz),$$

and the joint score by

$$\mathcal{S}_{n,R}(a, z) = s_n^{\text{loc}}(a, z) + \begin{cases} \bar{w}'_R(a)/\bar{w}_R(a), & \bar{w}_R(a) > 0, \\ 0, & \bar{w}_R(a) = 0. \end{cases}$$

First suppose $\mathcal{E}_{n,R} < \infty$. The likelihood score has mean zero at each path parameter. Hence its cross term with the prior score integrates to zero, and

$$\int \mathcal{S}_{n,R}^2 d\mathbb{B}_{n,R} = \int_{-R}^R \bar{w}_R(a) I_n^{\text{loc}}(a) da + \frac{40}{R^2}.$$

This quantity is positive and is at most $\mathcal{J}_{n,R} + 40/R^2$.

The target derivative along the path is

$$\frac{d}{da} \tau(\vartheta_n(a)) = \frac{c^\top v^*}{\sqrt{n}} = \frac{1}{\sqrt{n}}.$$

Integration by parts therefore gives

$$\int e_n \mathcal{S}_{n,R} d\mathbb{B}_{n,R} = \frac{1}{\sqrt{n}}.$$

Indeed, the derivative of the product of the error and the joint density is the error times the joint-density derivative minus $1/\sqrt{n}$ times that density. Its boundary integral vanishes

because $\bar{w}_R(\pm R) = 0$. Finite integrated squared error and finite joint score energy justify the integrals by Cauchy–Schwarz. Applying the same inequality to the displayed identity yields

$$\begin{aligned} n\mathcal{E}_{n,R} &\geq \frac{1}{\int_{-R}^R \bar{w}_R(a) I_n^{\text{loc}}(a) da + 40/R^2} \\ &\geq \frac{1}{\mathcal{J}_{n,R} + 40/R^2}. \end{aligned}$$

For every $|a| \leq R$,

$$av^* \in \mathcal{H}_{n,H(R)}(\theta_0), \quad \sqrt{n} e_n(a, z) = U_{n,av^*}^{\mathcal{P}}(z).$$

Thus the normalized prior average is bounded by the local worst risk:

$$\begin{aligned} \iota(n\mathcal{E}_{n,R}) &= \int_{-R}^R \bar{w}_R(a) \mathbb{E}_{\vartheta_n(a)}^{\mathcal{P}} [(U_{n,av^*}^{\mathcal{P}})^2] da \\ &\leq \text{Risk}_{n,H(R)}(\mathcal{P}). \end{aligned}$$

If instead $\mathcal{E}_{n,R} = \infty$, the same nonnegative-integral identity makes this prior average infinite. Its domination by the local worst risk then gives $\text{Risk}_{n,H(R)}(\mathcal{P}) = \infty$. Both cases establish the eventual bound

$$\iota\left(\frac{1}{\mathcal{J}_{n,R} + 40/R^2}\right) \leq \text{Risk}_{n,H(R)}(\mathcal{P}).$$

4. We record positivity of the optimized information before passing to reciprocal limits. Define

$$a_{\text{ctl}} = \frac{(e^\varepsilon - 1)^2(1-p)^2}{e^\varepsilon(e^\varepsilon + 3)}, \quad a_{\text{trt}} = \frac{(e^\varepsilon - 1)^2 p^2}{e^\varepsilon(e^\varepsilon + 3)}, \quad J_{\text{low}} = \frac{a_{\text{ctl}} a_{\text{trt}}}{a_{\text{ctl}} + a_{\text{trt}}}.$$

These constants are positive. The singleton staircase design

$$\alpha_S^{\text{RR}} = \begin{cases} (e^\varepsilon + 3)^{-1}, & S \text{ is a singleton,} \\ 0, & \text{otherwise} \end{cases}$$

belongs to $\mathcal{A}_\varepsilon$: each channel row contains one singleton entry with multiplier e^ε and three with multiplier one.

Using the singleton patterns $\{x_0\}$ and $\{x_3\}$, the upper bound $h_S(\theta') \leq e^\varepsilon$, and nonnegativity of the remaining terms gives, for every interior θ' and every $t \in \mathbb{R}$,

$$\begin{aligned} F_{\theta'}(\alpha^{\text{RR}}, t) &\geq a_{\text{ctl}} t^2 + a_{\text{trt}} (t+1)^2 \\ &= (a_{\text{ctl}} + a_{\text{trt}}) \left(t + \frac{a_{\text{trt}}}{a_{\text{ctl}} + a_{\text{trt}}} \right)^2 + J_{\text{low}} \geq J_{\text{low}}. \end{aligned}$$

The optimization in [Definition 5](#) consequently implies

$$J^*(\theta', p, \varepsilon) \geq J_{\text{low}} > 0, \quad J_0 > 0.$$

For each fixed $R > 0$, the convergence of $\mathcal{J}_{n,R}$ and the eventual lower bound therefore imply

$$\iota\left(\frac{1}{J_0 + 40/R^2}\right) \leq \liminf_{n \rightarrow \infty} \text{Risk}_{n,H(R)}(\mathcal{P}) \leq \text{Risk}_\infty(\mathcal{P}).$$

To remove the prior-information penalty, take

$$R_{\text{rad},\ell} = \ell + 1 \quad (\ell \in \mathbb{N}).$$

Then

$$\frac{40}{R_{\text{rad},\ell}^2} \rightarrow 0, \quad \frac{1}{J_0 + 40/R_{\text{rad},\ell}^2} \rightarrow \frac{1}{J_0} = V_0.$$

Passing to this limit proves

$$\iota(V_0) \leq \text{Risk}_\infty(\mathcal{P}).$$

Since $\mathcal{P}$ and its measurable output spaces were arbitrary, this establishes the universal lower bound.

5. Choose the schedule

$$m_n = \lfloor \sqrt{n} \rfloor \quad (n \in \mathbb{N}).$$

For every integer $b \geq 0$,

$$n \geq b^2 \implies m_n \geq b,$$

so $m_n \rightarrow \infty$. For $n \geq 1$,

$$0 \leq \frac{m_n}{n} \leq \frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}} \rightarrow 0.$$

Also,

$$1 \leq m_n < n \quad (n \geq 2).$$

Thus this schedule satisfies [Assumptions 10](#) and [11](#).

6. Let

$\mathcal{P}^*$ = the procedure in [Algorithm 1](#) with this schedule and the fixed strong-saddle selector.

Apply [Theorem 3](#) at θ_0 , with $\gamma = 1/2$. It supplies sequential privacy at every sample size, the common local limit

$$L_{\mathcal{P}^*} = N(0, V_0), \quad \int_{\mathbb{R}} u^2 L_{\mathcal{P}^*}(du) = V_0 < \infty,$$

and, for every $H > 0$,

$$\lim_{\substack{M \rightarrow \infty \\ M \in \mathbb{N}}} \limsup_{n \rightarrow \infty} \sup_{h \in \mathcal{H}_{n,H}(\theta_0)} \mathbb{E}_{\theta_{n,h}}^{\mathcal{P}^*} \left[(U_{n,h}^{\mathcal{P}^*})^2 \mathbf{1}\{|U_{n,h}^{\mathcal{P}^*}| > M\} \right] = 0.$$

Monotonicity in the threshold extends this limit to real $M \rightarrow \infty$. These are precisely the regularity requirements, so

$$\mathcal{P}^* \in \mathfrak{P}_\varepsilon^{\text{reg}}(\theta_0).$$

The conventions in [Algorithm 1](#) specify the procedure at every sample size. The following risk calculations use $n \geq 2$, when both sample portions are nonempty.

7. We establish exact risk attainment through the pilot-averaged conditional variance. First obtain uniform bounds for the selected correction. Define

$$C_{\text{zero}} = \sum_{S \in \mathcal{S}} (g_S^\top v(0))^2, \quad T_{\text{up}} = \sqrt{\frac{C_{\text{zero}}}{a_{\text{ctl}}}},$$

$$C_{\text{grad}} = \sum_{S \in \mathcal{S}} (|g_{S,0}| + |g_{S,1}|), \quad C_\phi = \frac{C_{\text{grad}}(T_{\text{up}} + 1)}{J_{\text{low}}}.$$

For an interior selector argument η , set

$$v_\eta = v(t_\eta), \quad \mathcal{I}_\eta = I_\eta(\alpha_\eta), \quad \phi_\eta(S) = \frac{g_S^\top v_\eta}{J_\eta h_S(\eta)} \quad (S \in \mathcal{S}).$$

The strong-saddle conditions in [Algorithm 1](#) and the reference-design bound give

$$J_{\text{low}} \leq a_{\text{ctl}} t_\eta^2 + a_{\text{trt}} (t_\eta + 1)^2 \leq F_\eta(\alpha^{\text{RR}}, t_\eta) \leq J_\eta.$$

On the other hand, minimizing in the profiling coordinate and using $\alpha_{\eta,S} \leq 1$ and $h_S(\eta) \geq 1$ gives

$$J_\eta = F_\eta(\alpha_\eta, t_\eta) \leq F_\eta(\alpha_\eta, 0) \leq C_{\text{zero}}.$$

It follows that

$$|t_\eta| \leq T_{\text{up}}, \quad J_\eta \geq J_{\text{low}}.$$

Finally,

$$|g_S^\top v_\eta| \leq (|g_{S,0}| + |g_{S,1}|)(|t_\eta| + 1) \leq C_{\text{grad}}(T_{\text{up}} + 1),$$

and hence

$$|\phi_\eta(S)| \leq C_\phi \quad (\eta \in (0, 1)^2, \quad S \in \mathcal{S}).$$

8. For interior true and pilot parameters θ' and η , define

$$\sigma^2(\theta', \eta) = \sum_{S \in \mathcal{S}} \alpha_{\eta,S} h_S(\theta') \phi_\eta(S)^2 - (\tau(\theta') - \tau(\eta))^2.$$

We derive the mean and second-moment identities entering this expression. Put

$$\mathbf{1}_2 = (1, 1)^\top.$$

Since $v(t) = v(0) + t\mathbf{1}_2$, minimization of the selected quadratic objective gives

$$0 = \left. \frac{d}{dt} F_\eta(\alpha_\eta, t) \right|_{t=t_\eta} = 2\mathbf{1}_2^\top \mathcal{I}_\eta v_\eta.$$

Thus $\mathcal{I}_\eta v_\eta$ lies in the span of c . Together with $c^\top v_\eta = 1$ and $v_\eta^\top \mathcal{I}_\eta v_\eta = J_\eta$, this yields

$$\mathcal{I}_\eta v_\eta = J_\eta c.$$

Channel normalization and the affine dependence of the pattern masses give

$$\sum_S \alpha_{\eta,S} h_S(\theta') = 1, \quad \sum_S \alpha_{\eta,S} g_S = 0, \quad h_S(\theta') = h_S(\eta) + g_S^\top (\theta' - \eta).$$

Therefore

$$\begin{aligned}\sum_S \alpha_{\eta,S} h_S(\theta') \phi_\eta(S) &= \frac{1}{J_\eta} \left[\sum_S \alpha_{\eta,S} g_S^\top v_\eta + (\theta' - \eta)^\top \mathcal{I}_\eta v_\eta \right] \\ &= c^\top (\theta' - \eta) = \tau(\theta') - \tau(\eta).\end{aligned}$$

At the diagonal,

$$\begin{aligned}\sum_S \alpha_{\eta,S} h_S(\eta) \phi_\eta(S)^2 &= \frac{1}{J_\eta^2} \sum_S \alpha_{\eta,S} \frac{(g_S^\top v_\eta)^2}{h_S(\eta)} \\ &= \frac{1}{J_\eta} = V^*(\eta, p, \varepsilon).\end{aligned}$$

Consequently,

$$\sigma^2(\theta', \eta) = \sum_S \alpha_{\eta,S} h_S(\theta') [\phi_\eta(S) - (\tau(\theta') - \tau(\eta))]^2,$$

and

$$0 \leq \sigma^2(\theta', \eta) \leq C_\phi^2, \quad \sigma^2(\eta, \eta) = V^*(\eta, p, \varepsilon).$$

The upper bound follows by bounding the uncentered second moment by C_ϕ^2 and subtracting the squared mean.

9. For $n \geq 2$, define the main-sample size and pilot-prefix probabilities by

$$N_n = n - m_n > 0,$$

$$\omega_{n,\theta'}(\mathbf{r}) = \prod_{i=1}^{m_n} \left[\sum_{j=0}^3 \pi_{\theta',j} Q_{\text{RR}}(r_i | x_j) \right], \quad \mathbf{r} \in \{0, 1, 2, 3\}^{m_n}, \quad \theta' \in (0, 1)^2.$$

They satisfy

$$\omega_{n,\theta'}(\mathbf{r}) \geq 0, \quad \sum_{\mathbf{r} \in \{0,1,2,3\}^{m_n}} \omega_{n,\theta'}(\mathbf{r}) = 1.$$

Let $\tilde{\theta}(\mathbf{r})$ be the clipped pilot estimate defined in [Algorithm 1](#). For a main-output string $\mathbf{s} \in \mathcal{S}^{N_n}$, factorization gives

$$\begin{aligned}\Pr_{\theta'}^*(H_{m_n} = \mathbf{r}, (S_{m_n+1}, \dots, S_n) = \mathbf{s}) \\ = \omega_{n,\theta'}(\mathbf{r}) \prod_{i=1}^{N_n} \alpha_{\tilde{\theta}(\mathbf{r}), s_i} h_{s_i}(\theta').\end{aligned}$$

Transcripts outside these pilot and main-output formats have probability zero.

Given the pilot prefix, the main corrections are independent with mean $\tau(\theta') - \tau(\tilde{\theta}(\mathbf{r}))$ and variance $\sigma^2(\theta', \tilde{\theta}(\mathbf{r}))$. The estimator formula therefore gives

$$\mathbb{E}_{\theta'}^* [\hat{\tau}^* - \tau(\theta') | H_{m_n} = \mathbf{r}] = 0.$$

Expanding the square of the centered sum, the off-diagonal terms vanish by independence and zero means, so

$$\mathbb{E}_{\theta'}^* [n(\hat{\tau}^* - \tau(\theta'))^2 | H_{m_n} = \mathbf{r}] = \frac{n}{N_n} \sigma^2(\theta', \tilde{\theta}(\mathbf{r})).$$

Define the prefix-averaged variance

$$\bar{\sigma}_n^2(\theta') = \sum_{\mathbf{r} \in \{0,1,2,3\}^{m_n}} \omega_{n,\theta'}(\mathbf{r}) \sigma^2(\theta', \tilde{\theta}(\mathbf{r})).$$

Summing the conditional identity over the finite prefix space gives, for every admissible h ,

$$\mathbb{E}_{\theta_{n,h}}^{\mathcal{P}^*} \left[(U_{n,h}^*)^2 \right] = \frac{n}{N_n} \bar{\sigma}_n^2(\theta_{n,h}), \quad 0 \leq \bar{\sigma}_n^2(\theta_{n,h}) \leq C_\phi^2.$$

10. We next show continuity of the variance at (θ_0, θ_0) . The uniform selected-coordinate bound supplies the compact interval

$$K_{\text{prof}} = [-T_{\text{up}}, T_{\text{up}}].$$

For every interior η , the selected saddle lies in $\mathcal{A}_\varepsilon \times K_{\text{prof}}$. Its saddle inequalities imply

$$J^*(\eta, p, \varepsilon) = \max_{\alpha \in \mathcal{A}_\varepsilon} \min_{t \in K_{\text{prof}}} F_\eta(\alpha, t).$$

Indeed, evaluating any feasible design at t_η bounds its restricted minimum above by J_η , while the selected design has restricted minimum J_η .

Joint continuity on the fixed compact weight-coordinate set gives

$$|J^*(\eta, p, \varepsilon) - J_0| \leq \sup_{\substack{\alpha \in \mathcal{A}_\varepsilon \\ t \in K_{\text{prof}}}} |F_\eta(\alpha, t) - F_{\theta_0}(\alpha, t)| \longrightarrow 0 \quad (\eta \rightarrow \theta_0).$$

Since $J_0 > 0$, reciprocation yields

$$V^*(\eta, p, \varepsilon) \longrightarrow V_0.$$

Subtracting the diagonal second-moment identity from the definition of σ^2 gives the exact expression

$$\begin{aligned} \sigma^2(\theta', \eta) - V^*(\eta, p, \varepsilon) &= \sum_S \alpha_{\eta,S} (h_S(\theta') - h_S(\eta)) \phi_\eta(S)^2 \\ &\quad - (\tau(\theta') - \tau(\eta))^2. \end{aligned}$$

Using $0 \leq \alpha_{\eta,S} \leq 1$ and $|\phi_\eta(S)| \leq C_\phi$, we obtain

$$\begin{aligned} |\sigma^2(\theta', \eta) - V^*(\eta, p, \varepsilon)| &\leq C_\phi^2 \sum_S |h_S(\theta') - h_S(\eta)| \\ &\quad + (\tau(\theta') - \tau(\eta))^2. \end{aligned}$$

The right-hand side tends to zero as $(\theta', \eta) \rightarrow (\theta_0, \theta_0)$. Hence

$$\sigma^2(\theta', \eta) \longrightarrow V_0 \quad \text{as } (\theta', \eta) \rightarrow (\theta_0, \theta_0).$$

11. Fix $H > 0$ and an error tolerance $\delta > 0$. By the preceding continuity, choose $r_{\text{cont}} > 0$ such that

$$\|\theta' - \theta_0\|_\infty < r_{\text{cont}}, \quad \|\eta - \theta_0\|_\infty < r_{\text{cont}} \implies |\sigma^2(\theta', \eta) - V_0| < \delta/2$$

for interior θ' and η . Define

$$d_{\text{int}} = \min\{\mu_0, 1 - \mu_0, \mu_1, 1 - \mu_1\} > 0, \quad r_{\text{pil}} = \min\{r_{\text{cont}}/3, d_{\text{int}}/8\} > 0,$$

where $\theta_0 = (\mu_0, \mu_1)^\top$.

Uniformly over the bounded local set,

$$\|\theta_{n,h} - \theta_0\|_\infty \leq \frac{H}{\sqrt{n}}.$$

Thus, eventually, this distance is less than r_{pil} , and

$$\delta_n = (m_n + 2)^{-1} < d_{\text{int}}/4.$$

For these n , every admissible local parameter satisfies

$$\delta_n + r_{\text{pil}} \leq \min\{(\theta_{n,h})_0, 1 - (\theta_{n,h})_0, (\theta_{n,h})_1, 1 - (\theta_{n,h})_1\}.$$

For an interior θ' , define the bad-prefix mass

$$\begin{aligned} B_n(\theta'; r) &= \sum_{\mathbf{r} \in \{0,1,2,3\}^{m_n}} \omega_{n,\theta'}(\mathbf{r}) \mathbf{1}\{\|\tilde{\theta}(\mathbf{r}) - \theta'\|_\infty \geq r\} \\ &= \Pr_{\theta'}^* \{\|\tilde{\theta} - \theta'\|_\infty \geq r\}, \quad r > 0. \end{aligned}$$

The randomized-response probabilities in [Definition 9](#) give

$$\mathbb{E}_{\theta'}[\hat{u}_1] = \frac{1 + (e^\varepsilon - 1)(1 - p)\theta'_0}{e^\varepsilon + 3}, \quad \mathbb{E}_{\theta'}[\hat{u}_3] = \frac{1 + (e^\varepsilon - 1)p\theta'_1}{e^\varepsilon + 3},$$

and independence of the pilot releases gives

$$\text{Var}_{\theta'}(\hat{u}_1) \leq \frac{1}{m_n}, \quad \text{Var}_{\theta'}(\hat{u}_3) \leq \frac{1}{m_n}.$$

Define the positive frequency thresholds

$$b_{\text{ctl}}(r) = \frac{r(1 - p)(e^\varepsilon - 1)}{e^\varepsilon + 3}, \quad b_{\text{trt}}(r) = \frac{rp(e^\varepsilon - 1)}{e^\varepsilon + 3}.$$

If both empirical-frequency deviations are smaller than their respective thresholds, the inverted arm estimates are within r of the true coordinates. Under the displayed interior-margin condition, clipping preserves these strict inequalities. Therefore Chebyshev's inequality and the union bound give

$$B_n(\theta'; r) \leq \frac{1}{m_n b_{\text{ctl}}(r)^2} + \frac{1}{m_n b_{\text{trt}}(r)^2}.$$

For fixed $r > 0$, this upper bound tends to zero as $m_n \rightarrow \infty$.

On a good prefix at $\theta' = \theta_{n,h}$, the pilot estimate satisfies

$$\|\tilde{\theta}(\mathbf{r}) - \theta_0\|_\infty < 2r_{\text{pil}} < r_{\text{cont}},$$

so its conditional variance differs from V_0 by less than $\delta/2$. On every prefix, the bound $0 \leq \sigma^2 \leq C_\phi^2$ gives

$$|\sigma^2(\theta_{n,h}, \tilde{\theta}(\mathbf{r})) - V_0| \leq C_\phi^2 + |V_0|.$$

Splitting the finite weighted average into good and bad prefixes consequently yields

$$\begin{aligned} |\bar{\sigma}_n^2(\theta_{n,h}) - V_0| &\leq \frac{\delta}{2} + (C_\phi^2 + |V_0|)B_n(\theta_{n,h}; r_{\text{pil}}) \\ &\leq \frac{\delta}{2} + (C_\phi^2 + |V_0|) \left[\frac{1}{m_n b_{\text{ctl}}(r_{\text{pil}})^2} + \frac{1}{m_n b_{\text{trt}}(r_{\text{pil}})^2} \right]. \end{aligned}$$

The last term is eventually less than $\delta/2$, uniformly over admissible h . Hence

$$\sup_{h \in \mathcal{H}_{n,H}(\theta_0)} |\bar{\sigma}_n^2(\theta_{n,h}) - V_0| \longrightarrow 0.$$

12. The sublinear schedule gives

$$\frac{n}{N_n} = \frac{1}{1 - m_n/n} \longrightarrow 1.$$

Using the bound on the averaged variance,

$$\left| \frac{n}{N_n} \bar{\sigma}_n^2(\theta_{n,h}) - V_0 \right| \leq \left| \frac{n}{N_n} - 1 \right| C_\phi^2 + |\bar{\sigma}_n^2(\theta_{n,h}) - V_0|.$$

Thus the exact risk identity implies

$$\sup_{h \in \mathcal{H}_{n,H}(\theta_0)} \left| \mathbb{E}_{\theta_{n,h}}^{\mathcal{P}^*} [(U_{n,h}^{\mathcal{P}^*})^2] - V_0 \right| \longrightarrow 0.$$

In particular, for every $\delta > 0$, eventually

$$\iota(V_0 - \delta) \leq \text{Risk}_{n,H}(\mathcal{P}^*) \leq \iota(V_0 + \delta).$$

For the lower inequality, use $h = 0$, which belongs to $\mathcal{H}_{n,H}(\theta_0)$; the upper inequality holds for every admissible perturbation and hence for their supremum. These bounds give convergence in the extended nonnegative reals:

$$\text{Risk}_{n,H}(\mathcal{P}^*) \longrightarrow \iota(V_0).$$

The upper bounds also ensure that the risks are eventually finite. Consequently,

$$\liminf_{n \rightarrow \infty} \text{Risk}_{n,H}(\mathcal{P}^*) = \iota(V_0) \quad (H > 0).$$

Taking the supremum over positive radii, a nonempty family whose values all equal $\iota(V_0)$, gives

$$\text{Risk}_\infty(\mathcal{P}^*) = \iota(V_0).$$

This is the required attaining witness for the fixed selector.

13. Finally, the universal lower bound applies to every member of $\mathfrak{P}_\varepsilon^{\text{reg}}(\theta_0)$, so

$$\iota(V_0) \leq \inf_{\mathcal{P} \in \mathfrak{P}_\varepsilon^{\text{reg}}(\theta_0)} \text{Risk}_\infty(\mathcal{P}).$$

The regular procedure $\mathcal{P}^*$ constructed above belongs to this class and has risk $\iota(V_0)$. Therefore

$$\inf_{\mathcal{P} \in \mathfrak{P}_\varepsilon^{\text{reg}}(\theta_0)} \text{Risk}_\infty(\mathcal{P}) \leq \text{Risk}_\infty(\mathcal{P}^*) = \iota(V_0).$$

The two inequalities establish the claimed infimum equality in $[0, \infty]$.

□

Proof of Theorem 3. Fix the selector and pilot schedule in the statement, and let $\mathcal{P}$ be the procedure in Algorithm 1. Use the staircase notation of Definition 4 and the contrast notation of Definition 8; thus

$$q = 1 - p, \quad d = e^\varepsilon - 1, \quad c = (-1, 1)^\top.$$

Write

$$N_n = \max\{n - m_n, 0\}.$$

Reciprocals of zero have the convention in Algorithm 1 and Definition 11. For $n \geq 2$, the schedule assumptions give

$$N_n = n - m_n > 0, \quad N_n \longrightarrow \infty, \quad \frac{n}{N_n} \longrightarrow 1.$$

We use the local alternatives and scaled errors of Definitions 6 and 7, with baseline θ .

1. *Bounds and centered conditional corrections.* Define the reference weights, supported on the four singleton patterns, by

$$\alpha_S^{\text{RR}} = \begin{cases} (e^\varepsilon + 3)^{-1}, & |S| = 1, \\ 0, & |S| \neq 1, \end{cases} \quad S \in \mathcal{S}.$$

Each record belongs to exactly one singleton pattern, so

$$\sum_{S \in \mathcal{S}} \alpha_S^{\text{RR}} b_S(j) = \frac{e^\varepsilon + 3}{e^\varepsilon + 3} = 1, \quad j \in \{0, 1, 2, 3\}.$$

Hence $\alpha^{\text{RR}} \in \mathcal{A}_\varepsilon$. For every interior parameter ϑ ,

$$1 \leq h_S(\vartheta) \leq e^\varepsilon, \quad 0 \leq \alpha_S \leq 1 \quad (\alpha \in \mathcal{A}_\varepsilon).$$

The first inequality follows because $h_S(\vartheta)$ is an average of entries of b_S , all of which lie in $[1, e^\varepsilon]$. The second follows from any channel-row normalization and $b_S(j) \geq 1$.

Define the constants

$$\underline{a}_0 = \frac{d^2 q^2}{e^\varepsilon (e^\varepsilon + 3)}, \quad \underline{a}_1 = \frac{d^2 p^2}{e^\varepsilon (e^\varepsilon + 3)}, \quad \underline{J} = \frac{\underline{a}_0 \underline{a}_1}{\underline{a}_0 + \underline{a}_1},$$

and

$$\bar{F}_0 = \sum_{S \in \mathcal{S}} (g_S^\top v(0))^2, \quad \bar{t} = \sqrt{\frac{\bar{F}_0}{\underline{a}_0}}, \quad \bar{g} = \sum_{S \in \mathcal{S}} (|g_{S,0}| + |g_{S,1}|), \quad \bar{\phi} = \frac{\bar{g}(\bar{t} + 1)}{\underline{J}}.$$

Here $\underline{a}_0, \underline{a}_1, \underline{J} > 0$. The contributions of the singleton patterns $\{0\}$ and $\{3\}$ give, for every $u \in \mathbb{R}$,

$$F_\vartheta(\alpha^{\text{RR}}, u) \geq \underline{a}_0 u^2 + \underline{a}_1 (u + 1)^2 = (\underline{a}_0 + \underline{a}_1) \left(u + \frac{\underline{a}_1}{\underline{a}_0 + \underline{a}_1} \right)^2 + \underline{J}.$$

Also, termwise use of $\alpha_S \leq 1$ and $h_S(\vartheta) \geq 1$ gives

$$F_\vartheta(\alpha, 0) \leq \bar{F}_0.$$

The two saddle inequalities therefore imply

$$\begin{aligned} \underline{J} &\leq F_\vartheta(\alpha^{\text{RR}}, t_\vartheta) \leq F_\vartheta(\alpha_\vartheta, t_\vartheta) = J_\vartheta & \underline{a}_0 t_\vartheta^2 &\leq J_\vartheta \leq \bar{F}_0. \\ &\leq F_\vartheta(\alpha_\vartheta, 0) \leq \bar{F}_0, \end{aligned}$$

Consequently,

$$J_\vartheta = J^*(\vartheta, p, \varepsilon) \geq \underline{J} > 0, \quad |t_\vartheta| \leq \bar{t}.$$

For $\vartheta \in (0, 1)^2$ and $S \in \mathcal{S}$, define the correction at the selection parameter by

$$\phi_\vartheta(S) = \frac{g_S^\top v(t_\vartheta)}{J_\vartheta h_S(\vartheta)}.$$

At $\vartheta = \tilde{\theta}$, this is the correction in [Definition 10](#). Since

$$|g_S^\top v(u)| \leq (|g_{S,0}| + |g_{S,1}|)(|u| + 1) \leq \bar{g}(|u| + 1),$$

the preceding bounds yield

$$|\phi_\vartheta(S)| \leq \bar{\phi} \quad (\vartheta \in (0, 1)^2, \ S \in \mathcal{S}).$$

We next establish the exact centering identities. Define

$$\mathbf{1}_2 = (1, 1)^\top.$$

Minimization in the profiling coordinate gives

$$0 = \frac{d}{du} F_\vartheta(\alpha_\vartheta, u) \Big|_{u=t_\vartheta} = 2\mathbf{1}_2^\top I_\vartheta(\alpha_\vartheta) v(t_\vartheta).$$

Thus $I_\vartheta(\alpha_\vartheta) v(t_\vartheta)$ is a multiple of c . Because

$$c^\top v(t_\vartheta) = 1, \quad v(t_\vartheta)^\top I_\vartheta(\alpha_\vartheta) v(t_\vartheta) = J_\vartheta,$$

that multiple is J_ϑ , and hence

$$I_\vartheta(\alpha_\vartheta) v(t_\vartheta) = J_\vartheta c.$$

Channel normalization and the affine formula for h_S also give

$$\sum_S \alpha_{\vartheta, S} g_S = 0, \quad h_S(\xi) = h_S(\vartheta) + g_S^\top (\xi - \vartheta).$$

For $\xi, \vartheta \in (0, 1)^2$, define the main-release probabilities and centered correction by

$$\omega_{\xi, \vartheta}(S) = \alpha_{\vartheta, S} h_S(\xi), \quad \zeta_{\xi, \vartheta}(S) = \phi_\vartheta(S) - c^\top (\xi - \vartheta).$$

These probabilities are nonnegative and sum to one. Expanding the affine denominator gives

$$\begin{aligned} \sum_S \omega_{\xi, \vartheta}(S) \phi_\vartheta(S) &= \frac{1}{J_\vartheta} \left\{ \sum_S \alpha_{\vartheta, S} g_S^\top v(t_\vartheta) + (\xi - \vartheta)^\top I_\vartheta(\alpha_\vartheta) v(t_\vartheta) \right\} \\ &= c^\top (\xi - \vartheta). \end{aligned}$$

At the diagonal,

$$\sum_S \omega_{\vartheta, \vartheta}(S) \phi_\vartheta(S)^2 = \frac{F_\vartheta(\alpha_\vartheta, t_\vartheta)}{J_\vartheta^2} = \frac{1}{J_\vartheta} = V^*(\vartheta, p, \varepsilon).$$

In particular,

$$\sum_S \omega_{\xi, \vartheta}(S) \zeta_{\xi, \vartheta}(S) = 0, \quad |\zeta_{\xi, \vartheta}(S)| \leq 2\bar{\phi}.$$

For the last inequality, the exact mean identity and the probability normalization imply

$$|c^\top (\xi - \vartheta)| = \left| \sum_S \omega_{\xi, \vartheta}(S) \phi_\vartheta(S) \right| \leq \bar{\phi}.$$

2. *Continuity of the conditional variance.* For $\xi, \vartheta \in (0, 1)^2$, define

$$\sigma^2(\xi, \vartheta) = \sum_S \omega_{\xi, \vartheta}(S) \zeta_{\xi, \vartheta}(S)^2 = \sum_S \alpha_{\vartheta, S} h_S(\xi) \phi_\vartheta(S)^2 - \{c^\top (\xi - \vartheta)\}^2.$$

The diagonal second-moment identity gives

$$\sigma^2(\vartheta, \vartheta) = V^*(\vartheta, p, \varepsilon).$$

Subtracting this identity from the preceding formula yields

$$\begin{aligned} \sigma^2(\xi, \vartheta) - V^*(\vartheta, p, \varepsilon) &= \sum_S \alpha_{\vartheta, S} \{h_S(\xi) - h_S(\vartheta)\} \phi_\vartheta(S)^2 \\ &\quad - \{c^\top (\xi - \vartheta)\}^2. \end{aligned}$$

Therefore,

$$|\sigma^2(\xi, \vartheta) - V^*(\vartheta, p, \varepsilon)| \leq \bar{\phi}^2 \sum_S |h_S(\xi) - h_S(\vartheta)| + \{c^\top (\xi - \vartheta)\}^2.$$

To establish continuity of the optimized value, define

$$K = [-\bar{t}, \bar{t}], \quad \mathcal{E}_\xi(u) = \max_{\alpha \in \mathcal{A}_\varepsilon} F_\xi(\alpha, u), \quad \xi \in (0, 1)^2, \quad u \in \mathbb{R}.$$

The saddle inequalities give

$$\mathcal{E}_\xi(t_\xi) = J_\xi, \quad \mathcal{E}_\xi(u) \geq F_\xi(\alpha_\xi, u) \geq J_\xi.$$

Since $t_\xi \in K$,

$$J^*(\xi, p, \varepsilon) = \min_{u \in K} \mathcal{E}_\xi(u).$$

The feasible set is nonempty, closed, and contained in $[0, 1]^{14}$, hence compact. The objective is jointly continuous on interior parameters, feasible weights, and K . The elementary bounds for maxima and minima give

$$|J^*(\xi, p, \varepsilon) - J^*(\theta, p, \varepsilon)| \leq \sup_{\substack{\alpha \in \mathcal{A}_\varepsilon \\ u \in K}} |F_\xi(\alpha, u) - F_\theta(\alpha, u)| \longrightarrow 0 \quad (\xi \rightarrow \theta).$$

The convergence follows from joint continuity on a compact parameter neighborhood times $\mathcal{A}_\varepsilon \times K$. Positivity of J^* then gives

$$V^*(\xi, p, \varepsilon) \longrightarrow V^*(\theta, p, \varepsilon).$$

Combining this with the displayed variance-error bound proves

$$\sigma^2(\xi, \vartheta) \longrightarrow V^*(\theta, p, \varepsilon) \quad \text{as } (\xi, \vartheta) \rightarrow (\theta, \theta) \text{ through interior pairs.}$$

3. *Gaussian convergence of deterministic conditional rows.* Fix $h \in \mathbb{R}^2$. To define valid row probabilities at every index, set

$$\xi_{n,h} = \begin{cases} \theta_{n,h}, & \theta_{n,h} \in (0, 1)^2, \\ \theta, & \theta_{n,h} \notin (0, 1)^2. \end{cases}$$

By [Definition 6](#), $\theta_{n,h} \rightarrow \theta$; thus it is eventually interior, and

$$\xi_{n,h} \rightarrow \theta, \quad \xi_{n,h} = \theta_{n,h} \quad \text{eventually.}$$

For $n \geq 2$ and $\vartheta \in (0, 1)^2$, let $\mathbb{P}_{n,h,\vartheta}^{\text{row}}$ be the product law of N_n independent pattern releases with

$$\mathbb{P}_{n,h,\vartheta}^{\text{row}}(S_j = S) = \omega_{\xi_{n,h},\vartheta}(S), \quad 1 \leq j \leq N_n, \quad S \in \mathcal{S}.$$

Define its total-sample-scaled distribution function by

$$\mathcal{C}_{n,h}(x; \vartheta) = \mathbb{P}_{n,h,\vartheta}^{\text{row}} \left\{ \frac{\sqrt{n}}{N_n} \sum_{j=1}^{N_n} \zeta_{\xi_{n,h},\vartheta}(S_j) \leq x \right\}, \quad x \in \mathbb{R}.$$

Also define the target distribution function

$$\mathcal{G}_\theta(x) = \Phi \left(\frac{x}{\sqrt{V^*(\theta, p, \varepsilon)}} \right).$$

Take any strictly increasing sequence of integers $n_k \geq 2$ and any interior sequence $\vartheta_k \rightarrow \theta$. Under the corresponding row law, put

$$Z_{k,j} = \zeta_{\xi_{n_k,h},\vartheta_k}(S_j), \quad 1 \leq j \leq N_{n_k},$$

and

$$\mathcal{D}_k(y) = \mathbb{P}_{n_k, h, \vartheta_k}^{\text{row}} \left\{ N_{n_k}^{-1/2} \sum_{j=1}^{N_{n_k}} Z_{k,j} \leq y \right\}, \quad y \in \mathbb{R}.$$

The rows are independent, centered, and bounded by $2\bar{\phi}$, and their single-release variances satisfy

$$\mathbb{E}^{\text{row}}[Z_{k,1}^2] = \sigma^2(\xi_{n_k, h}, \vartheta_k) \longrightarrow V^*(\theta, p, \varepsilon) > 0.$$

The increments normalized by $\sqrt{N_{n_k} V^*(\theta, p, \varepsilon)}$ therefore satisfy

$$\sum_{j=1}^{N_{n_k}} \mathbb{E}^{\text{row}} \left[\left(\frac{Z_{k,j}}{\sqrt{N_{n_k} V^*(\theta, p, \varepsilon)}} \right)^2 \right] = \frac{\sigma^2(\xi_{n_k, h}, \vartheta_k)}{V^*(\theta, p, \varepsilon)} \longrightarrow 1,$$

and

$$\sum_{j=1}^{N_{n_k}} \mathbb{E}^{\text{row}} \left[\left| \frac{Z_{k,j}}{\sqrt{N_{n_k} V^*(\theta, p, \varepsilon)}} \right|^4 \right] \leq \frac{(2\bar{\phi})^4}{N_{n_k} \{V^*(\theta, p, \varepsilon)\}^2} \longrightarrow 0.$$

The triangular-array Lyapunov central limit theorem consequently gives

$$\mathcal{D}_k(y) \longrightarrow \mathcal{G}_\theta(y) \quad (y \in \mathbb{R}).$$

Define

$$\lambda_k = \sqrt{\frac{n_k}{N_{n_k}}}, \quad \chi_k = \frac{x}{\lambda_k}.$$

Then $\lambda_k \rightarrow 1$, $\chi_k \rightarrow x$, and

$$\mathcal{C}_{n_k, h}(x; \vartheta_k) = \mathcal{D}_k(\chi_k).$$

For every $a > 0$, eventually

$$\mathcal{D}_k(x - a) \leq \mathcal{D}_k(\chi_k) \leq \mathcal{D}_k(x + a).$$

Pointwise convergence and continuity of $\mathcal{G}_\theta$, followed by $a \downarrow 0$, prove

$$\mathcal{C}_{n_k, h}(x; \vartheta_k) \longrightarrow \mathcal{G}_\theta(x).$$

4. *Averaging over the private pilot.* For $n \geq 2$ and a pilot release vector $\mathbf{r} = (r_1, \dots, r_{m_n}) \in \{0, 1, 2, 3\}^{m_n}$, define

$$\vartheta_n(\mathbf{r}) = \tilde{\theta} \quad \text{computed from } R_i = r_i \text{ by } \textit{Algorithm 1}.$$

Clipping ensures $\vartheta_n(\mathbf{r}) \in (0, 1)^2$ for every $\mathbf{r}$. Define the pilot weights

$$w_{n, h}(\mathbf{r}) = \prod_{i=1}^{m_n} \frac{1 + d \pi_{\xi_{n, h}, r_i}}{e^\varepsilon + 3}.$$

By [Definition 9](#), these weights are nonnegative and sum to one. Factorization of the successive release kernels gives, whenever $\theta_{n, h}$ is interior,

$$\Pr_{\theta_{n, h}} \{U_{n, h}^{\mathcal{P}} \leq x\} = \sum_{\mathbf{r}} w_{n, h}(\mathbf{r}) \mathcal{C}_{n, h}(x; \vartheta_n(\mathbf{r})).$$

Indeed, the exact mean identity gives, for each fixed pilot,

$$U_{n,h}^{\mathcal{P}} = \frac{\sqrt{n}}{N_n} \sum_{j=1}^{N_n} \zeta_{\theta_{n,h}, \vartheta_n(\mathbf{r})}(S_j).$$

We record the pilot concentration bound used in this averaging. Define

$$\ell_0 = \frac{qd}{e^\varepsilon + 3}, \quad \ell_1 = \frac{pd}{e^\varepsilon + 3}, \quad \mathcal{B}_n(\varrho) = \frac{1}{m_n(\varrho\ell_0)^2} + \frac{1}{m_n(\varrho\ell_1)^2}, \quad \varrho > 0, \quad n \geq 2.$$

For an interior parameter ξ , the means of the pilot indicators for categories 1 and 3 are

$$\frac{1 + dq\xi_0}{e^\varepsilon + 3}, \quad \frac{1 + dp\xi_1}{e^\varepsilon + 3}.$$

Each indicator has variance at most one, so independence and Chebyshev's inequality give, for $k \in \{1, 3\}$ and $b > 0$,

$$\Pr_{\xi}\{|\hat{u}_k - \mathbb{E}_{\xi}\hat{u}_k| \geq b\} \leq \frac{1}{m_n b^2}.$$

Frequency inversion scales the two coordinate errors by ℓ_0^{-1} and ℓ_1^{-1} . If

$$\delta_n + \varrho \leq \min\{\xi_0, 1 - \xi_0, \xi_1, 1 - \xi_1\},$$

then an inverted coordinate within ϱ of ξ lies inside the clipping interval. Taking the contrapositive coordinatewise and applying the union bound yields

$$\Pr_{\xi}\{\|\tilde{\theta} - \xi\|_{\infty} \geq \varrho\} \leq \mathcal{B}_n(\varrho), \quad \|\tilde{\theta} - \xi\|_{\infty} = \max_{k \in \{0,1\}} |\tilde{\theta}_k - \xi_k|.$$

Define, for $n \geq 2$,

$$\beta_n = m_n^{-1/4}, \quad \varrho_{n,h} = 2 \max \left\{ \beta_n, \sqrt{\|\theta_{n,h} - \theta\|_{\infty}} \right\},$$

and

$$\Delta_{\theta} = \min\{\theta_0, 1 - \theta_0, \theta_1, 1 - \theta_1\} > 0.$$

Pilot divergence and local convergence imply

$$\beta_n \rightarrow 0, \quad \varrho_{n,h} \rightarrow 0.$$

Eventually,

$$\beta_n < \Delta_{\theta}/4, \quad \delta_n < \Delta_{\theta}/4, \quad \|\theta_{n,h} - \theta\|_{\infty} < \Delta_{\theta}/4,$$

so the concentration bound applies at $\xi = \theta_{n,h}$ with radius β_n . Its right-hand side becomes

$$\mathcal{B}_n(\beta_n) = \frac{1}{\sqrt{m_n}} \left(\frac{1}{\ell_0^2} + \frac{1}{\ell_1^2} \right) \rightarrow 0.$$

Moreover, eventually

$$\|\theta_{n,h} - \theta\|_{\infty} \leq \sqrt{\|\theta_{n,h} - \theta\|_{\infty}}.$$

Hence

$$\|\vartheta_n(\mathbf{r}) - \theta_{n,h}\|_\infty < \beta_n \implies \|\vartheta_n(\mathbf{r}) - \theta\|_\infty \leq \varrho_{n,h}.$$

The deterministic-row convergence is uniform over these shrinking pilot neighborhoods. Explicitly, for every $a > 0$, eventually

$$\|\vartheta_n(\mathbf{r}) - \theta\|_\infty \leq \varrho_{n,h} \implies |\mathcal{C}_{n,h}(x; \vartheta_n(\mathbf{r})) - \mathcal{G}_\theta(x)| < a.$$

If this implication failed, one could choose strictly increasing n_k and pilot vectors $\mathbf{r}_k$ such that

$$\|\vartheta_{n_k}(\mathbf{r}_k) - \theta\|_\infty \leq \varrho_{n_k,h} \longrightarrow 0, \quad |\mathcal{C}_{n_k,h}(x; \vartheta_{n_k}(\mathbf{r}_k)) - \mathcal{G}_\theta(x)| \geq a.$$

This contradicts the deterministic-row limit.

Both distribution functions lie in $[0, 1]$. Splitting the pilot average into the displayed neighborhood and its complement therefore gives, eventually,

$$\left| \sum_{\mathbf{r}} w_{n,h}(\mathbf{r}) \mathcal{C}_{n,h}(x; \vartheta_n(\mathbf{r})) - \mathcal{G}_\theta(x) \right| \leq a + \mathcal{B}_n(\beta_n).$$

Letting $n \rightarrow \infty$, then $a \downarrow 0$, proves

$$\Pr_{\theta_{n,h}} \{U_{n,h}^{\mathcal{P}} \leq x\} \longrightarrow \mathcal{G}_\theta(x) \quad (h \in \mathbb{R}^2, x \in \mathbb{R}).$$

5. *The weak local limit.* For an interior parameter ξ , the transcript law is a probability law: it is a mixture of conditional transcript probability laws, with nonnegative input-path weights satisfying

$$\sum_{(j_1, \dots, j_n) \in \{0,1,2,3\}^n} \prod_{i=1}^n \pi_{\xi, j_i} = \left(\sum_{j=0}^3 \pi_{\xi, j} \right)^n = 1.$$

This also holds at $n = 0$, with the empty-product convention.

For each fixed h , define a Borel probability law on $\mathbb{R}$ by

$$\nu_{n,h} = \begin{cases} \mathcal{L}_{\theta_{n,h}}(U_{n,h}^{\mathcal{P}}), & \theta_{n,h} \in (0, 1)^2, \\ N(0, V^*(\theta, p, \varepsilon)), & \theta_{n,h} \notin (0, 1)^2. \end{cases}$$

The local parameter is eventually interior, so the preceding distribution-function limit applies to $\nu_{n,h}$. Convergence at every point of the continuous Gaussian distribution function implies weak convergence. Equivalently, for every bounded continuous $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$\int f(u) d\nu_{n,h}(u) \longrightarrow \int f(u) dN(0, V^*(\theta, p, \varepsilon))(u).$$

Eventually the left-hand side is $\mathbb{E}_{\theta_{n,h}}[f(U_{n,h}^{\mathcal{P}})]$, proving the asserted common weak local limit. Taking $h = 0$ gives the stated convergence at θ .

6. *Consistency of the empirical variance.* The empirical variance is the quantity in [Definition 11](#). Fix $\delta > 0$. Continuity of σ^2 at (θ, θ) supplies $R_{\text{var}} > 0$ such that

$$\|\vartheta - \theta\|_\infty < R_{\text{var}}, \quad \vartheta \in (0, 1)^2 \implies |\sigma^2(\theta, \vartheta) - V^*(\theta, p, \varepsilon)| < \delta/2.$$

Set

$$\varrho_{\text{var}} = \min\{R_{\text{var}}/2, \Delta_\theta/4\} > 0.$$

Eventually $\delta_n < \Delta_\theta/2$, so

$$\delta_n + \varrho_{\text{var}} \leq \Delta_\theta, \quad \Pr_\theta\{\|\tilde{\theta} - \theta\|_\infty \geq \varrho_{\text{var}}\} \leq \mathcal{B}_n(\varrho_{\text{var}}) \longrightarrow 0.$$

Condition on a pilot vector, and write its estimate as ϑ . Define the centered main-sample moments

$$\bar{\zeta}_n = \frac{1}{N_n} \sum_{j=1}^{N_n} \zeta_{\theta, \vartheta}(S_j), \quad \bar{\zeta}_n^2 = \frac{1}{N_n} \sum_{j=1}^{N_n} \zeta_{\theta, \vartheta}(S_j)^2.$$

Translation invariance of empirical variance gives

$$\hat{V}^* = \bar{\zeta}_n^2 - \bar{\zeta}_n^2.$$

Conditional independence, centering, and boundedness give

$$\mathbb{E}_\theta[\bar{\zeta}_n^2 \mid H_{m_n}] = \frac{\sigma^2(\theta, \vartheta)}{N_n} \leq \frac{(2\bar{\phi})^2}{N_n},$$

and

$$\text{Var}_\theta(\bar{\zeta}_n^2 \mid H_{m_n}) \leq \frac{(2\bar{\phi})^4}{N_n}.$$

On a pilot with $|\sigma^2(\theta, \vartheta) - V^*(\theta, p, \varepsilon)| < \delta/2$, the event $|\hat{V}^* - V^*(\theta, p, \varepsilon)| > \delta$ implies

$$|\bar{\zeta}_n^2 - \sigma^2(\theta, \vartheta)| \geq \delta/4 \quad \text{or} \quad \bar{\zeta}_n^2 \geq \delta/4.$$

Chebyshev's and Markov's inequalities consequently bound its conditional probability by

$$\frac{(2\bar{\phi})^4}{N_n(\delta/4)^2} + \frac{(2\bar{\phi})^2}{N_n(\delta/4)}.$$

On the complementary pilots the conditional probability is at most one. Averaging gives, eventually,

$$\begin{aligned} \Pr_\theta\{|\hat{V}^* - V^*(\theta, p, \varepsilon)| > \delta\} &\leq \mathcal{B}_n(\varrho_{\text{var}}) + \frac{(2\bar{\phi})^4}{N_n(\delta/4)^2} \\ &\quad + \frac{(2\bar{\phi})^2}{N_n(\delta/4)} \longrightarrow 0. \end{aligned}$$

7. *Uniform squared tails and regularity.* Fix $H > 0$, and use the admissible perturbation set of [Definition 8](#). For sufficiently large n ,

$$\frac{n}{N_n} \leq 2.$$

For every admissible h , condition on the pilot and denote the N_n centered corrections by

$$Z_j = \zeta_{\theta_{n,h}, \tilde{\theta}}(S_j), \quad 1 \leq j \leq N_n.$$

They are conditionally independent and identically distributed, with mean zero and

$$|Z_j| \leq 2\bar{\phi}, \quad U_{n,h}^{\mathcal{P}} = \frac{\sqrt{n}}{N_n} \sum_{j=1}^{N_n} Z_j.$$

Define their conditional moments by

$$\mathbf{m}_2 = \mathbb{E}_{\theta_{n,h}}[Z_1^2 \mid H_{m_n}], \quad \mathbf{m}_4 = \mathbb{E}_{\theta_{n,h}}[Z_1^4 \mid H_{m_n}].$$

Then

$$0 \leq \mathbf{m}_2 \leq (2\bar{\phi})^2, \quad 0 \leq \mathbf{m}_4 \leq (2\bar{\phi})^4.$$

In the fourth-power expansion, terms having an index that occurs exactly once have zero conditional expectation. The remaining terms give

$$\mathbb{E}_{\theta_{n,h}} \left[\left(\sum_{j=1}^{N_n} Z_j \right)^4 \middle| H_{m_n} \right] = N_n \mathbf{m}_4 + 3N_n(N_n - 1) \mathbf{m}_2^2.$$

Since $N_n \geq 1$,

$$\mathbb{E}_{\theta_{n,h}} \left[\left(N_n^{-1/2} \sum_{j=1}^{N_n} Z_j \right)^4 \middle| H_{m_n} \right] \leq 4(2\bar{\phi})^4.$$

Thus, uniformly over the pilot and admissible h ,

$$\mathbb{E}_{\theta_{n,h}}[(U_{n,h}^{\mathcal{P}})^4 \mid H_{m_n}] \leq 4 \left(\frac{n}{N_n} \right)^2 (2\bar{\phi})^4 \leq 16(2\bar{\phi})^4.$$

Averaging preserves this bound. For every integer $M > 0$, the pointwise inequality

$$u^2 \mathbf{1}\{|u| > M\} \leq \frac{u^4}{M^2}, \quad u \in \mathbb{R},$$

therefore yields

$$0 \leq \limsup_{n \rightarrow \infty} \sup_{h \in \mathcal{H}_{n,H}(\theta)} \mathbb{E}_{\theta_{n,h}} [(U_{n,h}^{\mathcal{P}})^2 \mathbf{1}\{|U_{n,h}^{\mathcal{P}}| > M\}] \leq \frac{16(2\bar{\phi})^4}{M^2}.$$

Letting $M \rightarrow \infty$ proves the required uniform squared-tail convergence.

The Gaussian law

$$L_{\mathcal{P}} = N(0, V^*(\theta, p, \varepsilon))$$

is a probability law with

$$\int u^2 dL_{\mathcal{P}}(u) < \infty.$$

Together with the common weak local limit and the squared-tail bound, this verifies the regularity conditions in [Definition 8](#).

8. *Standardizing the limiting distribution function.* The reference-design argument gives

$$J^*(\theta, p, \varepsilon) > 0, \quad V^*(\theta, p, \varepsilon) > 0, \quad \sqrt{V^*(\theta, p, \varepsilon)} > 0.$$

Define, for each sample size,

$$T_n = \frac{U_{n,0}^{\mathcal{P}}}{\sqrt{V^*(\theta, p, \varepsilon)}}.$$

Since $\theta_{n,0} = \theta$, positivity of the denominator gives, for every $x \in \mathbb{R}$,

$$\{T_n \leq x\} = \left\{U_{n,0}^{\mathcal{P}} \leq x\sqrt{V^*(\theta, p, \varepsilon)}\right\}.$$

The Gaussian scaling identity is

$$N(0, V^*(\theta, p, \varepsilon)) \left((-\infty, x\sqrt{V^*(\theta, p, \varepsilon)}] \right) = \Phi(x).$$

Applying the local distribution-function limit at this threshold proves

$$\Pr_{\theta}\{T_n \leq x\} \longrightarrow \Phi(x) \quad (x \in \mathbb{R}).$$

9. *Transfer to the Wald pivot.* First, consistency also controls non-strict deviation events. For every $b > 0$,

$$\{|\hat{V}^* - V^*(\theta, p, \varepsilon)| \geq b\} \subseteq \{|\hat{V}^* - V^*(\theta, p, \varepsilon)| > b/2\},$$

and hence

$$\Pr_{\theta}\{|\hat{V}^* - V^*(\theta, p, \varepsilon)| \geq b\} \longrightarrow 0.$$

The normal distribution function is continuous and strictly increasing, with limits zero and one at the two ends of the real line. The critical-value definition thus gives

$$\Phi(z_{1-\gamma/2}) = 1 - \gamma/2.$$

Normal symmetry gives $\Phi(0) = 1/2$. Since $1 - \gamma/2 > 1/2$,

$$z_{1-\gamma/2} > 0.$$

Define the estimation error and fallback pivot, for $n \geq 1$, by

$$\Delta_{\tau,n} = \hat{\tau}^* - \tau, \quad W_n = \begin{cases} 0, & \hat{V}^* = 0, \Delta_{\tau,n} = 0, \\ z_{1-\gamma/2} + 1, & \hat{V}^* = 0, \Delta_{\tau,n} \neq 0, \\ \sqrt{n} \Delta_{\tau,n} / \sqrt{\hat{V}^*}, & \hat{V}^* > 0. \end{cases}$$

The empirical variance is nonnegative on every transcript, since it is a nonnegative multiple of a sum of squares. When it is positive, ordinary square-root algebra gives

$$|W_n| \leq z_{1-\gamma/2} \iff |\Delta_{\tau,n}| \leq z_{1-\gamma/2} \sqrt{\hat{V}^*/n}.$$

When it is zero, both events hold precisely when $\Delta_{\tau,n} = 0$, by the two fallback branches. Thus the equivalence holds on every transcript.

We show that $W_n - T_n$ converges to zero in probability. The standardized distribution-function limit first gives tightness. Explicitly, for any $\eta > 0$, define

$$\eta_{\text{tail}} = \min\{\eta/8, 1/8\}.$$

Choose $R_{\text{tight}} > 0$ so that the standard normal probabilities of $(-\infty, -R_{\text{tight}}]$ and $[R_{\text{tight}}, \infty)$ are each at most η_{tail} . Distribution-function convergence gives weak convergence of T_n , and the two endpoints have zero normal mass. Consequently,

$$\Pr_{\theta}\{T_n \notin [-R_{\text{tight}}, R_{\text{tight}}]\} \longrightarrow 1 - \Phi(R_{\text{tight}}) + \Phi(-R_{\text{tight}}) \leq 2\eta_{\text{tail}} < \eta.$$

Taking

$$M_{\text{tight}} = R_{\text{tight}} + 1$$

therefore gives, eventually,

$$\Pr_{\theta}\{M_{\text{tight}} \leq |T_n|\} < \eta.$$

Now fix $\delta > 0$ and $\eta > 0$, and choose such a positive M_{tight} with tail probability eventually below $\eta/2$. Define

$$\mathfrak{g}_{\text{scale}}(u) = \frac{\sqrt{V^*(\theta, p, \varepsilon)}}{\sqrt{u}}, \quad u > 0.$$

This function is continuous at $V^*(\theta, p, \varepsilon)$, where it equals one. Choose $\varrho_{\text{scale},0} > 0$ so that

$$u > 0, \quad |u - V^*(\theta, p, \varepsilon)| < \varrho_{\text{scale},0} \implies |\mathfrak{g}_{\text{scale}}(u) - 1| < \delta/M_{\text{tight}},$$

and set

$$\varrho_{\text{scale}} = \min\{\varrho_{\text{scale},0}/2, V^*(\theta, p, \varepsilon)/2\} > 0.$$

On the event

$$|\widehat{V}^* - V^*(\theta, p, \varepsilon)| < \varrho_{\text{scale}}, \quad |T_n| < M_{\text{tight}},$$

the estimated variance is positive and

$$W_n = T_n \mathfrak{g}_{\text{scale}}(\widehat{V}^*), \quad |W_n - T_n| \leq M_{\text{tight}} |\mathfrak{g}_{\text{scale}}(\widehat{V}^*) - 1| < \delta.$$

It follows that

$$\begin{aligned} \Pr_{\theta}\{|W_n - T_n| \geq \delta\} &\leq \Pr_{\theta}\{|\widehat{V}^* - V^*(\theta, p, \varepsilon)| \geq \varrho_{\text{scale}}\} \\ &\quad + \Pr_{\theta}\{|T_n| \geq M_{\text{tight}}\} < \eta \end{aligned}$$

eventually. Thus this deviation probability tends to zero.

For every $x \in \mathbb{R}$ and $\delta > 0$,

$$\begin{aligned} \Pr_{\theta}\{T_n \leq x - \delta\} - \Pr_{\theta}\{|W_n - T_n| \geq \delta\} &\leq \Pr_{\theta}\{W_n \leq x\} \\ &\leq \Pr_{\theta}\{T_n \leq x + \delta\} + \Pr_{\theta}\{|W_n - T_n| \geq \delta\}. \end{aligned}$$

Taking limits and then letting $\delta \downarrow 0$ proves convergence of the pivot distribution functions to Φ . The closed interval $[-z_{1-\gamma/2}, z_{1-\gamma/2}]$ has zero normal boundary mass, so

$$\Pr_{\theta}\{|W_n| \leq z_{1-\gamma/2}\} \longrightarrow N(0, 1)([-z_{1-\gamma/2}, z_{1-\gamma/2}]).$$

The event equivalence transfers this limit to the stated Wald event.

10. *Privacy at every sample size and history.* The randomized-response probabilities in [Definition 9](#) are normalized, and for every pair of private records and every reported category,

$$Q_{\text{RR}}(k \mid x_j) \leq e^\varepsilon Q_{\text{RR}}(k \mid x_{j'}).$$

Summing over an output event proves the pilot privacy inequality.

At any main stage, including at an unattainable history, the history-based parameter estimate clips both coordinates to

$$[\delta_n, 1 - \delta_n] \subset (0, 1), \quad 0 < \delta_n = (m_n + 2)^{-1} \leq 1/2.$$

The selected weights are therefore feasible. For every pattern S ,

$$Q_{\alpha_\vartheta}(S \mid x_j) = \alpha_{\vartheta, S} b_S(j), \quad \sum_S Q_{\alpha_\vartheta}(S \mid x_j) = 1,$$

and

$$\alpha_{\vartheta, S} b_S(j) \leq e^\varepsilon \alpha_{\vartheta, S} b_S(j').$$

This inequality also holds for zero weights. Summing over patterns proves privacy for every measurable main-release event. At each fixed history, the same selected channel is used on the two compared private records. The two stage cases therefore verify the Markov property and the all-history inequality in [Assumption 9](#). At $n = 0$, the stage conditions are vacuous.

11. *Exact pilot-adapted unbiasedness.* Fix $n \geq 2$. Factorization gives, for a pilot vector $\mathbf{r}$ and a main pattern vector $\mathbf{s} = (s_1, \dots, s_{N_n}) \in \mathcal{S}^{N_n}$,

$$\Pr_\theta\{H_{m_n} = \mathbf{r}, (S_{m_n+1}, \dots, S_n) = \mathbf{s}\} = w_{n,0}(\mathbf{r}) \prod_{j=1}^{N_n} \omega_{\theta, \vartheta_n(\mathbf{r})}(s_j).$$

Transcripts with a release type incompatible with its stage have zero mass.

For each fixed pilot, product normalization and the exact correction mean give

$$\begin{aligned} \sum_{\mathbf{s} \in \mathcal{S}^{N_n}} \left(\prod_{j=1}^{N_n} \omega_{\theta, \vartheta_n(\mathbf{r})}(s_j) \right) \left\{ c^\top \vartheta_n(\mathbf{r}) + \frac{1}{N_n} \sum_{j=1}^{N_n} \phi_{\vartheta_n(\mathbf{r})}(s_j) - c^\top \theta \right\} \\ = c^\top \vartheta_n(\mathbf{r}) + \frac{N_n}{N_n} c^\top (\theta - \vartheta_n(\mathbf{r})) - c^\top \theta = 0. \end{aligned}$$

Multiplication by $w_{n,0}(\mathbf{r})$ shows that the integral of $\hat{\tau}^* - \tau$ over every pilot-prefix cell is zero. A prefix incompatible with the pilot release type has zero mass, so its cell integral is also zero.

A function f satisfying the statement is constant on each such cell. Summing the zero cell integrals, each multiplied by its corresponding value of f , gives

$$\mathbb{E}_\theta[(\hat{\tau}^* - \tau)f] = 0.$$

The transcript space is finite and the estimator and f are real-valued, so their product is integrable under the transcript probability law.

12. *Evaluation of the limiting coverage.* Normal symmetry and the critical-value identity give

$$\Phi(-z_{1-\gamma/2}) = 1 - \Phi(z_{1-\gamma/2}) = \gamma/2.$$

The standard normal law has zero mass at the interval endpoints. Hence

$$N(0, 1)([-z_{1-\gamma/2}, z_{1-\gamma/2}]) = \Phi(z_{1-\gamma/2}) - \Phi(-z_{1-\gamma/2}) = 1 - \gamma.$$

Substitution into the Wald-event limit proves

$$\Pr_{\theta} \left\{ |\widehat{\tau}^* - \tau| \leq z_{1-\gamma/2} \sqrt{\widehat{V}^*/n} \right\} \longrightarrow 1 - \gamma,$$

completing all the asserted conclusions. □

Proof of Theorem 4. 1. Use the record and pattern indexing of Definition 3. For every $S \in \mathcal{S}$, define

$$\begin{aligned} b_{S,j} &:= 1 + (e^{\varepsilon} - 1) \mathbf{1}_{\{j \in S\}}, & j &\in \{0, 1, 2, 3\}, \\ \mathbf{b}_S &:= (b_{S,0}, b_{S,1}, b_{S,2}, b_{S,3})^{\top}, \\ g_S &:= (e^{\varepsilon} - 1) \begin{pmatrix} (1-p)(\mathbf{1}_{\{1 \in S\}} - \mathbf{1}_{\{0 \in S\}}) \\ p(\mathbf{1}_{\{3 \in S\}} - \mathbf{1}_{\{2 \in S\}}) \end{pmatrix}, \\ h_S(\theta) &:= \sum_{j=0}^3 \pi_{\theta,j} b_{S,j}, & v(t) &:= (t, t+1)^{\top}, \quad t \in \mathbb{R}. \end{aligned}$$

The record probabilities are

$$(\pi_{\theta,0}, \pi_{\theta,1}, \pi_{\theta,2}, \pi_{\theta,3}) = ((1-p)(1-\mu_0), (1-p)\mu_0, p(1-\mu_1), p\mu_1).$$

The interior hypotheses make each probability positive, and $b_{S,j} \geq 1$. Consequently,

$$h_S(\theta) > 0 \quad (S \in \mathcal{S}).$$

Write

$$\begin{aligned} \mathcal{A}_{\varepsilon} &= \left\{ \beta \in \mathbb{R}^{\mathcal{S}} : \beta_S \geq 0, \sum_{S \in \mathcal{S}} \beta_S \mathbf{b}_S = \mathbf{1} \right\}, & \mathbf{1} &:= (1, 1, 1, 1)^{\top}, \\ f_S(t) &:= \frac{(g_S^{\top} v(t))^2}{h_S(\theta)}, & F_{\theta}(\beta, t) &= \sum_{S \in \mathcal{S}} \beta_S f_S(t), \\ E(t) &:= \sup_{\beta \in \mathcal{A}_{\varepsilon}} F_{\theta}(\beta, t), & t &\in \mathbb{R}. \end{aligned}$$

By Theorem 1, there is $t^* \in \mathbb{R}$ such that

$$E(t^*) = \inf_{t \in \mathbb{R}} E(t) = J^*(\theta, p, \varepsilon).$$

2. We first obtain a weight vector attaining the profiled optimum. The complementary patterns $S_1 = \{0\}$ and $S_{14} = \{1, 2, 3\}$ give the feasible vector

$$\alpha_S^{\text{base}} := \begin{cases} (e^\varepsilon + 1)^{-1}, & S \in \{S_1, S_{14}\}, \\ 0, & \text{otherwise,} \end{cases} \quad \sum_{S \in \mathcal{S}} \alpha_S^{\text{base}} \mathbf{b}_S = \frac{\mathbf{b}_{S_1} + \mathbf{b}_{S_{14}}}{e^\varepsilon + 1} = \mathbf{1}.$$

Thus $\mathcal{A}_\varepsilon$ is nonempty. It is closed, since its constraints are finitely many continuous equalities and inequalities. Moreover, for every $\beta \in \mathcal{A}_\varepsilon$ and $S \in \mathcal{S}$,

$$0 \leq \beta_S \leq \sum_{T \in \mathcal{S}} \beta_T \leq \sum_{T \in \mathcal{S}} \beta_T b_{T,0} = 1.$$

Hence $\mathcal{A}_\varepsilon$ is compact.

Define the profiled objective on this set by

$$\Phi(\beta) := \inf_{t \in \mathbb{R}} F_\theta(\beta, t), \quad \beta \in \mathcal{A}_\varepsilon.$$

Nonnegativity of the weights and positivity of the denominators give

$$0 \leq \Phi(\beta) \leq F_\theta(\beta, 0) < \infty.$$

For each fixed t , $F_\theta(\cdot, t)$ is continuous. Its infimum over t is therefore upper semicontinuous. The maximum of Φ on the nonempty compact feasible set is attained. Using the optimized value in [Definition 5](#), choose $\alpha^{\text{opt}} \in \mathcal{A}_\varepsilon$ with

$$\Phi(\alpha^{\text{opt}}) = J^*(\theta, p, \varepsilon).$$

In particular,

$$J^*(\theta, p, \varepsilon) \leq F_\theta(\alpha^{\text{opt}}, t) \quad (t \in \mathbb{R}).$$

3. Continuity and compactness also supply a maximizer at the fixed coordinate t^* : choose

$$\alpha^{\text{env}} \in \arg \max_{\beta \in \mathcal{A}_\varepsilon} F_\theta(\beta, t^*).$$

Then

$$E(t^*) = F_\theta(\alpha^{\text{env}}, t^*),$$

and the preceding lower bound yields

$$J^*(\theta, p, \varepsilon) \leq F_\theta(\alpha^{\text{opt}}, t^*) \leq F_\theta(\alpha^{\text{env}}, t^*) = E(t^*) = J^*(\theta, p, \varepsilon).$$

Therefore

$$F_\theta(\alpha^{\text{opt}}, t^*) = J^*(\theta, p, \varepsilon), \quad F_\theta(\beta, t^*) \leq F_\theta(\alpha^{\text{opt}}, t^*) \quad (\beta \in \mathcal{A}_\varepsilon).$$

4. Define, for $S \in \mathcal{S}$, $t \in \mathbb{R}$, and $\beta \in \mathbb{R}^\mathcal{S}$,

$$\dot{f}_S(t) := \frac{2(g_S^\top v(t))(g_{S,0} + g_{S,1})}{h_S(\theta)},$$

$$D_\beta(t) := \sum_{S \in \mathcal{S}} \beta_S \dot{f}_S(t),$$

$$C_\beta := \sum_{S \in \mathcal{S}} \beta_S \frac{(g_{S,0} + g_{S,1})^2}{h_S(\theta)}.$$

Expanding each square gives the identity

$$F_\theta(\beta, t + \lambda) - F_\theta(\beta, t) = \lambda D_\beta(t) + \lambda^2 C_\beta \quad (t, \lambda \in \mathbb{R}).$$

The preceding step shows that t^* minimizes $F_\theta(\alpha^{\text{opt}}, \cdot)$. If $D_{\alpha^{\text{opt}}}(t^*) \neq 0$, take

$$\lambda_{\text{stat}} := -\frac{D_{\alpha^{\text{opt}}}(t^*)}{2(|C_{\alpha^{\text{opt}}}| + 1)}.$$

The quadratic identity then gives

$$\begin{aligned} 0 &\leq F_\theta(\alpha^{\text{opt}}, t^* + \lambda_{\text{stat}}) - F_\theta(\alpha^{\text{opt}}, t^*) \\ &= \frac{D_{\alpha^{\text{opt}}}(t^*)^2}{4(|C_{\alpha^{\text{opt}}}| + 1)^2} [C_{\alpha^{\text{opt}}} - 2(|C_{\alpha^{\text{opt}}}| + 1)] < 0, \end{aligned}$$

where the strict inequality uses $C_{\alpha^{\text{opt}}} \leq |C_{\alpha^{\text{opt}}}|$. This contradiction proves

$$D_{\alpha^{\text{opt}}}(t^*) = 0.$$

5. For every real weight vector, define its indexed support by

$$\text{supp}(\beta) := \{S \in \mathcal{S} : \beta_S \neq 0\}, \quad \beta \in \mathbb{R}^{\mathcal{S}}.$$

Consider

$$\mathcal{W} := \{\beta \in \mathcal{A}_\varepsilon : D_\beta(t^*) = D_{\alpha^{\text{opt}}}(t^*), F_\theta(\beta, t^*) = F_\theta(\alpha^{\text{opt}}, t^*)\}.$$

This set contains α^{opt} . Choose a member of minimum support cardinality:

$$\alpha^{\text{sp}} \in \arg \min_{\beta \in \mathcal{W}} |\text{supp}(\beta)|, \quad \mathcal{K}_{\text{sp}} := \text{supp}(\alpha^{\text{sp}}).$$

Such a choice exists because support cardinalities are nonnegative integers. By construction,

$$D_{\alpha^{\text{sp}}}(t^*) = 0, \quad F_\theta(\alpha^{\text{sp}}, t^*) = J^*(\theta, p, \varepsilon),$$

and

$$F_\theta(\beta, t^*) \leq F_\theta(\alpha^{\text{sp}}, t^*) \quad (\beta \in \mathcal{A}_\varepsilon).$$

Suppose that $|\mathcal{K}_{\text{sp}}| > 5$. The more than five columns

$$\begin{pmatrix} \mathbf{b}_S \\ f_S(t^*) \end{pmatrix} \in \mathbb{R}^5, \quad S \in \mathcal{K}_{\text{sp}},$$

are linearly dependent. Extending their dependence coefficients by zero outside $\mathcal{K}_{\text{sp}}$ gives

$$\begin{aligned} \Delta &= (\Delta_S)_{S \in \mathcal{S}} \in \mathbb{R}^{\mathcal{S}} \setminus \{0\}, \quad \Delta_S = 0 \quad (S \notin \mathcal{K}_{\text{sp}}), \\ \sum_{S \in \mathcal{S}} \Delta_S \mathbf{b}_S &= 0, \quad \sum_{S \in \mathcal{S}} \Delta_S f_S(t^*) = 0. \end{aligned}$$

6. The objective moment of this direction also vanishes. To see this, define

$$\delta_S := \begin{cases} 1, & \alpha_S^{\text{sp}} = 0, \\ \frac{\alpha_S^{\text{sp}}}{|\Delta_S| + 1}, & \alpha_S^{\text{sp}} > 0, \end{cases} \quad \delta := \min_{S \in \mathcal{S}} \delta_S.$$

Each δ_S is positive. Since $\mathcal{S}$ is finite and nonempty, and $\Delta_S = 0$ wherever $\alpha_S^{\text{sp}} = 0$,

$$\delta > 0, \quad \delta |\Delta_S| \leq \alpha_S^{\text{sp}} \quad (S \in \mathcal{S}).$$

Thus both perturbations are feasible:

$$\alpha^{\text{sp}} \pm \delta \Delta \geq 0, \quad \sum_{S \in \mathcal{S}} (\alpha_S^{\text{sp}} \pm \delta \Delta_S) \mathbf{b}_S = \mathbf{1}.$$

Their objective values satisfy

$$F_\theta(\alpha^{\text{sp}} \pm \delta \Delta, t^*) = F_\theta(\alpha^{\text{sp}}, t^*) \pm \delta \sum_{S \in \mathcal{S}} \Delta_S f_S(t^*) \leq F_\theta(\alpha^{\text{sp}}, t^*).$$

Using both signs and $\delta > 0$ proves

$$\sum_{S \in \mathcal{S}} \Delta_S f_S(t^*) = 0.$$

7. We now move to a boundary point with smaller support. Choose $S_{\text{pivot}} \in \mathcal{S}$ with $\Delta_{S_{\text{pivot}}} \neq 0$, and define

$$\sigma := \begin{cases} 1, & \Delta_{S_{\text{pivot}}} > 0, \\ -1, & \Delta_{S_{\text{pivot}}} < 0, \end{cases} \quad \Delta_S^{\text{mov}} := \sigma \Delta_S, \quad \mathcal{K}_+ := \{S \in \mathcal{S} : \Delta_S^{\text{mov}} > 0\}.$$

Then $\mathcal{K}_+$ is nonempty. Define

$$\lambda_{\text{bdry}} := \min_{S \in \mathcal{K}_+} \frac{\alpha_S^{\text{sp}}}{\Delta_S^{\text{mov}}} \geq 0, \quad \alpha_S^{\text{bdry}} := \alpha_S^{\text{sp}} - \lambda_{\text{bdry}} \Delta_S^{\text{mov}}.$$

For $S \in \mathcal{K}_+$, the defining minimum gives

$$\lambda_{\text{bdry}} \Delta_S^{\text{mov}} \leq \alpha_S^{\text{sp}}.$$

For $S \notin \mathcal{K}_+$,

$$\alpha_S^{\text{bdry}} \geq \alpha_S^{\text{sp}} \geq 0.$$

Hence $\alpha^{\text{bdry}} \geq 0$. Choose an index attaining the minimum, so that

$$S_{\text{hit}} \in \mathcal{K}_+, \quad \lambda_{\text{bdry}} = \frac{\alpha_{S_{\text{hit}}}^{\text{sp}}}{\Delta_{S_{\text{hit}}}^{\text{mov}}}.$$

The support condition on Δ implies $\alpha_{S_{\text{hit}}}^{\text{sp}} > 0$, whereas

$$\alpha_{S_{\text{hit}}}^{\text{bdry}} = 0.$$

Coordinates outside $\mathcal{K}_{\text{sp}}$ remain zero. Consequently,

$$\text{supp}(\alpha^{\text{bdry}}) \subsetneq \text{supp}(\alpha^{\text{sp}}).$$

The three vanishing moments established above give

$$\begin{aligned} \sum_{S \in \mathcal{S}} \alpha_S^{\text{bdry}} \mathbf{b}_S &= \mathbf{1}, \\ D_{\alpha^{\text{bdry}}}(t^*) &= D_{\alpha^{\text{sp}}}(t^*) = 0, \\ F_{\theta}(\alpha^{\text{bdry}}, t^*) &= F_{\theta}(\alpha^{\text{sp}}, t^*) = J^*(\theta, p, \varepsilon). \end{aligned}$$

Thus $\alpha^{\text{bdry}} \in \mathcal{W}$, contradicting the minimum-support choice. We conclude that

$$|\text{supp}(\alpha^{\text{sp}})| \leq 5.$$

8. Nonnegative weights and positive denominators imply

$$C_{\alpha^{\text{sp}}} = \sum_{S \in \mathcal{S}} \alpha_S^{\text{sp}} \frac{(g_{S,0} + g_{S,1})^2}{h_S(\theta)} \geq 0.$$

Since $D_{\alpha^{\text{sp}}}(t^*) = 0$, the quadratic identity yields

$$F_{\theta}(\alpha^{\text{sp}}, t) - F_{\theta}(\alpha^{\text{sp}}, t^*) = (t - t^*)^2 C_{\alpha^{\text{sp}}} \geq 0 \quad (t \in \mathbb{R}).$$

Therefore t^* is a global minimizer, including when the quadratic coefficient is zero, and

$$\inf_{t \in \mathbb{R}} F_{\theta}(\alpha^{\text{sp}}, t) = F_{\theta}(\alpha^{\text{sp}}, t^*) = J^*(\theta, p, \varepsilon).$$

9. Finally, set

$$\alpha := \alpha^{\text{sp}}, \quad \mathcal{S}_{\text{out}} := \{S \in \mathcal{S} : (P_{\theta}Q_{\alpha})(\{S\}) \neq 0\}.$$

For these feasible weights, the staircase construction gives

$$Q_{\alpha}(\{S\} \mid x_j) = \alpha_S b_{S,j}, \quad (P_{\theta}Q_{\alpha})(\{S\}) = \sum_{j=0}^3 \pi_{\theta,j} \alpha_S b_{S,j}.$$

In particular,

$$\alpha_S = 0 \implies Q_{\alpha}(\{S\} \mid x_j) = 0 \text{ for every } j \implies (P_{\theta}Q_{\alpha})(\{S\}) = 0.$$

Thus

$$\mathcal{S}_{\text{out}} \subseteq \text{supp}(\alpha), \quad \kappa(Q_{\alpha}) = |\mathcal{S}_{\text{out}}| \leq |\text{supp}(\alpha)| \leq 5.$$

Together with feasibility and the profiled equality from the preceding step, this proves the claim. □

Proof of Theorem 5. 1. *Certificate coordinates and profiling equations.* Let $\mathcal{K}_5$ be the active-pattern set and let $\mathcal{U}$ be the interior parameter domain:

$$\mathcal{K}_5 = \{S_3, S_4, S_6, S_8, S_9\}, \quad \mathcal{U} = (0, 1)^3 \times (0, \infty).$$

For $\xi = (p, \mu_0, \mu_1, \varepsilon) \in \mathcal{U}$, set

$$\theta(\xi) = (\mu_0, \mu_1)^\top, \quad q = 1 - p, \quad r_\varepsilon = e^\varepsilon, \quad d_\varepsilon = r_\varepsilon - 1.$$

Use the pattern masses, gradients, profiling directions, and projected scores from Lemma 1. Write their dependence on ξ explicitly as

$$h_S(\xi) = h_S(\theta(\xi)), \quad g_S(\xi) = \nabla_\theta h_S(\theta(\xi)), \quad f_S(\xi, t) = \frac{(g_S(\xi)^\top v(t))^2}{h_S(\xi)}, \quad \rho_S(\xi, t) = \frac{g_S(\xi)^\top v(t)}{h_S(\xi)}.$$

Here p, ε are fixed in the gradient. In particular,

$$v(t) = \begin{pmatrix} t \\ t+1 \end{pmatrix}, \quad e_{\text{prof}} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad F_\theta(\alpha, t) = \sum_{S \in \mathcal{S}} \alpha_S f_S(\xi, t).$$

Index the four records by x_0, x_1, x_2, x_3 . Their probabilities, in this order, are

$$(\pi_{\theta,0}, \pi_{\theta,1}, \pi_{\theta,2}, \pi_{\theta,3}) = (q(1 - \mu_0), q\mu_0, p(1 - \mu_1), p\mu_1).$$

The pattern enumeration is specified by

$$x_j \in S_m \iff \left\lfloor \frac{m}{2^j} \right\rfloor \equiv 1 \pmod{2}, \quad 0 \leq j \leq 3, \quad 1 \leq m \leq 14.$$

Define the ray vectors and ray matrix by

$$(b_S(\varepsilon))_j = \begin{cases} r_\varepsilon, & x_j \in S, \\ 1, & x_j \notin S, \end{cases} \quad B_\varepsilon = (b_{S_1}(\varepsilon), \dots, b_{S_{14}}(\varepsilon)).$$

All record probabilities and ray coordinates are positive, so

$$h_S(\xi) = \sum_{j=0}^3 \pi_{\theta,j} (b_S(\varepsilon))_j > 0.$$

The stationarity, dual, and score conditions described in Definition 13 enter the following full certificate definition. A point $\xi \in \mathbb{R}^4$ belongs to $\mathcal{R}_5$ if and only if $\xi \in \mathcal{U}$ and there exist $\alpha \in \mathbb{R}^{14}$, $t \in \mathbb{R}$, and $\eta \in \mathbb{R}^4$ satisfying

$$\begin{aligned} \alpha &\geq 0, & B_\varepsilon \alpha &= \mathbf{1}, & \alpha_S &> 0 \ (S \in \mathcal{K}_5), & \alpha_S &= 0 \ (S \notin \mathcal{K}_5), \\ & & \partial_t F_\theta(\alpha, t) &= 0, \\ & & b_S(\varepsilon)^\top \eta &= f_S(\xi, t) \quad (S \in \mathcal{K}_5), \\ & & f_S(\xi, t) &< b_S(\varepsilon)^\top \eta \quad (S \notin \mathcal{K}_5), \\ & & \rho_S(\xi, t) &\neq \rho_T(\xi, t) \quad (S, T \in \mathcal{K}_5, \ S \neq T). \end{aligned}$$

The derivative appearing here is

$$\partial_t F_\theta(\alpha, t) = 2 \sum_S \alpha_S \frac{(g_S(\xi)^\top v(t))(g_S(\xi)^\top e_{\text{prof}})}{h_S(\xi)}.$$

The five active columns of the ray matrix are

$$B_\varepsilon|_{\mathcal{K}_5} = \begin{pmatrix} r_\varepsilon & 1 & 1 & 1 & r_\varepsilon \\ r_\varepsilon & 1 & r_\varepsilon & 1 & 1 \\ 1 & r_\varepsilon & r_\varepsilon & 1 & 1 \\ 1 & 1 & 1 & r_\varepsilon & r_\varepsilon \end{pmatrix},$$

with columns ordered as S_3, S_4, S_6, S_8, S_9 . Subtracting consecutive normalization equations, and using $r_\varepsilon - 1 > 0$, gives

$$\alpha_{S_3} = \alpha_{S_4} = \alpha_{S_8}, \quad \alpha_{S_6} = \alpha_{S_9}, \quad (r_\varepsilon + 2)\alpha_{S_3} + (r_\varepsilon + 1)\alpha_{S_6} = 1.$$

Thus every feasible vector supported within $\mathcal{K}_5$ has the form

$$\alpha_S^u(\varepsilon) = \begin{cases} u/(r_\varepsilon + 1), & S \in \{S_6, S_9\}, \\ (1 - u)/(r_\varepsilon + 2), & S \in \{S_3, S_4, S_8\}, \\ 0, & S \notin \mathcal{K}_5, \end{cases} \quad 0 \leq u \leq 1.$$

Conversely, the displayed matrix verifies feasibility of this vector. Its five active weights are positive precisely when $0 < u < 1$.

Define the two normalized constituent objectives and their coefficients by

$$\begin{aligned} \Phi_{\text{bin}}(\xi, t) &= \frac{f_{S_6}(\xi, t) + f_{S_9}(\xi, t)}{r_\varepsilon + 1}, \\ \Phi_{\text{ter}}(\xi, t) &= \frac{f_{S_3}(\xi, t) + f_{S_4}(\xi, t) + f_{S_8}(\xi, t)}{r_\varepsilon + 2}, \\ a_{\text{bin}}(\xi) &= \frac{d_\varepsilon^2}{r_\varepsilon + 1} \left(\frac{1}{h_{S_6}(\xi)} + \frac{1}{h_{S_9}(\xi)} \right), \\ a_{\text{ter}}(\xi) &= \frac{d_\varepsilon^2 p^2}{r_\varepsilon + 2} \left(\frac{1}{h_{S_4}(\xi)} + \frac{1}{h_{S_8}(\xi)} \right). \end{aligned}$$

Both coefficients are positive throughout $\mathcal{U}$. The active gradients are

$$g_{S_3} = 0, \quad g_{S_4} = \begin{pmatrix} 0 \\ -d_\varepsilon p \end{pmatrix}, \quad g_{S_6} = \begin{pmatrix} d_\varepsilon q \\ -d_\varepsilon p \end{pmatrix}, \quad g_{S_8} = \begin{pmatrix} 0 \\ d_\varepsilon p \end{pmatrix}, \quad g_{S_9} = -g_{S_6}.$$

Consequently,

$$\begin{aligned} \Phi_{\text{bin}}(\xi, t) &= a_{\text{bin}}(\xi) \{p - (1 - 2p)t\}^2, \\ \Phi_{\text{ter}}(\xi, t) &= a_{\text{ter}}(\xi) (t + 1)^2, \\ F_\theta(\alpha^u, t) &= u\Phi_{\text{bin}}(\xi, t) + (1 - u)\Phi_{\text{ter}}(\xi, t). \end{aligned}$$

The binary rays sum to $(r_\varepsilon + 1)\mathbf{1}$, and the ternary rays sum to $(r_\varepsilon + 2)\mathbf{1}$. Averaging the active dual equalities over these two families therefore yields

$$\Phi_{\text{bin}}(\xi, t) = \sum_{j=0}^3 \eta_j = \Phi_{\text{ter}}(\xi, t).$$

Finally, the certificate stationarity equation becomes

$$u \partial_t \Phi_{\text{bin}}(\xi, t) + (1 - u) \partial_t \Phi_{\text{ter}}(\xi, t) = 0.$$

2. *Continuous continuation and openness.* Fix

$$\xi_{\circ} = (p_{\circ}, \mu_{0,\circ}, \mu_{1,\circ}, \varepsilon_{\circ}) \in \mathcal{R}_5$$

and a certificate $(\alpha_{\circ}, t_{\circ}, \eta_{\circ})$. Define, for $\xi \in \mathcal{U}$,

$$s_{\text{loc}}(\xi) = \sqrt{\frac{a_{\text{ter}}(\xi)}{a_{\text{bin}}(\xi)}}.$$

The constituent equality implies

$$\{p_{\circ} - (1 - 2p_{\circ})t_{\circ}\}^2 = \{s_{\text{loc}}(\xi_{\circ})(t_{\circ} + 1)\}^2.$$

Splitting equality of squares gives a sign $\sigma \in \{-1, 1\}$ such that

$$p_{\circ} - (1 - 2p_{\circ})t_{\circ} = \sigma s_{\text{loc}}(\xi_{\circ})(t_{\circ} + 1).$$

Its rearrangement is

$$\{1 - 2p_{\circ} + \sigma s_{\text{loc}}(\xi_{\circ})\}t_{\circ} = p_{\circ} - \sigma s_{\text{loc}}(\xi_{\circ}).$$

If the coefficient on the left were zero, the right side would also be zero, giving $p_{\circ} = 1$. Thus, on a neighborhood where its denominator is nonzero, define

$$t_{\sigma}(\xi) = \frac{p - \sigma s_{\text{loc}}(\xi)}{1 - 2p + \sigma s_{\text{loc}}(\xi)}.$$

This function is continuous at $\xi_{\circ}$, equals $t_{\circ}$ there, and satisfies

$$\Phi_{\text{bin}}(\xi, t_{\sigma}(\xi)) = \Phi_{\text{ter}}(\xi, t_{\sigma}(\xi)).$$

At the original certificate,

$$\partial_t \Phi_{\text{ter}}(\xi_{\circ}, t_{\circ}) = 2a_{\text{ter}}(\xi_{\circ})(t_{\circ} + 1) \neq 0.$$

Indeed, vanishing would give $t_{\circ} = -1$; the constituent equality would then give

$$a_{\text{bin}}(\xi_{\circ})(1 - p_{\circ})^2 = 0,$$

contrary to $p_{\circ} < 1$. If the two constituent derivatives were equal, stationarity would make this common derivative zero. Their difference is therefore nonzero, and stationarity uniquely determines the mixture coordinate:

$$u_{\circ} = (r_{\varepsilon_{\circ}} + 1)\alpha_{\circ, S_6} = -\frac{\partial_t \Phi_{\text{ter}}(\xi_{\circ}, t_{\circ})}{\partial_t \Phi_{\text{bin}}(\xi_{\circ}, t_{\circ}) - \partial_t \Phi_{\text{ter}}(\xi_{\circ}, t_{\circ})}.$$

On a neighborhood where the derivative difference remains nonzero, set

$$u_{\sigma}(\xi) = -\frac{\partial_t \Phi_{\text{ter}}(\xi, t_{\sigma}(\xi))}{\partial_t \Phi_{\text{bin}}(\xi, t_{\sigma}(\xi)) - \partial_t \Phi_{\text{ter}}(\xi, t_{\sigma}(\xi))}.$$

It is continuous at ξ_o , equals u_o there, and remains in $(0, 1)$ after shrinking the neighborhood. These arguments also apply when $p_o = 1/2$.

We next continue the dual certificate while preserving its value at the original point. Define

$$D_\varepsilon = \begin{pmatrix} r_\varepsilon & r_\varepsilon & 1 & 1 \\ 1 & 1 & r_\varepsilon & 1 \\ 1 & r_\varepsilon & r_\varepsilon & 1 \\ 1 & 1 & 1 & r_\varepsilon \end{pmatrix}, \quad \mathbf{f}_{\text{act}}(\xi) = \begin{pmatrix} f_{S_3}(\xi, t_\sigma(\xi)) \\ f_{S_4}(\xi, t_\sigma(\xi)) \\ f_{S_6}(\xi, t_\sigma(\xi)) \\ f_{S_8}(\xi, t_\sigma(\xi)) \end{pmatrix}.$$

For $\varepsilon > 0$, an explicit right inverse is

$$N_\varepsilon = \begin{pmatrix} \frac{r_\varepsilon + 1}{d_\varepsilon(r_\varepsilon + 2)} & \frac{r_\varepsilon + 1}{d_\varepsilon(r_\varepsilon + 2)} & -\frac{1}{d_\varepsilon} & -\frac{1}{d_\varepsilon(r_\varepsilon + 2)} \\ 0 & -\frac{1}{d_\varepsilon} & \frac{1}{d_\varepsilon} & 0 \\ -\frac{1}{d_\varepsilon(r_\varepsilon + 2)} & \frac{r_\varepsilon + 1}{d_\varepsilon(r_\varepsilon + 2)} & 0 & -\frac{1}{d_\varepsilon(r_\varepsilon + 2)} \\ -\frac{1}{d_\varepsilon(r_\varepsilon + 2)} & -\frac{1}{d_\varepsilon(r_\varepsilon + 2)} & 0 & \frac{r_\varepsilon + 1}{d_\varepsilon(r_\varepsilon + 2)} \end{pmatrix}.$$

Direct multiplication gives

$$D_\varepsilon N_\varepsilon = \text{Id}_4.$$

Set

$$N_o = N_{\varepsilon_o}.$$

At ξ_o , the matrix $D_{\varepsilon_o} N_o$ is the identity. Its determinant remains nonzero nearby. On that neighborhood define

$$\eta(\xi) = \eta_o + N_o(D_\varepsilon N_o)^{-1}(\mathbf{f}_{\text{act}}(\xi) - D_\varepsilon \eta_o).$$

Matrix inversion is continuous there. The original active equalities make the parenthesized residual zero at ξ_o , and multiplication by D_ε gives

$$\eta(\xi_o) = \eta_o, \quad D_\varepsilon \eta(\xi) = \mathbf{f}_{\text{act}}(\xi).$$

Thus the four active equalities indexed by S_3, S_4, S_6, S_8 continue.

The fifth equality follows from the constituent equality. Indeed,

$$\frac{b_{S_6}^\top \eta + b_{S_9}^\top \eta}{r_\varepsilon + 1} = \frac{b_{S_3}^\top \eta + b_{S_4}^\top \eta + b_{S_8}^\top \eta}{r_\varepsilon + 2},$$

because both sides equal $\mathbf{1}^\top \eta$. Subtracting this identity from $\Phi_{\text{bin}}(\xi, t_\sigma) = \Phi_{\text{ter}}(\xi, t_\sigma)$, using the four continued equalities, gives

$$b_{S_9}(\varepsilon)^\top \eta(\xi) = f_{S_9}(\xi, t_\sigma(\xi)).$$

All pattern masses stay positive nearby. Hence the finitely many information functions, projected scores, and dual-ray values are continuous at ξ_o . The strict inactive slacks and nonzero score differences therefore persist simultaneously:

$$\begin{aligned} f_S(\xi, t_\sigma(\xi)) &< b_S(\varepsilon)^\top \eta(\xi) \quad (S \notin \mathcal{K}_5), \\ \rho_S(\xi, t_\sigma(\xi)) &\neq \rho_T(\xi, t_\sigma(\xi)) \quad (S, T \in \mathcal{K}_5, S \neq T). \end{aligned}$$

The strict interior inequalities, the branch denominator, and the derivative difference likewise persist. For every point in the resulting neighborhood, choose

$$\alpha(\xi) = \alpha^{u_\sigma(\xi)}(\varepsilon).$$

The normalization calculation gives feasibility and five positive weights, while the defining formula for u_σ gives

$$\partial_t F_{\theta(\xi)}(\alpha(\xi), t_\sigma(\xi)) = 0.$$

Together with the continued dual equalities, strict slacks, and score separation, these are all the certificate conditions. Every point of $\mathcal{R}_5$ consequently has a neighborhood in $\mathbb{R}^4$ contained in $\mathcal{R}_5$, proving openness.

3. *The centered certificate and nonemptiness.* Fix $0 < p < 1/2$, and define

$$\begin{aligned} \theta_{\text{cen}} &= \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}, & \xi_{\text{cen}}(p) &= (p, 1/2, 1/2, \log 3), \\ s_{\text{cen}} &= \sqrt{\frac{8p^2}{5(1+p)}}, & D_{\text{cen}} &= 1 - 2p + s_{\text{cen}}, \\ t_{\text{cen}} &= \frac{p - s_{\text{cen}}}{D_{\text{cen}}}, & \lambda_{\text{cen}} &= \frac{s_{\text{cen}}}{D_{\text{cen}}}, \\ J_{\text{cen}} &= \frac{s_{\text{cen}}^2(1-p)^2}{D_{\text{cen}}^2}, & \eta_{\text{cen}} &= \frac{J_{\text{cen}}}{4}(-1, -1, 3, 3)^\top. \end{aligned}$$

The identity

$$5(1+p)s_{\text{cen}}^2 = 8p^2$$

and $0 < p < 1/2$ give

$$p < s_{\text{cen}} < 2p, \quad D_{\text{cen}} > 0, \quad 0 < \lambda_{\text{cen}} < 1, \quad t_{\text{cen}} < 0.$$

Set

$$\alpha_{\text{cen}} = \alpha^{\lambda_{\text{cen}}}(\log 3).$$

Thus

$$\alpha_{\text{cen}, S_6} = \alpha_{\text{cen}, S_9} = \frac{\lambda_{\text{cen}}}{4}, \quad \alpha_{\text{cen}, S_3} = \alpha_{\text{cen}, S_4} = \alpha_{\text{cen}, S_8} = \frac{1 - \lambda_{\text{cen}}}{5},$$

and all other weights vanish. Its row sums are

$$5 \frac{1 - \lambda_{\text{cen}}}{5} + 4 \frac{\lambda_{\text{cen}}}{4} = 1,$$

so it is feasible and has precisely the five stated positive weights.

At these parameters the constituent objectives reduce to

$$\Phi_{\text{bin}}(\xi_{\text{cen}}(p), t) = \{p - (1 - 2p)t\}^2, \quad \Phi_{\text{ter}}(\xi_{\text{cen}}(p), t) = s_{\text{cen}}^2(t + 1)^2.$$

Define the positive score magnitudes

$$\rho_{\text{bin}} = p - (1 - 2p)t_{\text{cen}} = \frac{s_{\text{cen}} q}{D_{\text{cen}}}, \quad \rho_{\text{ter}} = \frac{2p(t_{\text{cen}} + 1)}{1 + p}.$$

Direct rearrangement gives

$$t_{\text{cen}} + 1 = \frac{q}{D_{\text{cen}}}, \quad s_{\text{cen}}(t_{\text{cen}} + 1) = \rho_{\text{bin}}, \quad J_{\text{cen}} = \rho_{\text{bin}}^2 > 0.$$

Consequently,

$$\begin{aligned} \partial_t F_{\theta_{\text{cen}}}(\alpha_{\text{cen}}, t_{\text{cen}}) &= -2\lambda_{\text{cen}}(1 - 2p)\rho_{\text{bin}} + 2(1 - \lambda_{\text{cen}})s_{\text{cen}}\rho_{\text{bin}} \\ &= 0. \end{aligned}$$

For completeness, the information and dual calculations over all fourteen patterns are

patterns	$b_S(\log 3)^\top \eta_{\text{cen}} / J_{\text{cen}}$	$f_S(\xi_{\text{cen}}(p), t_{\text{cen}}) / J_{\text{cen}}$
S_1, S_2	1/2	$\frac{4q^2 t_{\text{cen}}^2}{(2-p)J_{\text{cen}}}$
S_3	0	0
S_4, S_8	5/2	5/2
S_5, S_{10}	2	$\frac{2(p + t_{\text{cen}})^2}{J_{\text{cen}}}$
S_6, S_9	2	$\frac{2}{J_{\text{cen}}}$
S_7, S_{11}	3/2	$\frac{5(1+p)}{2(3-p)}$
S_{12}	4	0
S_{13}, S_{14}	7/2	$\frac{4q^2 t_{\text{cen}}^2}{(2+p)J_{\text{cen}}}$

This table follows by inserting the displayed record probabilities and binary-mask enumeration into h_S , g_S , and $b_S^\top \eta_{\text{cen}}$. For example, for $S_4 = \{x_2\}$,

$$h_{S_4}(\xi_{\text{cen}}(p)) = 1 + p, \quad g_{S_4}(\xi_{\text{cen}}(p)) = (0, -2p)^\top,$$

so

$$f_{S_4}(\xi_{\text{cen}}(p), t_{\text{cen}}) = \frac{4p^2(t_{\text{cen}} + 1)^2}{1 + p} = \frac{5}{2}J_{\text{cen}},$$

and its ray gives the same dual value.

We verify strictness in the five inactive families. First,

$$5s_{\text{cen}}^2 < 8p^2 \implies s_{\text{cen}} < \frac{4p}{3}.$$

Together with $p < s_{\text{cen}}$, this yields

$$8(p - s_{\text{cen}})^2 < \frac{8}{9}p^2 < \frac{3}{2}p^2 < (2 - p)s_{\text{cen}}^2.$$

Multiplying by $q^2/D_{\text{cen}}^2 > 0$ and dividing by $2(2 - p)$ proves

$$\frac{4q^2 t_{\text{cen}}^2}{2 - p} < \frac{1}{2}J_{\text{cen}},$$

the strict slack for S_1, S_2 .

Next,

$$p + t_{\text{cen}} = \frac{q(2p - s_{\text{cen}})}{D_{\text{cen}}}, \quad 0 < p + t_{\text{cen}} < \frac{qs_{\text{cen}}}{D_{\text{cen}}} = \rho_{\text{bin}}.$$

Squaring proves

$$2(p + t_{\text{cen}})^2 < 2J_{\text{cen}},$$

which handles S_5, S_{10} .

For S_7, S_{11} , the same square identity gives

$$3s_{\text{cen}}^2(3 - p) - 8p^2 = \frac{8p^2(4 - 8p)}{5(1 + p)} > 0.$$

Using $t_{\text{cen}} + 1 = q/D_{\text{cen}}$, this is exactly

$$\frac{4p^2(t_{\text{cen}} + 1)^2}{3 - p} < \frac{3}{2}J_{\text{cen}}.$$

For S_{12} , strictness is $0 < 4J_{\text{cen}}$. Finally,

$$8(p - s_{\text{cen}})^2 < (2 - p)s_{\text{cen}}^2 < 7(2 + p)s_{\text{cen}}^2$$

gives

$$\frac{4q^2t_{\text{cen}}^2}{2 + p} < \frac{7}{2}J_{\text{cen}},$$

as required for S_{13}, S_{14} .

The active projected scores, in the order S_3, S_4, S_6, S_8, S_9 , are

$$(0, -\rho_{\text{ter}}, -\rho_{\text{bin}}, \rho_{\text{ter}}, \rho_{\text{bin}}).$$

Both magnitudes are positive, and

$$\left(\frac{\rho_{\text{ter}}}{\rho_{\text{bin}}}\right)^2 = \frac{5}{2(1 + p)} > 1.$$

Hence all five scores are distinct: the comparisons between positive magnitudes are strict, and the remaining comparisons follow from their signs. Reversing the order of a pair gives the reverse comparison.

We have verified feasibility, positive active weights, stationarity, all active dual equalities, all strict inactive slacks, and score separation. Therefore

$$\xi_{\text{cen}}(p) \in \mathcal{R}_5 \quad (0 < p < 1/2).$$

Taking $p = 1/4$ proves that $\mathcal{R}_5$ is nonempty.

4. *Centered optimality and uniqueness.* Stationarity and the nonnegative quadratic coefficients give

$$\begin{aligned} F_{\theta_{\text{cen}}}(\alpha_{\text{cen}}, t) - F_{\theta_{\text{cen}}}(\alpha_{\text{cen}}, t_{\text{cen}}) \\ = \{\lambda_{\text{cen}}(1 - 2p)^2 + (1 - \lambda_{\text{cen}})s_{\text{cen}}^2\}(t - t_{\text{cen}})^2 \geq 0. \end{aligned}$$

Its value at t_{cen} is J_{cen} . For every feasible β , nonnegativity of its weights and the dual bounds give

$$\begin{aligned} \inf_{t \in \mathbb{R}} F_{\theta_{\text{cen}}}(\beta, t) &\leq F_{\theta_{\text{cen}}}(\beta, t_{\text{cen}}) \\ &\leq \sum_S \beta_S b_S (\log 3)^\top \eta_{\text{cen}} \\ &= (B_{\log 3} \beta)^\top \eta_{\text{cen}} = \mathbf{1}^\top \eta_{\text{cen}} = J_{\text{cen}}. \end{aligned}$$

The infima here are bounded below by zero and are taken over nonempty sets. The feasible vector α_{cen} reaches the upper bound, so the optimization in [Definition 5](#) yields

$$J^*(\theta_{\text{cen}}, p, \log 3) = J_{\text{cen}} = \inf_t F_{\theta_{\text{cen}}}(\alpha_{\text{cen}}, t).$$

Suppose a feasible β has this optimized profile. Every inequality in the preceding bound is then an equality. In particular,

$$\sum_S \beta_S \{b_S (\log 3)^\top \eta_{\text{cen}} - f_S(\xi_{\text{cen}}(p), t_{\text{cen}})\} = 0.$$

Each summand is nonnegative, and every inactive slack is strictly positive. Thus

$$\beta_S = 0 \quad (S \notin \mathcal{K}_5).$$

Also,

$$F_{\theta_{\text{cen}}}(\beta, t_{\text{cen}}) = J_{\text{cen}} \leq F_{\theta_{\text{cen}}}(\beta, t) \quad (t \in \mathbb{R}),$$

so differentiation of this quadratic at its minimum gives

$$\partial_t F_{\theta_{\text{cen}}}(\beta, t_{\text{cen}}) = 0.$$

The row constraints already show that

$$\beta_{S_3} = \beta_{S_4} = \beta_{S_8} = \frac{1 - 4\beta_{S_6}}{5}, \quad \beta_{S_9} = \beta_{S_6}.$$

To solve the remaining stationarity equation, define its reduced coefficient

$$C_{\text{cen}} = 10s_{\text{cen}}p^2 + 5s_{\text{cen}}p - 5s_{\text{cen}} - 8p^2.$$

Substitute these row coordinates into the displayed derivative formula, use $t_{\text{cen}} = (p - s_{\text{cen}})/D_{\text{cen}}$, and clear the positive denominators. The resulting equation is

$$(p - 1)(C_{\text{cen}}\beta_{S_6} + 2p^2) = 0.$$

Since $p < 1/2$, it follows that

$$C_{\text{cen}}\beta_{S_6} = -2p^2.$$

The square identity for s_{cen} further gives

$$\begin{aligned} C_{\text{cen}}s_{\text{cen}} &= 5(2p - 1)(1 + p)s_{\text{cen}}^2 - 8p^2s_{\text{cen}} \\ &= -8p^2D_{\text{cen}} < 0. \end{aligned}$$

Thus $C_{\text{cen}} \neq 0$, and

$$C_{\text{cen}}(4D_{\text{cen}}\beta_{S_6}) = -8p^2D_{\text{cen}} = C_{\text{cen}}s_{\text{cen}}.$$

Cancellation proves

$$\beta_{S_6} = \frac{s_{\text{cen}}}{4D_{\text{cen}}} = \alpha_{\text{cen}, S_6}.$$

The row-coordinate identities determine the other four active weights, and both vectors vanish elsewhere. Hence

$$\beta = \alpha_{\text{cen}},$$

proving the centered uniqueness assertion.

5. *Attainment, refinement, and the centered cardinality bound.* We first give the directional inequality used for attaining channels. For every $\xi \in \mathcal{U}$, define the singleton design

$$\alpha_S^{\text{sgl}} = \begin{cases} 1/(r_\varepsilon + 3), & S \in \{S_1, S_2, S_4, S_8\}, \\ 0, & \text{otherwise.} \end{cases}$$

These four patterns are the four singleton subsets. Their rays sum to $(r_\varepsilon + 3)\mathbf{1}$, proving feasibility. Its objective is a quadratic with nonnegative terms, and in particular

$$F_\theta(\alpha^{\text{sgl}}, t) \geq \frac{d_\varepsilon^2 q^2}{(r_\varepsilon + 3)h_{S_1}(\xi)} t^2, \quad F_\theta(\alpha^{\text{sgl}}, t) \geq \frac{d_\varepsilon^2 p^2}{(r_\varepsilon + 3)h_{S_8}(\xi)} (t + 1)^2.$$

Both coefficients are positive. This quadratic has a minimizer, and a zero minimum would force both $t = 0$ and $t + 1 = 0$. Therefore

$$0 < \inf_t F_\theta(\alpha^{\text{sgl}}, t) \leq J^*(\theta, p, \varepsilon),$$

where the final comparison is supplied by the optimization in [Definition 5](#).

For a stationary private channel Q , write

$$P_Q = P_\theta Q, \quad \nu_Q = \sum_{j=0}^3 Q(x_j, \cdot),$$

and define its output score by

$$s_Q(z) = \begin{cases} \frac{\nabla_\theta \{dP_Q/d\nu_Q\}(z)}{\{dP_Q/d\nu_Q\}(z)}, & \{dP_Q/d\nu_Q\}(z) > 0, \\ 0, & \{dP_Q/d\nu_Q\}(z) = 0. \end{cases}$$

The score representation of the information matrix in [Lemma 1](#) gives, with $I_Q = I_\theta(Q)$,

$$I_Q = \int s_Q(z)s_Q(z)^\top dP_Q(z), \quad v_{\text{sol}}^\top I_Q v_{\text{sol}} = \int (v_{\text{sol}}^\top s_Q(z))^2 dP_Q(z) \geq 0 \quad (v_{\text{sol}} \in \mathbb{R}^2).$$

Thus I_Q is positive semidefinite.

Let $\mathcal{V}(I)$ denote the extended contrast variance:

$$\mathcal{V}(I) = \begin{cases} \max \left\{ \inf_{\{v_{\text{sol}} \in \mathbb{R}^2: Iv_{\text{sol}} = c\}} c^\top v_{\text{sol}}, 0 \right\}, & c \in \text{range}(I), \\ +\infty, & c \notin \text{range}(I), \end{cases} \quad c = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

If Q attains the variance in the statement, its variance is finite, so there is a vector $v_{\text{sol}} \in \mathbb{R}^2$ satisfying

$$I_Q v_{\text{sol}} = c.$$

Define

$$V_{\text{sol}} = c^\top v_{\text{sol}} = v_{\text{sol}}^\top I_Q v_{\text{sol}} > 0.$$

To justify strict positivity, zero quadratic value for a positive semidefinite matrix implies $I_Q v_{\text{sol}} = 0$, contradicting $c \neq 0$. Any other solution has the same contrast pairing, since

$$I_Q v'_{\text{sol}} = c \implies c^\top v'_{\text{sol}} = v_{\text{sol}}^\top I_Q v'_{\text{sol}} = v_{\text{sol}}^\top c = V_{\text{sol}}.$$

Consequently attainment and positivity of J^* imply

$$V_{\text{sol}} = \mathcal{V}(I_Q) = V^*(\theta, p, \varepsilon) = \frac{1}{J^*(\theta, p, \varepsilon)}.$$

Here both values are positive, so the nonnegative-part convention preserves their real values.

For arbitrary $t \in \mathbb{R}$, define the residual vector

$$v_{\text{res}} = V_{\text{sol}} v(t) - v_{\text{sol}}.$$

Since $c^\top v(t) = 1$, positive semidefiniteness gives

$$\begin{aligned} 0 &\leq v_{\text{res}}^\top I_Q v_{\text{res}} \\ &= V_{\text{sol}}^2 v(t)^\top I_Q v(t) - V_{\text{sol}}. \end{aligned}$$

Dividing by $V_{\text{sol}} > 0$ and substituting its value proves

$$J^*(\theta, p, \varepsilon) \leq v(t)^\top I_Q v(t) \quad (t \in \mathbb{R}).$$

Return to the centered parameters and an attaining Q . Use the constant privacy sequence $\varepsilon_n = \log 3$; it satisfies [Assumption 8](#). The centered parameters also satisfy [Assumptions 6](#) and [7](#). By [Lemma 1](#), there are feasible weights β and a probability kernel K such that

$$Q = K \circ Q_\beta, \quad I_{\theta_{\text{cen}}}(\beta) - I_{\theta_{\text{cen}}}(Q) \succeq 0.$$

For every t , the directional lower bound and this matrix order imply

$$J_{\text{cen}} \leq v(t)^\top I_{\theta_{\text{cen}}}(Q) v(t) \leq v(t)^\top I_{\theta_{\text{cen}}}(\beta) v(t) = F_{\theta_{\text{cen}}}(\beta, t).$$

The centered dual bound supplies the reverse inequality at t_{cen} . Hence

$$F_{\theta_{\text{cen}}}(\beta, t_{\text{cen}}) = J_{\text{cen}}.$$

Equality in the strict dual bound forces β to vanish outside $\mathcal{K}_5$. The preceding inequality for every t makes t_{cen} a minimizer for β , so its derivative there is zero. The supported stationary calculation above therefore gives

$$\beta = \alpha_{\text{cen}}, \quad \beta_S > 0 \quad (S \in \mathcal{K}_5).$$

Moreover, the directional inequalities at t_{cen} give

$$v(t_{\text{cen}})^\top I_{\theta_{\text{cen}}}(\beta)v(t_{\text{cen}}) = v(t_{\text{cen}})^\top I_{\theta_{\text{cen}}}(Q)v(t_{\text{cen}}) = J_{\text{cen}}.$$

The score-equality conclusion of [Lemma 1](#) now applies. If two distinct active conditional output measures were not mutually singular, their projected scores would agree. The centered score calculation proves that they differ. Thus, writing $K_S = K(S, \cdot)$,

$$K_S \perp K_T \quad (S, T \in \mathcal{K}_5, S \neq T).$$

We spell out how these five singular probability measures give the cardinality bound. Evaluations of output-space measures on arbitrary subsets use their outer-measure extensions. For any probability measure P_{out} on Z , define

$$P_{\text{out}}^{\text{out}}(A) = \inf\{P_{\text{out}}(E) : A \subseteq E \subseteq Z, E \text{ measurable}\}, \quad A \subseteq Z.$$

We also write $P_{\text{out}}(A)$ for this extension; it agrees with the original measure on measurable sets. In particular, singleton evaluations of K_S and P_Q below use this convention, as does the output cardinality:

$$\kappa(Q) = \begin{cases} \#\{z \in Z : P_Q^{\text{out}}(\{z\}) \neq 0\}, & Z \text{ finite,} \\ +\infty, & Z \text{ infinite.} \end{cases}$$

These extensions are monotone and finitely subadditive. Absolute continuity transfers to their evaluations on arbitrary sets: a set of outer measure zero is contained in a measurable null set, obtained by intersecting measurable supersets whose masses tend to zero; the dominated measure also vanishes on that measurable null set and hence on the original set.

For every measurable output event A , factorization and the positive record probability $\pi_{\theta,0} > 0$ give

$$\beta_S(b_S(\varepsilon))_0 K_S(A) \leq Q(x_0, A), \quad \pi_{\theta,0} Q(x_0, A) \leq P_Q(A).$$

The coefficients on the left are positive for active S . Therefore

$$K_S \ll Q(x_0, \cdot) \ll P_Q \quad (S \in \mathcal{K}_5).$$

Suppose first that the output space Z is finite. For each $S \in \mathcal{K}_5$, choose a point $z_S \in Z$ such that

$$K_S(\{z_S\}) > 0.$$

Such a point exists because otherwise finite subadditivity would give

$$1 = K_S(Z) = K_S\left(\bigcup_{z \in Z} \{z\}\right) \leq \sum_{z \in Z} K_S(\{z\}) = 0.$$

This contradiction also covers an empty output space, with the sum interpreted as zero. Absolute continuity then gives

$$P_Q(\{z_S\}) > 0 \quad (S \in \mathcal{K}_5),$$

since a zero output-law mass would force $K_S(\{z_S\}) = 0$.

To prove that the chosen points are distinct, fix distinct $S, T \in \mathcal{K}_5$. Mutual singularity supplies a measurable set $E_{S,T} \subseteq Z$ with

$$K_S(E_{S,T}) = 0, \quad K_T(Z \setminus E_{S,T}) = 0.$$

Suppose $z_S = z_T$. If $z_S \in E_{S,T}$, measure monotonicity gives

$$0 < K_S(\{z_S\}) \leq K_S(E_{S,T}) = 0.$$

If $z_S \notin E_{S,T}$, then $z_T \in Z \setminus E_{S,T}$, and

$$0 < K_T(\{z_T\}) \leq K_T(Z \setminus E_{S,T}) = 0.$$

Both cases are contradictory. Thus the five selected points are distinct and have positive output-law mass, proving

$$5 = |\mathcal{K}_5| = \#\{z_S : S \in \mathcal{K}_5\} \leq \#\{z \in Z : \mathbb{P}_Q(\{z\}) \neq 0\} = \kappa(Q).$$

For an infinite output space, the stated convention gives $\kappa(Q) = +\infty$, and the same bound follows.

6. *Exchange of the treatment labels.* Suppose $1/2 < p < 1$ and put $q = 1 - p$. Then

$$0 < q < 1/2, \quad (q, 1/2, 1/2, \log 3) \in \mathcal{R}_5.$$

Define the input exchange, its action on patterns and weights, and the coordinate-exchange matrix by

$$\begin{aligned} \mathfrak{s}(x_0) &= x_2, & \mathfrak{s}(x_1) &= x_3, & \mathfrak{s}(x_2) &= x_0, & \mathfrak{s}(x_3) &= x_1, \\ \mathfrak{s}S &= \{\mathfrak{s}x : x \in S\}, & \alpha_S^{\text{sw}} &= \alpha_{\mathfrak{s}S}, & \Pi &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \end{aligned}$$

Both exchanges are involutions. The ray identity

$$(b_{\mathfrak{s}S}(\varepsilon))_j = b_S(\varepsilon)_{\mathfrak{s}(j)}$$

shows, by reindexing the normalization sums, that

$$\alpha \in \mathcal{A}_\varepsilon \iff \alpha^{\text{sw}} \in \mathcal{A}_\varepsilon.$$

On the centered slice, exchanging the record probabilities gives

$$h_{\mathfrak{s}S}(\xi_{\text{cen}}(p)) = h_S(\xi_{\text{cen}}(q)), \quad g_{\mathfrak{s}S}(\xi_{\text{cen}}(p)) = \Pi g_S(\xi_{\text{cen}}(q)).$$

Since

$$\Pi v(t) = -v(-1 - t),$$

the information contributions satisfy

$$f_{\mathfrak{s}S}(\xi_{\text{cen}}(p), t) = f_S(\xi_{\text{cen}}(q), -1 - t).$$

Reindexing the sum and using the bijection $t \mapsto -1 - t$ consequently gives

$$\inf_t \sum_S \alpha_S^{\text{sw}} f_S(\xi_{\text{cen}}(p), t) = \inf_t \sum_S \alpha_S f_S(\xi_{\text{cen}}(q), t).$$

The exchange is also a bijection of feasible weights. Taking the optimized profiles on both sides proves

$$J^*(\theta_{\text{cen}}, p, \log 3) = J^*(\theta_{\text{cen}}, q, \log 3), \quad V^*(\theta_{\text{cen}}, p, \log 3) = V^*(\theta_{\text{cen}}, q, \log 3).$$

Choose the centered optimizer at q and exchange its weights. This produces a feasible optimizer at p with support

$$\mathfrak{s}\mathcal{K}_5 = \{\mathfrak{s}S : S \in \mathcal{K}_5\}.$$

If another feasible vector β is optimal at p , then β^{sw} is optimal at q . Centered uniqueness gives

$$\beta^{\text{sw}} = \alpha_{\text{cen}}(q).$$

Applying the involution again proves

$$\beta = \alpha_{\text{cen}}(q)^{\text{sw}},$$

the claimed uniqueness at the original assignment probability.

For the channel assertion, define

$$Q^{\text{sw}}(x, A) = Q(\mathfrak{s}x, A).$$

This is a probability kernel and satisfies the same privacy inequalities as Q , since those inequalities hold for every input pair. In the next identities, quantities for Q^{sw} are evaluated at assignment probability q , and those for Q at p . Reindexing the four record probabilities gives equality of the output laws; reindexing their derivatives exchanges the two parameter coordinates. Therefore

$$P_{\theta_{\text{cen}}} Q^{\text{sw}} = P_{\theta_{\text{cen}}} Q, \quad I_{\theta_{\text{cen}}}(Q^{\text{sw}}) = \Pi I_{\theta_{\text{cen}}}(Q) \Pi.$$

The contrast variance is preserved by this matrix exchange. Indeed, $\Pi c = -c$, and for every $v_{\text{sol}} \in \mathbb{R}^2$,

$$I v_{\text{sol}} = c \quad \Longleftrightarrow \quad (\Pi / \Pi)(-\Pi v_{\text{sol}}) = c, \quad c^\top (-\Pi v_{\text{sol}}) = c^\top v_{\text{sol}}.$$

Thus the solution-value sets coincide; if they are empty, both contrast variances equal $+\infty$. In either case,

$$\mathcal{V}(\Pi / \Pi) = \mathcal{V}(I).$$

An attaining Q at p consequently gives an attaining Q^{sw} at q . The centered bound yields $\kappa(Q^{\text{sw}}) \geq 5$. Equality of their output laws, on the same output space, gives

$$\kappa(Q^{\text{sw}}) = \kappa(Q),$$

and hence the desired bound at p .

7. *Uniqueness for a general five-pattern certificate.* Fix $\xi = (p, \mu_0, \mu_1, \varepsilon) \in \mathcal{R}_5$, put $\theta = \theta(\xi)$, and choose a certificate (α_0, t_0, η) . For any feasible weight vector, its objective is nonnegative. For the certificate vector, expansion of the quadratic gives

$$\begin{aligned} F_\theta(\alpha_0, t) - F_\theta(\alpha_0, t_0) &= (t - t_0) \partial_t F_\theta(\alpha_0, t_0) \\ &\quad + (t - t_0)^2 \sum_S \alpha_{0,S} \frac{(g_S(\xi)^\top e_{\text{prof}})^2}{h_S(\xi)} \\ &\geq 0, \end{aligned}$$

where stationarity makes the linear term zero. The active equalities, the vanishing inactive weights, and normalization give

$$F_\theta(\alpha_0, t_0) = \sum_S \alpha_{0,S} b_S(\varepsilon)^\top \eta = \sum_{j=0}^3 \eta_j.$$

For every feasible β ,

$$\inf_t F_\theta(\beta, t) \leq F_\theta(\beta, t_0) \leq \sum_S \beta_S b_S(\varepsilon)^\top \eta = \sum_{j=0}^3 \eta_j.$$

The certificate vector attains this upper bound. Hence

$$J^*(\theta, p, \varepsilon) = \inf_t F_\theta(\alpha_0, t) = F_\theta(\alpha_0, t_0) = \sum_{j=0}^3 \eta_j.$$

Moreover, equality of another feasible profile with J^* forces equality in the dual bound and therefore

$$\beta_S = 0 \quad (S \notin \mathcal{K}_5),$$

by the same nonnegative slack-sum argument.

The constituent coefficients are positive, and the active dual equalities give

$$a_{\text{bin}}(\xi) \{p - (1 - 2p)t_0\}^2 = a_{\text{ter}}(\xi)(t_0 + 1)^2.$$

As already derived, $t_0 = -1$ would force $p = 1$, so

$$\partial_t \Phi_{\text{ter}}(\xi, t_0) \neq 0.$$

Define the certificate mixture coordinate by

$$u_0 = (r_\varepsilon + 1) \alpha_{0, S_6}.$$

The normalization and stationarity calculations give

$$u_0 = -\frac{\partial_t \Phi_{\text{ter}}(\xi, t_0)}{\partial_t \Phi_{\text{bin}}(\xi, t_0) - \partial_t \Phi_{\text{ter}}(\xi, t_0)}, \quad \partial_t \Phi_{\text{bin}}(\xi, t_0) - \partial_t \Phi_{\text{ter}}(\xi, t_0) \neq 0.$$

Now let β be any feasible optimized vector. Its nonnegative objective has profile J^* , and the dual upper bound at t_0 gives

$$J^* \leq F_\theta(\beta, t_0) \leq J^*.$$

It follows that

$$F_\theta(\beta, t_0) = J^* \leq F_\theta(\beta, t) \quad (t \in \mathbb{R}), \quad \partial_t F_\theta(\beta, t_0) = 0.$$

Since β vanishes outside $\mathcal{K}_5$, define

$$u_\beta = (r_\varepsilon + 1)\beta_{S_6}.$$

Its stationarity equation yields

$$u_\beta = -\frac{\partial_t \Phi_{\text{ter}}(\xi, t_0)}{\partial_t \Phi_{\text{bin}}(\xi, t_0) - \partial_t \Phi_{\text{ter}}(\xi, t_0)} = u_0.$$

The supported normalization formula determines every weight from this coordinate. Hence $\beta = \alpha_0$. In particular, the unique optimized vector has positive weight on all five active patterns.

8. *The cardinality bound throughout the region.* Keep the general certificate from the preceding argument and let Q be any stationary ε -LDP channel attaining the specified extended contrast variance. The certificate includes the interior conditions and $\varepsilon > 0$. With the constant sequence $\varepsilon_n = \varepsilon$, all hypotheses of [Lemma 1](#) hold. It supplies feasible weights β and a probability kernel K with

$$Q = K \circ Q_\beta, \quad I_\theta(\beta) - I_\theta(Q) \succeq 0.$$

The directional attainment bound already derived applies to these parameters. Consequently, for every $t \in \mathbb{R}$,

$$J^*(\theta, p, \varepsilon) \leq v(t)^\top I_\theta(Q)v(t) \leq v(t)^\top I_\theta(\beta)v(t) = F_\theta(\beta, t).$$

At the certificate direction, the dual bound is

$$F_\theta(\beta, t_0) \leq \sum_{j=0}^3 \eta_j = J^*(\theta, p, \varepsilon).$$

Thus

$$F_\theta(\beta, t_0) = J^*(\theta, p, \varepsilon) \leq F_\theta(\beta, t) \quad (t \in \mathbb{R}),$$

and

$$\inf_t F_\theta(\beta, t) = J^*(\theta, p, \varepsilon).$$

The general certificate uniqueness just proved gives

$$\beta = \alpha_0, \quad \beta_S > 0 \quad (S \in \mathcal{K}_5).$$

The same upper and lower bounds at t_0 also give directional equality:

$$v(t_0)^\top I_\theta(\beta)v(t_0) = v(t_0)^\top I_\theta(Q)v(t_0) = J^*(\theta, p, \varepsilon).$$

By the score-equality conclusion of [Lemma 1](#), overlapping active conditional measures would have equal projected scores. The region's certificate separates every pair, so

$$K(S, \cdot) \perp K(T, \cdot) \quad (S, T \in \mathcal{K}_5, S \neq T).$$

Factorization, positive active weights, positive ray coordinates, and $\pi_{\theta,0} = q(1 - \mu_0) > 0$ give

$$K(S, \cdot) \ll Q(x_0, \cdot) \ll P_\theta Q \quad (S \in \mathcal{K}_5).$$

Suppose first that the output space Z is finite. With the outer-measure convention above, the conditional-atom argument chooses, for every $S \in \mathcal{K}_5$, a point $z_S \in Z$ satisfying

$$K(S, \{z_S\}) > 0.$$

Indeed, if all singleton masses of this probability measure were zero, finite subadditivity would give

$$1 = K(S, Z) \leq \sum_{z \in Z} K(S, \{z\}) = 0,$$

including the empty-space case. Absolute continuity transfers positivity:

$$(P_\theta Q)(\{z_S\}) > 0 \quad (S \in \mathcal{K}_5).$$

For distinct $S, T \in \mathcal{K}_5$, choose a measurable separating set $E_{S,T} \subseteq Z$ such that

$$K(S, E_{S,T}) = 0, \quad K(T, Z \setminus E_{S,T}) = 0.$$

If $z_S = z_T$, membership in this set gives the two cases

$$\begin{aligned} z_S \in E_{S,T} &\implies 0 < K(S, \{z_S\}) \leq K(S, E_{S,T}) = 0, \\ z_S \notin E_{S,T} &\implies 0 < K(T, \{z_T\}) \leq K(T, Z \setminus E_{S,T}) = 0. \end{aligned}$$

Hence the selected points are distinct, and

$$5 = |\mathcal{K}_5| = \#\{z_S : S \in \mathcal{K}_5\} \leq \#\{z \in Z : (P_\theta Q)(\{z\}) \neq 0\} = \kappa(Q).$$

For an infinite output space the cardinality convention gives $\kappa(Q) = +\infty$. Thus in both cases

$$\kappa(Q) \geq |\mathcal{K}_5| = 5.$$

This proves the universal assertion throughout $\mathcal{R}_5$, and completes all the claims. $\square$

Proof of Theorem 6. 1. *Certificate notation.* Write the parameter domain and the active pattern set as

$$\mathcal{D} := (0, 1)^3 \times (0, \infty), \quad \mathcal{K} := \{S_1, S_6, S_8\}.$$

For a parameter vector, use the coordinate notation

$$\xi = (p(\xi), \mu_0(\xi), \mu_1(\xi), \varepsilon(\xi)), \quad \theta(\xi) := (\mu_0(\xi), \mu_1(\xi))^\top.$$

The fixed pattern enumeration is

$$\mathcal{S} := \{S_k : 1 \leq k \leq 14\}, \quad S_k := \{j \in \{0, 1, 2, 3\} : \text{the } j\text{-th binary digit of } k \text{ is } 1\}.$$

In particular,

$$S_1 = \{0\}, \quad S_6 = \{1, 2\}, \quad S_8 = \{3\}.$$

For $S \in \mathcal{S}$, define its staircase ray by

$$b_{S,j}(\varepsilon) := \begin{cases} e^\varepsilon, & j \in S, \\ 1, & j \notin S, \end{cases} \quad j \in \{0, 1, 2, 3\}, \quad b_S(\varepsilon) := (b_{S,0}(\varepsilon), \dots, b_{S,3}(\varepsilon))^\top.$$

The record probabilities, pattern gradients, and pattern masses are

$$\pi_\theta(p) := \begin{pmatrix} (1-p)(1-\mu_0) \\ (1-p)\mu_0 \\ p(1-\mu_1) \\ p\mu_1 \end{pmatrix}, \quad g_S(p, \varepsilon) := \begin{pmatrix} (1-p)(b_{S,1}(\varepsilon) - b_{S,0}(\varepsilon)) \\ p(b_{S,3}(\varepsilon) - b_{S,2}(\varepsilon)) \end{pmatrix},$$

$$h_S(\theta; p, \varepsilon) := \sum_{j=0}^3 \pi_{\theta,j}(p) b_{S,j}(\varepsilon), \quad v(t) := (t, t+1)^\top, \quad t \in \mathbb{R}.$$

The additional arguments on h_S display its dependence on assignment and privacy. For $\xi \in \mathcal{D}$, define the pattern information and its profiling slope by

$$f_S(\xi, t) := \frac{(g_S(p(\xi), \varepsilon(\xi))^\top v(t))^2}{h_S(\theta(\xi); p(\xi), \varepsilon(\xi))},$$

$$\sigma_S(\xi, t) := \frac{2(g_S(p(\xi), \varepsilon(\xi))^\top v(t))(g_{S,0}(p(\xi), \varepsilon(\xi)) + g_{S,1}(p(\xi), \varepsilon(\xi)))}{h_S(\theta(\xi); p(\xi), \varepsilon(\xi))}.$$

The masses are positive on this domain, as shown below, so these quotients are well defined. The feasible set and objective are

$$\mathcal{A}_\varepsilon := \left\{ \beta \in [0, \infty)^{14} : \sum_{S \in \mathcal{S}} \beta_S b_{S,j}(\varepsilon) = 1 \text{ for every } j \in \{0, 1, 2, 3\} \right\},$$

$$F_{\theta(\xi)}(\beta, t) = \sum_{S \in \mathcal{S}} \beta_S f_S(\xi, t),$$

where the objective in this display is evaluated at $p(\xi)$ and $\varepsilon(\xi)$. These are the quantities entering [Definition 5](#). The three-pattern certificate of [Definition 13](#) has support $\mathcal{K}$, positive active weights, zero weighted profiling slope, active dual equalities, and strict inactive dual inequalities.

2. *Canonical weights for an arbitrary certificate.* Fix $\xi_0 \in \mathcal{R}_3$, and choose witnessing quantities

$$\beta^{(0)} \in \mathcal{A}_{\varepsilon(\xi_0)}, \quad t_{\text{wit}} \in \mathbb{R}, \quad \eta^{(0)} \in \mathbb{R}^4.$$

Their support is $\mathcal{K}$, so

$$\beta_S^{(0)} = 0 \quad (S \notin \mathcal{K}).$$

Put

$$r_0 := e^{\varepsilon(\xi_0)} > 1.$$

Normalization at input coordinates 0, 1, 3 gives

$$\begin{aligned} r_0 \beta_{S_1}^{(0)} + \beta_{S_6}^{(0)} + \beta_{S_8}^{(0)} &= 1, \\ \beta_{S_1}^{(0)} + r_0 \beta_{S_6}^{(0)} + \beta_{S_8}^{(0)} &= 1, \\ \beta_{S_1}^{(0)} + \beta_{S_6}^{(0)} + r_0 \beta_{S_8}^{(0)} &= 1. \end{aligned}$$

Subtracting consecutive equations yields

$$(r_0 - 1)(\beta_{S_1}^{(0)} - \beta_{S_6}^{(0)}) = 0, \quad (r_0 - 1)(\beta_{S_6}^{(0)} - \beta_{S_8}^{(0)}) = 0.$$

Consequently,

$$\beta^{(0)} = \alpha^{(3)}(\varepsilon(\xi_0)), \quad \alpha_S^{(3)}(\varepsilon) := \begin{cases} (e^\varepsilon + 2)^{-1}, & S \in \mathcal{K}, \\ 0, & S \notin \mathcal{K}, \end{cases} \quad \varepsilon > 0.$$

For every $\varepsilon > 0$, the active weights are positive. Since the three active patterns partition the inputs,

$$\sum_{S \in \mathcal{K}} b_{S,j}(\varepsilon) = e^\varepsilon + 2 \quad (j = 0, 1, 2, 3).$$

Thus

$$\alpha^{(3)}(\varepsilon) \in \mathcal{A}_\varepsilon, \quad \{S : \alpha_S^{(3)}(\varepsilon) \neq 0\} = \mathcal{K}.$$

3. *A continuous stationary direction.* For every $\xi \in \mathcal{D}$, all record probabilities are nonnegative and all staircase-ray entries are positive. In particular,

$$h_S(\theta(\xi); p(\xi), \varepsilon(\xi)) \geq (1 - p(\xi))(1 - \mu_0(\xi))b_{S,0}(\varepsilon(\xi)) > 0 \quad (S \in \mathcal{S}).$$

Define the canonical stationarity function and its linear coefficient by

$$L_3(\xi, t) := \sum_{S \in \mathcal{S}} \alpha_S^{(3)}(\varepsilon(\xi)) \sigma_S(\xi, t), \quad a_3(\xi) := L_3(\xi, 1) - L_3(\xi, 0).$$

The identity

$$g_S^\top v(t) = g_{S,1} + t(g_{S,0} + g_{S,1})$$

gives, with all quantities evaluated at ξ ,

$$\sigma_S(\xi, t) = \sigma_S(\xi, 0) + t \frac{2(g_{S,0} + g_{S,1})^2}{h_S(\theta(\xi))}.$$

Hence

$$L_3(\xi, t) = L_3(\xi, 0) + t a_3(\xi),$$

$$a_3(\xi) = \sum_{S \in \mathcal{S}} \alpha_S^{(3)}(\varepsilon(\xi)) \frac{2(g_{S,0} + g_{S,1})^2}{h_S(\theta(\xi))} > 0.$$

Every summand is nonnegative, and the S_1 summand is strictly positive because

$$g_{S_1,0} + g_{S_1,1} = -(e^{\varepsilon(\xi)} - 1)(1 - p(\xi)) \neq 0.$$

We may therefore define, throughout $\mathcal{D}$,

$$t_3(\xi) := -\frac{L_3(\xi, 0)}{a_3(\xi)}, \quad L_3(\xi, t_3(\xi)) = 0.$$

The witnessing stationarity equation and the canonical-weight identity give

$$0 = L_3(\xi_0, t_{\text{wit}}) = L_3(\xi_0, 0) + t_{\text{wit}} a_3(\xi_0), \quad t_{\text{wit}} = t_3(\xi_0).$$

For each fixed pattern and input coordinate, the ray entry is either e^ε or 1, and is continuous. The masses are finite sums of products of continuous functions; the gradients are continuous as well. Their positive denominators show that $L_3(\xi, 0)$ and $L_3(\xi, 1)$ are continuous at ξ_0 . Since $a_3(\xi_0) > 0$, the quotient defining t_3 is continuous there. It follows that

$$\xi \mapsto f_S(\xi, t_3(\xi)) \text{ is continuous at } \xi_0 \quad (S \in \mathcal{S}).$$

4. *A local dual solution anchored at the witness.* Define the active dual matrix and target vector on $\mathcal{D}$ by

$$r(\xi) := e^{\varepsilon(\xi)}, \quad A_3(\xi) := \begin{pmatrix} r(\xi) & 1 & 1 & 1 \\ 1 & r(\xi) & r(\xi) & 1 \\ 1 & 1 & 1 & r(\xi) \end{pmatrix}, \quad \mathbf{f}_3(\xi) := \begin{pmatrix} f_{S_1}(\xi, t_3(\xi)) \\ f_{S_6}(\xi, t_3(\xi)) \\ f_{S_8}(\xi, t_3(\xi)) \end{pmatrix}.$$

The active equalities at ξ_0 , together with $t_{\text{wit}} = t_3(\xi_0)$, say

$$A_3(\xi_0)\eta^{(0)} = \mathbf{f}_3(\xi_0).$$

Use the fixed right inverse

$$R_0 := \frac{1}{(r_0 - 1)(r_0 + 2)} \begin{pmatrix} r_0 + 1 & -1 & -1 \\ -1 & r_0 + 1 & -1 \\ 0 & 0 & 0 \\ -1 & -1 & r_0 + 1 \end{pmatrix}.$$

Its denominator is nonzero. Multiplication gives

$$A_3(\xi_0)R_0 = \frac{1}{(r_0 - 1)(r_0 + 2)} \begin{pmatrix} r_0(r_0 + 1) - 2 & 0 & 0 \\ 0 & r_0(r_0 + 1) - 2 & 0 \\ 0 & 0 & r_0(r_0 + 1) - 2 \end{pmatrix} = \text{diag}(1, 1, 1).$$

The matrix A_3 and the target $\mathbf{f}_3$ are continuous at ξ_0 . Since

$$\det(A_3(\xi_0)R_0) = 1,$$

continuity of the determinant gives a neighborhood of ξ_0 within

$$\mathcal{U}_{\text{inv}} := \{\xi \in \mathcal{D} : \det(A_3(\xi)R_0) \neq 0\}.$$

On this set define

$$\eta_{\text{loc}}(\xi) := \eta^{(0)} + R_0(A_3(\xi)R_0)^{-1}(\mathbf{f}_3(\xi) - A_3(\xi)\eta^{(0)}).$$

The inverse in this formula is taken on the stated invertible domain. Continuity of matrix inversion and the displayed formula imply continuity of η_{loc} at ξ_0 . The residual vanishes at the base point, so

$$\eta_{\text{loc}}(\xi_0) = \eta^{(0)}.$$

Moreover, for every $\xi \in \mathcal{U}_{\text{inv}}$,

$$\begin{aligned} A_3(\xi)\eta_{\text{loc}}(\xi) &= A_3(\xi)\eta^{(0)} + A_3(\xi)R_0(A_3(\xi)R_0)^{-1}(\mathbf{f}_3(\xi) - A_3(\xi)\eta^{(0)}) \\ &= \mathbf{f}_3(\xi). \end{aligned}$$

Thus the three active dual equalities hold locally, and the original inactive strict inequalities hold at the base point:

$$f_S(\xi_0, t_3(\xi_0)) < b_S(\varepsilon(\xi_0))^\top \eta_{\text{loc}}(\xi_0) \quad (S \notin \mathcal{K}).$$

5. *Persistence of the certificate.* For every $S \in \mathcal{S}$, the dual value

$$b_S(\varepsilon(\xi))^\top \eta_{\text{loc}}(\xi) = \sum_{j=0}^3 b_{S,j}(\varepsilon(\xi)) \eta_{\text{loc},j}(\xi)$$

is continuous at ξ_0 , being a finite sum of products of continuous functions. Consequently, each inactive strict inequality persists in a neighborhood of ξ_0 . Intersect these finitely many neighborhoods with the neighborhood on which the active equations hold. The strict coordinate inequalities

$$0 < p(\xi) < 1, \quad 0 < \mu_0(\xi) < 1, \quad 0 < \mu_1(\xi) < 1, \quad 0 < \varepsilon(\xi)$$

also persist locally. We obtain a neighborhood $\mathcal{N}$ of ξ_0 such that, for every $\xi \in \mathcal{N}$,

$$\begin{aligned} \xi &\in \mathcal{D}, \quad \alpha^{(3)}(\varepsilon(\xi)) \in \mathcal{A}_{\varepsilon(\xi)}, \\ \{S : \alpha_S^{(3)}(\varepsilon(\xi)) \neq 0\} &= \mathcal{K}, \quad \alpha_S^{(3)}(\varepsilon(\xi)) > 0 \quad (S \in \mathcal{K}), \\ \sum_{S \in \mathcal{S}} \alpha_S^{(3)}(\varepsilon(\xi)) \sigma_S(\xi, t_3(\xi)) &= 0, \\ b_S(\varepsilon(\xi))^\top \eta_{\text{loc}}(\xi) &= f_S(\xi, t_3(\xi)) \quad (S \in \mathcal{K}), \\ f_S(\xi, t_3(\xi)) &< b_S(\varepsilon(\xi))^\top \eta_{\text{loc}}(\xi) \quad (S \notin \mathcal{K}). \end{aligned}$$

These quantities constitute the certificate in [Definition 13](#), so

$$\mathcal{N} \subseteq \mathcal{R}_3.$$

As ξ_0 was arbitrary, $\mathcal{R}_3$ is open in $\mathbb{R}^4$.

6. *Primal feasibility and stationarity at the stated point.* Define

$$\xi_{\text{low}} := \left(\frac{3}{10}, \frac{1}{10}, \frac{1}{10}, \log 3 \right), \quad \theta_{\text{low}} := \theta(\xi_{\text{low}}) = \left(\frac{1}{10}, \frac{1}{10} \right)^\top, \quad t_{\text{low}} := -\frac{339}{9985},$$

$$\alpha := \alpha^{(3)}(\log 3), \quad \alpha_S = \begin{cases} 1/5, & S \in \mathcal{K}, \\ 0, & S \notin \mathcal{K}. \end{cases}$$

Since $\log 3 > 0$ and $e^{\log 3} = 3$, the canonical-weight calculation yields

$$\alpha \in \mathcal{A}_{\log 3}, \quad \{S : \alpha_S \neq 0\} = \mathcal{K}, \quad \alpha_{S_1} = \alpha_{S_6} = \alpha_{S_8} = \frac{1}{5}.$$

The record probabilities are

$$\pi_{\theta_{\text{low}}}(3/10) = \frac{1}{100} (63, 7, 27, 3)^\top.$$

Substitution into the pattern formulas gives

S	$h_S(\theta_{\text{low}})$	g_S
S_1	$113/50$	$(-7/5, 0)^\top$
S_6	$42/25$	$(7/5, -3/5)^\top$
S_8	$53/50$	$(0, 3/5)^\top$

and hence, for every $t \in \mathbb{R}$,

$$F_{\theta_{\text{low}}}(\alpha, t) = \frac{1}{5} \left[\frac{98}{113} t^2 + \frac{(4t-3)^2}{42} + \frac{18}{53} (t+1)^2 \right].$$

Here and below objectives at θ_{low} are evaluated at $p = 3/10$ and $\varepsilon = \log 3$. The weighted slope at t_{low} is therefore

$$\sum_{S \in \mathcal{S}} \alpha_S \sigma_S(\xi_{\text{low}}, t_{\text{low}}) = \frac{1}{5} \left[\frac{196}{113} t_{\text{low}} + \frac{8(4t_{\text{low}}-3)}{42} + \frac{36}{53} (t_{\text{low}}+1) \right] = 0.$$

This establishes the required profiling stationarity.

7. *The exact dual certificate and region membership.* Define the four-coordinate dual vector by

$$\eta_{\text{low}} := \begin{pmatrix} -21817593/398800900 \\ 32820291/8773619800 \\ 509870781/8773619800 \\ 41183667/398800900 \end{pmatrix}.$$

Its coordinate sum is

$$\sum_{j=0}^3 \eta_{\text{low},j} = -\frac{21817593}{398800900} + \frac{32820291}{8773619800} + \frac{509870781}{8773619800} + \frac{41183667}{398800900} = \frac{441}{3994}.$$

For every $S \in \mathcal{S}$, define its slack and the specialized ray entries by

$$\Delta_S := b_S(\log 3)^\top \eta_{\text{low}} - f_S(\xi_{\text{low}}, t_{\text{low}}), \quad b_{S,j}^{\text{low}} := b_{S,j}(\log 3) = \begin{cases} 3, & j \in S, \\ 1, & j \notin S. \end{cases}$$

The slack is computed directly as

$$\Delta_S = \sum_{j=0}^3 b_{S,j}^{\text{low}} \eta_{\text{low},j} - \frac{\left[\frac{7}{10} (b_{S,1}^{\text{low}} - b_{S,0}^{\text{low}}) t_{\text{low}} + \frac{3}{10} (b_{S,3}^{\text{low}} - b_{S,2}^{\text{low}}) (t_{\text{low}} + 1) \right]^2}{(63b_{S,0}^{\text{low}} + 7b_{S,1}^{\text{low}} + 27b_{S,2}^{\text{low}} + 3b_{S,3}^{\text{low}})/100}.$$

For example, $S_2 = \{1\}$ has ray $(1, 3, 1, 1)^\top$, so

$$\Delta_{S_2} = \frac{441}{3994} + 2 \frac{32820291}{8773619800} - \frac{98}{57} \left(\frac{339}{9985} \right)^2 = \frac{1932296079}{16669877620}.$$

Applying the same displayed formula to each of the fourteen binary patterns gives

S	Δ_S	S	Δ_S
S_1	0	S_8	0
S_2	1932296079/16669877620	S_9	4394565/115652261
S_3	7441119/877361980	S_{10}	77641557/877361980
S_4	7441119/877361980	S_{11}	2821826721/35971841180
S_5	14134743/877361980	S_{12}	380056761/877361980
S_6	0	S_{13}	3683761767/11405705740
S_7	41610/3988009	S_{14}	50813490/115652261

Thus the active and inactive cases satisfy, respectively,

$$b_S(\log 3)^\top \eta_{\text{low}} = f_S(\xi_{\text{low}}, t_{\text{low}}) \quad (S \in \mathcal{K}),$$

$$f_S(\xi_{\text{low}}, t_{\text{low}}) < b_S(\log 3)^\top \eta_{\text{low}} \quad (S \notin \mathcal{K}).$$

The point ξ_{low} belongs to $\mathcal{D}$. Together with the feasible positive active weights and stationarity established above, these equalities and strict inequalities supply its certificate in [Definition 13](#). Therefore

$$\xi_{\text{low}} \in \mathcal{R}_3.$$

8. *A uniform upper bound on profiled information.* Let $\beta \in \mathcal{A}_{\log 3}$. Positive pattern masses and nonnegative weights imply

$$F_{\theta_{\text{low}}}(\beta, t) = \sum_{S \in \mathcal{S}} \beta_S \frac{(g_S^\top v(t))^2}{h_S(\theta_{\text{low}})} \geq 0 \quad (t \in \mathbb{R}).$$

Thus its set of objective values,

$$\mathcal{U}_\beta := \{F_{\theta_{\text{low}}}(\beta, t) : t \in \mathbb{R}\},$$

is nonempty and bounded below by zero. In particular,

$$\inf_{t \in \mathbb{R}} F_{\theta_{\text{low}}}(\beta, t) \leq F_{\theta_{\text{low}}}(\beta, t_{\text{low}}).$$

The active equality and inactive strict-inequality cases both give

$$f_S(\xi_{\text{low}}, t_{\text{low}}) \leq b_S(\log 3)^\top \eta_{\text{low}} \quad (S \in \mathcal{S}).$$

Multiplication by $\beta_S \geq 0$ and summation yield

$$F_{\theta_{\text{low}}}(\beta, t_{\text{low}}) \leq \sum_{S \in \mathcal{S}} \beta_S b_S(\log 3)^\top \eta_{\text{low}}.$$

Normalization identifies the latter sum:

$$\begin{aligned} \sum_{S \in \mathcal{S}} \beta_S b_S(\log 3)^\top \eta_{\text{low}} &= \sum_{j=0}^3 \eta_{\text{low},j} \left(\sum_{S \in \mathcal{S}} \beta_S b_{S,j}(\log 3) \right) \\ &= \sum_{j=0}^3 \eta_{\text{low},j} = \frac{441}{3994}. \end{aligned}$$

Consequently, for every feasible β ,

$$\inf_{t \in \mathbb{R}} F_{\theta_{\text{low}}}(\beta, t) \leq F_{\theta_{\text{low}}}(\beta, t_{\text{low}}) \leq \frac{441}{3994}.$$

9. *The candidate infimum and optimized value.* Expanding the candidate objective gives

$$F_{\theta_{\text{low}}}(\alpha, t) = \frac{39940}{125769} t^2 + \frac{8}{371} t + \frac{411}{3710} = \frac{441}{3994} + \frac{39940}{125769} \left(t + \frac{339}{9985} \right)^2.$$

The coefficient of the square is positive. Therefore

$$F_{\theta_{\text{low}}}(\alpha, t) \geq \frac{441}{3994} \quad (t \in \mathbb{R}), \quad F_{\theta_{\text{low}}}(\alpha, t_{\text{low}}) = \frac{441}{3994},$$

and hence

$$\inf_{t \in \mathbb{R}} F_{\theta_{\text{low}}}(\alpha, t) = \frac{441}{3994}.$$

Define the set of feasible profiled values by

$$\mathcal{E}_{\text{prof}} := \left\{ \inf_{t \in \mathbb{R}} F_{\theta_{\text{low}}}(\beta, t) : \beta \in \mathcal{A}_{\log 3} \right\}.$$

The uniform bound gives

$$z \leq \frac{441}{3994} \quad (z \in \mathcal{E}_{\text{prof}}),$$

while the feasible candidate gives

$$\frac{441}{3994} \in \mathcal{E}_{\text{prof}}.$$

Thus this nonempty, bounded-above set has supremum $441/3994$. By the optimized-value definition in [Definition 5](#),

$$J^*(\theta_{\text{low}}, 3/10, \log 3) = \sup \mathcal{E}_{\text{prof}} = \frac{441}{3994} = \inf_{t \in \mathbb{R}} F_{\theta_{\text{low}}}(\alpha, t).$$

This proves attainment by the stated weight vector.

10. *Uniqueness of the optimizing weights.* Suppose that $\beta \in \mathcal{A}_{\log 3}$ satisfies

$$\inf_{t \in \mathbb{R}} F_{\theta_{\text{low}}}(\beta, t) = J^*(\theta_{\text{low}}, 3/10, \log 3).$$

Its objective-value set is bounded below by zero, so the infimum is bounded above by its value at t_{low} . The dual upper bound then gives

$$\frac{441}{3994} = \inf_{t \in \mathbb{R}} F_{\theta_{\text{low}}}(\beta, t) \leq F_{\theta_{\text{low}}}(\beta, t_{\text{low}}) \leq \frac{441}{3994}.$$

All these values are equal. In particular, normalization of the dual sum yields

$$\sum_{S \in \mathcal{S}} \beta_S f_S(\xi_{\text{low}}, t_{\text{low}}) = \frac{441}{3994} = \sum_{S \in \mathcal{S}} \beta_S b_S(\log 3)^\top \eta_{\text{low}}.$$

For an active pattern the corresponding terms are equal. For an inactive pattern, nonnegative multiplication of its strict slack gives

$$\beta_S f_S(\xi_{\text{low}}, t_{\text{low}}) \leq \beta_S b_S(\log 3)^\top \eta_{\text{low}}.$$

Since these finitely many termwise inequalities have equal sums, equality holds in every term:

$$\beta_S f_S(\xi_{\text{low}}, t_{\text{low}}) = \beta_S b_S(\log 3)^\top \eta_{\text{low}} \quad (S \in \mathcal{S}).$$

If $S \notin \mathcal{K}$ and $\beta_S \neq 0$, feasibility implies $\beta_S > 0$. Its strict slack would then give

$$\beta_S f_S(\xi_{\text{low}}, t_{\text{low}}) < \beta_S b_S(\log 3)^\top \eta_{\text{low}},$$

a contradiction. Hence

$$\beta_S = 0 \quad (S \notin \mathcal{K}).$$

The normalization equations at input coordinates 0, 1, 3 now reduce to

$$\begin{aligned} 3\beta_{S_1} + \beta_{S_6} + \beta_{S_8} &= 1, \\ \beta_{S_1} + 3\beta_{S_6} + \beta_{S_8} &= 1, \\ \beta_{S_1} + \beta_{S_6} + 3\beta_{S_8} &= 1. \end{aligned}$$

Their differences give

$$2(\beta_{S_1} - \beta_{S_6}) = 0, \quad 2(\beta_{S_6} - \beta_{S_8}) = 0,$$

and substitution gives

$$\beta_{S_1} = \beta_{S_6} = \beta_{S_8} = \frac{1}{5}.$$

For $S \in \mathcal{K}$, these weights equal α_S ; for $S \notin \mathcal{K}$, both weights are zero. Therefore $\beta = \alpha$.

11. *Reciprocal variance.* Finally, the reciprocal definition in [Definition 5](#) and the positive optimized value give

$$V^*(\theta_{\text{low}}, 3/10, \log 3) = (J^*(\theta_{\text{low}}, 3/10, \log 3))^{-1} = \left(\frac{441}{3994} \right)^{-1} = \frac{3994}{441}.$$

□

Proof of [Theorem 7](#). Throughout, fix $p = 3/10$ and $\varepsilon = \log 3$. Set

$$A_5 = \{S_3, S_4, S_6, S_8, S_9\}, \quad A_3 = \{S_1, S_6, S_8\}, \quad \iota : \mathbb{R}^2 \longrightarrow \mathbb{R}^4, \quad \iota(\theta) = \left(\frac{3}{10}, \mu_0, \mu_1, \log 3 \right),$$

where $\theta = (\mu_0, \mu_1)^\top$. We use the objective, rays, and certificates of [Definitions 4](#) and [12](#), with the pattern enumeration in [Definition 3](#).

1. *Construct the open slices.* By [Theorems 5](#) and [6](#),

$$\mathcal{R}_5, \mathcal{R}_3 \text{ are open,} \quad \iota((1/2, 1/2)^\top) \in \mathcal{R}_5, \quad \iota((1/10, 1/10)^\top) \in \mathcal{R}_3.$$

Define

$$\mathcal{V}_5 = \iota^{-1}(\mathcal{R}_5) \cap \{\theta : \mu_0 > 3/10\}, \quad \mathcal{V}_3 = \iota^{-1}(\mathcal{R}_3) \cap \{\theta : \mu_0 < 3/10\}.$$

The map ι is continuous, so both sets are open. Their respective centers belong to them because

$$\frac{1}{10} < \frac{3}{10} < \frac{1}{2}.$$

They are disjoint: membership in their intersection would require both $\mu_0 > 3/10$ and $\mu_0 < 3/10$. Moreover,

$$\theta \in \mathcal{V}_5 \implies \iota(\theta) \in \mathcal{R}_5, \quad \theta \in \mathcal{V}_3 \implies \iota(\theta) \in \mathcal{R}_3.$$

2. *Preserve the three distinct scores near the low-mean point.* For this neighborhood construction, let

$$\mathcal{O} = (0, 1)^3 \times (0, \infty), \quad \xi = (p_\xi, \mu_{0,\xi}, \mu_{1,\xi}, \varepsilon_\xi) \in \mathcal{O}, \quad \theta_\xi = (\mu_{0,\xi}, \mu_{1,\xi})^\top.$$

Write the record probabilities, rays, masses, and gradients at these parameters as

$$\begin{aligned} \pi_\xi &= ((1 - p_\xi)(1 - \mu_{0,\xi}), (1 - p_\xi)\mu_{0,\xi}, p_\xi(1 - \mu_{1,\xi}), p_\xi\mu_{1,\xi})^\top, \\ b_S(\xi) &= \mathbf{1}_4 + (e^{\varepsilon_\xi} - 1)\mathbf{1}_S, \quad h_S(\xi) = \pi_\xi^\top b_S(\xi), \\ g_S(\xi) &= (e^{\varepsilon_\xi} - 1) \begin{pmatrix} (1 - p_\xi)\{\mathbf{1}_S(1) - \mathbf{1}_S(0)\} \\ p_\xi\{\mathbf{1}_S(3) - \mathbf{1}_S(2)\} \end{pmatrix}, \quad \mathbf{1}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \end{aligned}$$

Here $\mathbf{1}_S$ is the indicator vector of the pattern S . In particular, $h_S(\xi) > 0$ for every S .

Define the canonical three-pattern weights and their stationarity coefficients by

$$\begin{aligned} \alpha_S^{\text{can}}(\xi) &= \begin{cases} (e^{\varepsilon_\xi} + 2)^{-1}, & S \in A_3, \\ 0, & S \notin A_3, \end{cases} \\ a_3(\xi) &= \frac{2}{e^{\varepsilon_\xi} + 2} \sum_{S \in A_3} \frac{(g_S(\xi)^\top \mathbf{1}_2)^2}{h_S(\xi)}, \quad \ell_3(\xi) = \frac{2}{e^{\varepsilon_\xi} + 2} \sum_{S \in A_3} \frac{(g_S(\xi)^\top v(0))(g_S(\xi)^\top \mathbf{1}_2)}{h_S(\xi)}. \end{aligned}$$

Expansion of the quadratic objective gives

$$\frac{\partial}{\partial t} F_{\theta_\xi}(\alpha^{\text{can}}(\xi), t) = \ell_3(\xi) + a_3(\xi)t,$$

where the objective is evaluated at p_ξ, ε_ξ . The S_1 summand yields

$$a_3(\xi) \geq \frac{2(e^{\varepsilon_\xi} - 1)^2(1 - p_\xi)^2}{(e^{\varepsilon_\xi} + 2)h_{S_1}(\xi)} > 0.$$

Thus the unique stationary coordinate is the function

$$t_3 : \mathcal{O} \longrightarrow \mathbb{R}, \quad t_3(\xi) = -\frac{\ell_3(\xi)}{a_3(\xi)}.$$

All denominators in these formulas are positive, so t_3 and the three functions

$$r_{3,S}(\xi) = \frac{g_S(\xi)^\top v(t_3(\xi))}{h_S(\xi)}, \quad S \in A_3,$$

are continuous at

$$\xi_{\text{low}} = \left(\frac{3}{10}, \frac{1}{10}, \frac{1}{10}, \log 3 \right).$$

To calculate their values, the record probabilities at this point are

$$\pi_{\xi_{\text{low}}} = \left(\frac{63}{100}, \frac{7}{100}, \frac{27}{100}, \frac{3}{100} \right)^\top.$$

Using $S_1 = \{0\}$, $S_6 = \{1, 2\}$, and $S_8 = \{3\}$, the definitions give

S	$h_S(\xi_{\text{low}})$	$g_S(\xi_{\text{low}})$
S_1	$113/50$	$(-7/5, 0)^\top$
S_6	$42/25$	$(7/5, -3/5)^\top$
S_8	$53/50$	$(0, 3/5)^\top$

and hence

$$\ell_3(\xi_{\text{low}}) = \frac{8}{371}, \quad a_3(\xi_{\text{low}}) = \frac{79880}{125769}, \quad t_3(\xi_{\text{low}}) = -\frac{339}{9985}.$$

Substitution into the projected-score formula gives

$$r_{3,S_1}(\xi_{\text{low}}) = \frac{42}{1997}, \quad r_{3,S_6}(\xi_{\text{low}}) = -\frac{1491}{3994}, \quad r_{3,S_8}(\xi_{\text{low}}) = \frac{1092}{1997}.$$

These values are pairwise distinct: the middle value is negative, the others are positive, and

$$\frac{1092}{1997} - \frac{42}{1997} = \frac{1050}{1997} > 0.$$

For each ordered pair $S, T \in A_3$ with $S \neq T$, continuity of their difference therefore supplies an open neighborhood $D_{S,T}$ satisfying

$$\xi_{\text{low}} \in D_{S,T} \subseteq \{\xi \in \mathcal{O} : r_{3,S}(\xi) - r_{3,T}(\xi) \neq 0\}.$$

Define

$$D_3 = \bigcap_{\substack{S, T \in A_3 \\ S \neq T}} D_{S,T}.$$

This finite intersection is open, contains ξ_{low} , and satisfies

$$\xi \in D_3, \quad S, T \in A_3, \quad S \neq T \implies r_{3,S}(\xi) \neq r_{3,T}(\xi).$$

3. *Choose the final neighborhoods.* Set

$$U_5 = \mathcal{V}_5, \quad U_3 = \mathcal{V}_3 \cap \iota^{-1}(D_3).$$

Continuity of ι makes U_3 open. The center conditions established above give

$$(1/2, 1/2)^\top \in U_5, \quad (1/10, 1/10)^\top \in U_3.$$

Since $U_3 \subseteq \mathcal{V}_3$, the sets U_5, U_3 are disjoint. Every point in either neighborhood has interior arm means, by its corresponding region membership.

4. *Obtain the five-pattern optimum and support rigidity.* Fix $\theta \in U_5$. Membership in $\mathcal{R}_5$, together with [Definitions 12](#) and [13](#), supplies

$$\alpha^{(5)} \in \mathcal{A}_\epsilon, \quad t_{\text{cert},5} \in \mathbb{R}, \quad \eta^{(5)} \in \mathbb{R}^4,$$

with

$$\alpha_S^{(5)} > 0 \quad (S \in A_5), \quad \alpha_S^{(5)} = 0 \quad (S \notin A_5), \quad \frac{\partial}{\partial t} F_\theta(\alpha^{(5)}, t_{\text{cert},5}) = 0,$$

and

$$f_S(t_{\text{cert},5}) = b_S^\top \eta^{(5)} \quad (S \in A_5), \quad f_S(t_{\text{cert},5}) < b_S^\top \eta^{(5)} \quad (S \notin A_5).$$

Here $f_S(t) = (g_S^\top v(t))^2 / h_S(\theta)$, as in [Definition 12](#). Stationarity gives the exact expansion

$$F_\theta(\alpha^{(5)}, t) - F_\theta(\alpha^{(5)}, t_{\text{cert},5}) = (t - t_{\text{cert},5})^2 \sum_{S \in \mathcal{S}} \alpha_S^{(5)} \frac{(g_S^\top \mathbf{1}_2)^2}{h_S(\theta)} \geq 0.$$

For every feasible β , normalization and the dual inequalities give

$$\begin{aligned} \inf_{t \in \mathbb{R}} F_\theta(\beta, t) &\leq F_\theta(\beta, t_{\text{cert},5}) \\ &\leq \sum_{S \in \mathcal{S}} \beta_S b_S^\top \eta^{(5)} = \mathbf{1}_4^\top \eta^{(5)} = F_\theta(\alpha^{(5)}, t_{\text{cert},5}). \end{aligned}$$

The last equality uses the active dual equalities and the vanishing inactive weights. Since $\alpha^{(5)}$ attains its minimum at the certifying coordinate,

$$J^*(\theta, p, \varepsilon) = \inf_t F_\theta(\alpha^{(5)}, t) = F_\theta(\alpha^{(5)}, t_{\text{cert},5}) = \mathbf{1}_4^\top \eta^{(5)}.$$

If a feasible β satisfies $J^* = \inf_t F_\theta(\beta, t)$, the preceding inequalities are equalities. Consequently,

$$0 = \sum_{S \in \mathcal{S}} \beta_S (b_S^\top \eta^{(5)} - f_S(t_{\text{cert},5})).$$

Every summand is nonnegative, and every inactive slack is strictly positive. Therefore

$$\beta_S = 0 \quad (S \notin A_5).$$

5. *Determine the five-pattern mixture uniquely.* For a feasible β vanishing outside A_5 , the four normalization equations, using $e^\varepsilon = 3$, are

$$\begin{aligned} 3\beta_{S_3} + \beta_{S_4} + \beta_{S_6} + \beta_{S_8} + 3\beta_{S_9} &= 1, \\ 3\beta_{S_3} + \beta_{S_4} + 3\beta_{S_6} + \beta_{S_8} + \beta_{S_9} &= 1, \\ \beta_{S_3} + 3\beta_{S_4} + 3\beta_{S_6} + \beta_{S_8} + \beta_{S_9} &= 1, \\ \beta_{S_3} + \beta_{S_4} + \beta_{S_6} + 3\beta_{S_8} + 3\beta_{S_9} &= 1. \end{aligned}$$

Subtracting these equations yields

$$\beta_{S_3} = \beta_{S_4} = \beta_{S_8}, \quad \beta_{S_6} = \beta_{S_9}, \quad 5\beta_{S_3} + 4\beta_{S_6} = 1.$$

Thus, with the mixture coordinate defined by

$$\lambda(\beta) = 4\beta_{S_6},$$

the weights satisfy

$$\beta_{S_6} = \beta_{S_9} = \frac{\lambda(\beta)}{4}, \quad \beta_{S_3} = \beta_{S_4} = \beta_{S_8} = \frac{1 - \lambda(\beta)}{5}.$$

Define the two normalized component objectives and their coefficients by

$$B_5(t) = \frac{f_{S_6}(t) + f_{S_9}(t)}{4}, \quad T_5(t) = \frac{f_{S_3}(t) + f_{S_4}(t) + f_{S_8}(t)}{5},$$

$$C_{\text{bin}} = \frac{1}{100} \left(\frac{1}{h_{S_6}(\theta)} + \frac{1}{h_{S_9}(\theta)} \right) > 0, \quad C_{\text{tri}} = \frac{9}{125} \left(\frac{1}{h_{S_4}(\theta)} + \frac{1}{h_{S_8}(\theta)} \right) > 0.$$

The pattern gradients are

$$g_{S_3} = 0, \quad g_{S_4} = (0, -3/5)^\top, \quad g_{S_6} = (7/5, -3/5)^\top, \quad g_{S_8} = (0, 3/5)^\top, \quad g_{S_9} = (-7/5, 3/5)^\top,$$

so direct substitution gives

$$B_5(t) = C_{\text{bin}}(4t - 3)^2, \quad T_5(t) = C_{\text{tri}}(t + 1)^2.$$

Also,

$$\frac{b_{S_6} + b_{S_9}}{4} = \frac{b_{S_3} + b_{S_4} + b_{S_8}}{5} = \mathbf{1}_4.$$

The active dual equalities therefore imply

$$B_5(t_{\text{cert},5}) = T_5(t_{\text{cert},5}) = \mathbf{1}_4^\top \eta^{(5)}.$$

In particular,

$$T_5'(t_{\text{cert},5}) = 2C_{\text{tri}}(t_{\text{cert},5} + 1) \neq 0.$$

Indeed, a zero derivative would give $t_{\text{cert},5} = -1$, whereas the component equality would then require $49C_{\text{bin}} = 0$, contradicting $C_{\text{bin}} > 0$.

For the certificate weights, the mixture representation gives

$$F_\theta(\alpha^{(5)}, t) = \lambda(\alpha^{(5)})B_5(t) + (1 - \lambda(\alpha^{(5)}))T_5(t).$$

Its stationarity consequently implies

$$\lambda(\alpha^{(5)})B_5'(t_{\text{cert},5}) + (1 - \lambda(\alpha^{(5)}))T_5'(t_{\text{cert},5}) = 0.$$

The derivative difference is nonzero: if the two derivatives were equal, this equation would force $T_5'(t_{\text{cert},5}) = 0$. Solving gives

$$\lambda(\alpha^{(5)}) = -\frac{T_5'(t_{\text{cert},5})}{B_5'(t_{\text{cert},5}) - T_5'(t_{\text{cert},5})}.$$

For any feasible optimal β , the preceding support-rigidity argument applies, and

$$F_\theta(\beta, t_{\text{cert},5}) = J^* = \inf_t F_\theta(\beta, t).$$

Hence β is stationary at the same coordinate and satisfies the same mixture equation. Thus

$$\lambda(\beta) = \lambda(\alpha^{(5)}), \quad \beta = \alpha^{(5)}.$$

This proves the required uniqueness among all feasible optimal weights.

6. *Establish exact cardinality five.* We first record positivity of the optimized information throughout the interior fixed-parameter slice. Define the feasible singleton design by

$$\alpha_S^{\text{rr}} = \begin{cases} 1/6, & S \in \{S_1, S_2, S_4, S_8\}, \\ 0, & \text{otherwise.} \end{cases}$$

Each input row sums to $(3 + 1 + 1 + 1)/6 = 1$. Its objective satisfies, for every $t \in \mathbb{R}$,

$$F_\theta(\alpha^{\text{rr}}, t) \geq \frac{49}{150h_{S_1}(\theta)}t^2, \quad F_\theta(\alpha^{\text{rr}}, t) \geq \frac{3}{50h_{S_8}(\theta)}(t+1)^2.$$

It is a coercive quadratic, so choose

$$t_{\text{rr}} \in \arg \min_{t \in \mathbb{R}} F_\theta(\alpha^{\text{rr}}, t).$$

A zero minimum would force both $t_{\text{rr}} = 0$ and $t_{\text{rr}} = -1$. Therefore

$$0 < F_\theta(\alpha^{\text{rr}}, t_{\text{rr}}) \leq J^*(\theta, p, \varepsilon).$$

For the five-pattern certificate, define

$$I_5 = I_\theta(\alpha^{(5)}), \quad c = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad w_5 = \frac{v(t_{\text{cert},5})}{J^*(\theta, p, \varepsilon)}.$$

The matrix I_5 is positive semidefinite, since it is a sum of nonnegative multiples of $g_S g_S^\top$. Stationarity and the certificate value give

$$\mathbf{1}_2^\top I_5 v(t_{\text{cert},5}) = 0, \quad v(t_{\text{cert},5})^\top I_5 v(t_{\text{cert},5}) = J^*.$$

The first identity makes the two coordinates of $I_5 v(t_{\text{cert},5})$ opposite; the second makes its second coordinate J^* . Hence

$$I_5 v(t_{\text{cert},5}) = J^* c, \quad I_5 w_5 = c, \quad c^\top w_5 = \frac{1}{J^*} = V^*.$$

Every other solution $w_{\text{alt}} \in \mathbb{R}^2$ of $I_5 w_{\text{alt}} = c$ has the same contrast value, because

$$c^\top (w_{\text{alt}} - w_5) = w_5^\top I_5 (w_{\text{alt}} - w_5) = 0.$$

The contrast-variance convention in [Theorem 5](#) therefore gives variance $V^* > 0$, equal to its nonnegative part.

For the associated staircase channel,

$$(P_\theta Q_{\alpha^{(5)}})(\{S\}) = \alpha_S^{(5)} h_S(\theta).$$

The positive masses $h_S(\theta)$ make precisely the five patterns in A_5 active, so

$$\kappa(Q_{\alpha^{(5)}}) = 5.$$

Feasibility supplies unit row sums, and the ray entries 1, 3 supply the privacy inequalities. Finally, [Theorem 5](#) gives

$$\kappa(Q) \geq 5$$

for every attaining stationary ε -LDP channel on any measurable output space, since $\iota(\theta) \in \mathcal{R}_5$.

7. *Obtain the unique three-pattern optimum.* Fix $\theta \in U_3$. Its membership in $\mathcal{R}_3$ supplies certificate data

$$\alpha^{(3)} \in \mathcal{A}_\varepsilon, \quad t_{\text{cert},3} \in \mathbb{R}, \quad \eta^{(3)} \in \mathbb{R}^4,$$

with support A_3 , positive active weights, stationarity, active dual equalities, and strict inactive dual inequalities. The quadratic expansion and normalization calculation used above give

$$J^* = \inf_t F_\theta(\alpha^{(3)}, t) = F_\theta(\alpha^{(3)}, t_{\text{cert},3}) = \mathbf{1}_4^\top \eta^{(3)},$$

and, for every feasible β ,

$$F_\theta(\beta, t_{\text{cert},3}) \leq \mathbf{1}_4^\top \eta^{(3)} = J^*.$$

If $J^* = \inf_t F_\theta(\beta, t)$, equality follows and

$$0 = \sum_{S \in \mathcal{S}} \beta_S (b_S^\top \eta^{(3)} - f_S(t_{\text{cert},3})), \quad \beta_S = 0 \quad (S \notin A_3).$$

For any feasible weights supported on A_3 , the normalization rows are

$$3\beta_{S_1} + \beta_{S_6} + \beta_{S_8} = 1,$$

$$\beta_{S_1} + 3\beta_{S_6} + \beta_{S_8} = 1,$$

$$\beta_{S_1} + \beta_{S_6} + 3\beta_{S_8} = 1,$$

with the second row also applying to the other record in S_6 . Subtraction and substitution give

$$\beta_{S_1} = \beta_{S_6} = \beta_{S_8} = \frac{1}{5}.$$

These equations apply both to $\alpha^{(3)}$ and to every feasible optimal β . Consequently,

$$\alpha_S^{(3)} = \begin{cases} 1/5, & S \in A_3, \\ 0, & S \notin A_3, \end{cases} \quad \beta = \alpha^{(3)}.$$

8. *Construct an attaining channel with three active outputs.* The canonical weights just obtained agree with $\alpha^{\text{can}}(\iota(\theta))$. Their stationarity equation is

$$0 = \ell_3(\iota(\theta)) + a_3(\iota(\theta))t_{\text{cert},3}.$$

Since $a_3(\iota(\theta)) > 0$,

$$t_{\text{cert},3} = t_3(\iota(\theta)).$$

The information matrix is positive semidefinite, and its certifying minimum equals $J^* > 0$. The same two-coordinate calculation therefore gives

$$I_\theta(\alpha^{(3)})v(t_{\text{cert},3}) = J^*c, \quad I_\theta(\alpha^{(3)})\frac{v(t_{\text{cert},3})}{J^*} = c, \quad c^\top \frac{v(t_{\text{cert},3})}{J^*} = V^* > 0.$$

Thus the feasible staircase channel attains the required contrast variance. Its output probabilities satisfy

$$(P_\theta Q_{\alpha^{(3)}})(\{S\}) = \alpha_S^{(3)} h_S(\theta), \quad \kappa(Q_{\alpha^{(3)}}) = |A_3| = 3.$$

9. *Prove the universal three-output lower bound.* Let Q be any attaining stationary ε -LDP channel on a measurable output space Z , and define

$$I_Q = I_\theta(Q).$$

Its contrast variance is the finite positive value $V^* = 1/J^*$. Hence the contrast is in the range of I_Q , and we may choose

$$w_Q \in \mathbb{R}^2, \quad I_Q w_Q = c, \quad c^\top w_Q = \frac{1}{J^*}.$$

Positive semidefiniteness and Cauchy–Schwarz give

$$1 = (v(t_{\text{cert},3})^\top I_Q w_Q)^2 \leq (v(t_{\text{cert},3})^\top I_Q v(t_{\text{cert},3}))(w_Q^\top I_Q w_Q).$$

Since $w_Q^\top I_Q w_Q = c^\top w_Q = 1/J^*$, this implies

$$J^* \leq v(t_{\text{cert},3})^\top I_Q v(t_{\text{cert},3}).$$

Apply [Lemma 1](#), using the interior parameters and the constant positive privacy budget. It supplies feasible weights $\bar{\alpha}$ and a probability kernel K with

$$Q = K \circ Q_{\bar{\alpha}}, \quad I_\theta(\bar{\alpha}) - I_Q \succeq 0.$$

The directional lower bound, refinement inequality, and certificate upper bound form the chain

$$J^* \leq v(t_{\text{cert},3})^\top I_Q v(t_{\text{cert},3}) \leq F_\theta(\bar{\alpha}, t_{\text{cert},3}) \leq J^*.$$

Thus every inequality is an equality. Equality in the dual bound gives

$$0 = \sum_{S \in \mathcal{S}} \bar{\alpha}_S (b_S^\top \eta^{(3)} - f_S(t_{\text{cert},3})), \quad \bar{\alpha}_S = 0 \quad (S \notin A_3).$$

The normalization equations above then force

$$\bar{\alpha}_{S_1} = \bar{\alpha}_{S_6} = \bar{\alpha}_{S_8} = \frac{1}{5}.$$

Because $\iota(\theta) \in D_3$ and $t_{\text{cert},3} = t_3(\iota(\theta))$, the active projected scores satisfy

$$\rho_S(\theta, t_{\text{cert},3}) \neq \rho_T(\theta, t_{\text{cert},3}) \quad (S, T \in A_3, S \neq T).$$

Equality in the refinement direction, together with [Lemma 1](#), consequently implies

$$K(S, \cdot) \perp K(T, \cdot) \quad (S, T \in A_3, S \neq T).$$

Define the output law by

$$\nu_Q = P_\theta Q = \sum_{S \in A_3} \frac{h_S(\theta)}{5} K(S, \cdot).$$

In particular, for every measurable output event E and every $S \in A_3$,

$$\nu_Q(E) \geq \frac{h_S(\theta)}{5} K(S, E), \quad K(S, \cdot) \ll \nu_Q.$$

If Z is infinite, the cardinality convention gives $\kappa(Q) = +\infty \geq 3$.

If Z is finite, each probability measure $K(S, \cdot)$, for $S \in A_3$, satisfies

$$1 = K(S, Z) \leq \sum_{z \in Z} K(S, \{z\}).$$

Thus choose, for every $S \in A_3$, a point

$$z_S \in Z, \quad K(S, \{z_S\}) > 0.$$

Absolute continuity transfers positivity to the output law: indeed,

$$\nu_Q(\{z_S\}) = 0 \implies K(S, \{z_S\}) = 0,$$

so $\nu_Q(\{z_S\}) > 0$.

To prove that these points are distinct, fix distinct $S, T \in A_3$. Mutual singularity supplies a measurable set $E_{S,T} \subseteq Z$ with

$$K(S, E_{S,T}) = 0, \quad K(T, Z \setminus E_{S,T}) = 0.$$

Suppose that $z_S = z_T$. If $z_S \in E_{S,T}$, monotonicity gives

$$0 < K(S, \{z_S\}) \leq K(S, E_{S,T}) = 0.$$

If $z_S \notin E_{S,T}$, then $z_T \in Z \setminus E_{S,T}$, and

$$0 < K(T, \{z_T\}) \leq K(T, Z \setminus E_{S,T}) = 0.$$

Both cases yield a contradiction. Hence the three selected points are distinct. Hence

$$\kappa(Q) = \#\{z \in Z : \nu_Q(\{z\}) \neq 0\} \geq |A_3| = 3.$$

Together with the attaining three-output staircase channel, this proves exact optimal cardinality three throughout U_3 , and completes the result. □

Proof of Theorem 8. Throughout, $p = 1/2$. We use the staircase notation of Definitions 3 to 5.

1. Define the binary privacy contraction and the corresponding information value by

$$\kappa_\varepsilon := \frac{e^\varepsilon - 1}{e^\varepsilon + 1}, \quad J_{\text{bin}} := \frac{\kappa_\varepsilon^2}{1 - \kappa_\varepsilon^2 \tau^2}.$$

Since $\varepsilon > 0$, we have $e^\varepsilon > 1$, and hence

$$0 < \kappa_\varepsilon < 1.$$

The interior-mean inequalities give

$$-1 < \mu_1 - \mu_0 < 1, \quad |\tau| < 1.$$

Consequently,

$$0 \leq \kappa_\varepsilon^2 \tau^2 \leq \kappa_\varepsilon^2 < 1, \quad 1 - \kappa_\varepsilon^2 \tau^2 > 0, \quad J_{\text{bin}} > 0.$$

2. We first establish positivity of the optimized information. For every staircase pattern,

$$1 \leq b_S(j) \leq e^\varepsilon, \quad \sum_{j=0}^3 \pi_{\theta,j} = 1,$$

so

$$1 \leq h_S(\theta) = \sum_{j=0}^3 \pi_{\theta,j} b_S(j) \leq e^\varepsilon.$$

Define the singleton design by

$$\alpha_S^{\text{sing}} := \begin{cases} (e^\varepsilon + 3)^{-1}, & |S| = 1, \\ 0, & |S| \neq 1, \end{cases} \quad S \in \mathcal{S}.$$

For each record j , exactly one singleton pattern contributes e^ε and the other three contribute 1. Thus

$$\sum_{S \in \mathcal{S}} \alpha_S^{\text{sing}} b_S(j) = \frac{e^\varepsilon + 3}{e^\varepsilon + 3} = 1, \quad \alpha^{\text{sing}} \in \mathcal{A}_\varepsilon.$$

For completeness, its objective attains a minimum. Define

$$a_{\text{sing}} := 2 \sum_{S \in \mathcal{S}} \alpha_S^{\text{sing}} \frac{(g_S^\top v(0))(g_{S,0} + g_{S,1})}{h_S(\theta)},$$

$$b_{\text{sing}} := \sum_{S \in \mathcal{S}} \alpha_S^{\text{sing}} \frac{(g_{S,0} + g_{S,1})^2}{h_S(\theta)} \geq 0.$$

Expanding $v(t) = (t, t+1)^\top$ gives

$$F_\theta(\alpha^{\text{sing}}, t) = F_\theta(\alpha^{\text{sing}}, 0) + a_{\text{sing}} t + b_{\text{sing}} t^2, \quad t \in \mathbb{R}.$$

If $b_{\text{sing}} = 0$, nonnegativity of the objective forces $a_{\text{sing}} = 0$: otherwise, at

$$t_{\text{bad}} := -\frac{F_\theta(\alpha^{\text{sing}}, 0) + 1}{a_{\text{sing}}},$$

the objective would equal -1 . In this case $t = 0$ is a minimizer. If $b_{\text{sing}} > 0$, define

$$t_{\text{sing}} := -\frac{a_{\text{sing}}}{2b_{\text{sing}}}.$$

Then

$$F_\theta(\alpha^{\text{sing}}, t) - F_\theta(\alpha^{\text{sing}}, t_{\text{sing}}) = b_{\text{sing}}(t - t_{\text{sing}})^2 \geq 0.$$

In the constant case, set $t_{\text{sing}} := 0$.

The contributions from $S = \{0\}$ and $S = \{3\}$ satisfy

$$0 \leq \frac{(e^\varepsilon - 1)^2 t_{\text{sing}}^2}{4(e^\varepsilon + 3)h_{\{0\}}(\theta)} \leq F_\theta(\alpha^{\text{sing}}, t_{\text{sing}}),$$

$$0 \leq \frac{(e^\varepsilon - 1)^2 (t_{\text{sing}} + 1)^2}{4(e^\varepsilon + 3)h_{\{3\}}(\theta)} \leq F_\theta(\alpha^{\text{sing}}, t_{\text{sing}}).$$

Both coefficients are strictly positive. A zero minimum would therefore force $t_{\text{sing}} = 0$ and $t_{\text{sing}} + 1 = 0$, a contradiction. By the definition of J^* ,

$$0 < \min_{t \in \mathbb{R}} F_\theta(\alpha^{\text{sing}}, t) \leq J^*(\theta, 1/2, \varepsilon).$$

3. We next bound the balanced objective. Fix $\alpha \in \mathcal{A}_\varepsilon$, and write its staircase channel as

$$Q_\alpha(x_j, \{S\}) := \alpha_S b_S(j), \quad j \in \{0, 1, 2, 3\}, \quad S \in \mathcal{S}.$$

The feasibility equations make its rows probability measures. Since $b_S(j) \leq e^\varepsilon b_S(j')$, summing the atomic inequalities gives

$$Q_\alpha(x_j, A) \leq e^\varepsilon Q_\alpha(x_{j'}, A), \quad A \subseteq \mathcal{S}.$$

Define the grouped binary channel by

$$\begin{aligned} \overline{Q}_\alpha(+1, A) &:= \frac{Q_\alpha(x_0, A) + Q_\alpha(x_3, A)}{2}, \\ \overline{Q}_\alpha(-1, A) &:= \frac{Q_\alpha(x_1, A) + Q_\alpha(x_2, A)}{2}. \end{aligned}$$

Its rows are probability measures. Pairing the original row inequalities gives

$$\begin{aligned} \overline{Q}_\alpha(+1, A) &\leq \frac{e^\varepsilon}{2} \{Q_\alpha(x_1, A) + Q_\alpha(x_2, A)\} = e^\varepsilon \overline{Q}_\alpha(-1, A), \\ \overline{Q}_\alpha(-1, A) &\leq e^\varepsilon \overline{Q}_\alpha(+1, A). \end{aligned}$$

The inequalities with identical binary inputs follow from $1 \leq e^\varepsilon$.

Introduce the reference measure and its nonnegative row densities:

$$\nu_\alpha := \overline{Q}_\alpha(+1, \cdot) + \overline{Q}_\alpha(-1, \cdot), \quad f_{\alpha, \pm} := \frac{d\overline{Q}_\alpha(\pm 1, \cdot)}{d\nu_\alpha}.$$

Define, on the entire output space,

$$a_\alpha(z) := \frac{f_{\alpha,+}(z) + f_{\alpha,-}(z)}{2}, \quad \delta_\alpha(z) := \begin{cases} \frac{f_{\alpha,+}(z) - f_{\alpha,-}(z)}{f_{\alpha,+}(z) + f_{\alpha,-}(z)}, & f_{\alpha,+}(z) + f_{\alpha,-}(z) > 0, \\ 0, & f_{\alpha,+}(z) + f_{\alpha,-}(z) = 0. \end{cases}$$

Row normalization yields

$$\int a_\alpha d\nu_\alpha = 1, \quad a_\alpha \delta_\alpha = \frac{f_{\alpha,+} - f_{\alpha,-}}{2}, \quad \int a_\alpha \delta_\alpha d\nu_\alpha = 0.$$

Privacy implies, almost everywhere,

$$f_{\alpha,+} \leq e^\varepsilon f_{\alpha,-}, \quad f_{\alpha,-} \leq e^\varepsilon f_{\alpha,+}.$$

At a point where their sum is positive, these inequalities imply

$$(e^\varepsilon + 1)|f_{\alpha,+} - f_{\alpha,-}| \leq (e^\varepsilon - 1)(f_{\alpha,+} + f_{\alpha,-}),$$

and therefore $|\delta_\alpha| \leq \kappa_\varepsilon$. Where the sum vanishes, both densities vanish and the same bound follows from $\delta_\alpha = 0$.

The grouped mixture density and its derivative with respect to the binary sign parameter are

$$m_{\alpha,\tau} := \frac{1+\tau}{2}f_{\alpha,+} + \frac{1-\tau}{2}f_{\alpha,-} = a_\alpha(1+\tau\delta_\alpha), \quad \partial_\tau m_{\alpha,\tau} = a_\alpha\delta_\alpha.$$

Thus its Fisher information is

$$\mathcal{I}_\alpha := \int \frac{a_\alpha\delta_\alpha^2}{1+\tau\delta_\alpha} d\nu_\alpha.$$

Zero-density outputs contribute zero. Since $|\tau| < 1$ and $|\delta_\alpha| \leq \kappa_\varepsilon < 1$, the remaining denominators are positive. The pointwise inequality needed below follows from the identity

$$J_{\text{bin}}(1-\tau\delta_\alpha) - \frac{\delta_\alpha^2}{1+\tau\delta_\alpha} = \frac{\kappa_\varepsilon^2 - \delta_\alpha^2}{(1-\kappa_\varepsilon^2\tau^2)(1+\tau\delta_\alpha)} \geq 0.$$

This identity also applies when $\tau = 0$. Multiplying by $a_\alpha \geq 0$ and integrating gives

$$\begin{aligned} \mathcal{I}_\alpha &\leq J_{\text{bin}} \int (a_\alpha - \tau a_\alpha \delta_\alpha) d\nu_\alpha \\ &= J_{\text{bin}}. \end{aligned}$$

The bound $|a_\alpha\delta_\alpha| \leq \kappa_\varepsilon a_\alpha$ ensures the integrability of the displayed upper bound.

We now identify this information with the staircase objective. Balance gives

$$\tau = 1 - 2\mu_0, \quad \pi_\theta = \frac{1}{2}(1 - \mu_0, \mu_0, \mu_0, 1 - \mu_0)^\top.$$

Hence, for every $S \in \mathcal{S}$,

$$\begin{aligned} h_S(\theta) &= \frac{(1-\mu_0)\{b_S(0) + b_S(3)\} + \mu_0\{b_S(1) + b_S(2)\}}{2}, \\ g_S^\top v(-1/2) &= \frac{b_S(0) + b_S(3) - b_S(1) - b_S(2)}{4}. \end{aligned}$$

Define the grouped atomic probabilities by

$$q_{\alpha,+}(S) := \frac{\alpha_S\{b_S(0) + b_S(3)\}}{2}, \quad q_{\alpha,-}(S) := \frac{\alpha_S\{b_S(1) + b_S(2)\}}{2}.$$

The discrete Fisher-information formula therefore gives

$$\begin{aligned} \mathcal{I}_\alpha &= \sum_{S \in \mathcal{S}} \frac{\{(q_{\alpha,+}(S) - q_{\alpha,-}(S))/2\}^2}{\{(1+\tau)q_{\alpha,+}(S) + (1-\tau)q_{\alpha,-}(S)\}/2} \\ &= \sum_{S \in \mathcal{S}} \alpha_S \frac{(g_S^\top v(-1/2))^2}{h_S(\theta)} = F_\theta(\alpha, -1/2). \end{aligned}$$

Here a summand with $\alpha_S = 0$ is defined as zero; both grouped probabilities then vanish. When $\alpha_S > 0$, its mixture probability is $\alpha_S h_S(\theta) > 0$, and the displayed equality follows by cancellation. We have proved

$$F_\theta(\alpha, -1/2) \leq J_{\text{bin}} \quad \text{for every } \alpha \in \mathcal{A}_\varepsilon.$$

4. The feasible set is nonempty by the singleton construction above. It is closed, and its row constraints imply

$$0 \leq \alpha_S \leq \sum_{S' \in \mathcal{S}} \alpha_{S'} \leq 1.$$

Thus it is compact. Since the objective at a fixed profiling coordinate is continuous and linear in the weights, choose

$$\alpha^{\text{env}} \in \arg \max_{\alpha \in \mathcal{A}_\varepsilon} F_\theta(\alpha, -1/2).$$

For each feasible α , its profiled minimum is at most its value at $-1/2$. Consequently,

$$\begin{aligned} J^*(\theta, 1/2, \varepsilon) &\leq \max_{\alpha \in \mathcal{A}_\varepsilon} F_\theta(\alpha, -1/2) \\ &= F_\theta(\alpha^{\text{env}}, -1/2) \leq J_{\text{bin}}. \end{aligned}$$

Both information values are positive, so taking reciprocals yields

$$J_{\text{bin}}^{-1} \leq V^*(\theta, 1/2, \varepsilon).$$

5. We establish the channel properties needed for the matching bound. For the ordered record alphabet in [Definition 3](#), define

$$(\sigma_0, \sigma_1, \sigma_2, \sigma_3) := (1, -1, -1, 1).$$

These are the values of $(2W_i - 1)(2Y_i - 1)$. The randomized-response channel has atomic probabilities

$$Q(x_j, \{z\}) = \begin{cases} \frac{e^\varepsilon}{e^\varepsilon + 1}, & z = \sigma_j, \\ \frac{1}{e^\varepsilon + 1}, & z = -\sigma_j, \end{cases} \quad z \in \{-1, 1\}.$$

They are positive and satisfy

$$\sum_{z \in \{-1, 1\}} Q(x_j, \{z\}) = \frac{e^\varepsilon + 1}{e^\varepsilon + 1} = 1.$$

Thus Q is a Markov channel, and using this same kernel at each release makes it stationary. For every pair of rows and every output,

$$\frac{Q(x_j, \{z\})}{Q(x_{j'}, \{z\})} \in \{e^{-\varepsilon}, 1, e^\varepsilon\},$$

so

$$Q(x_j, \{z\}) \leq e^\varepsilon Q(x_{j'}, \{z\}).$$

Summing over the outputs in any measurable event gives

$$Q(x_j, A) \leq e^\varepsilon Q(x_{j'}, A).$$

6. Group this channel in the same way:

$$\overline{Q}_{\text{RR}}(+1, \cdot) := \frac{Q(x_0, \cdot) + Q(x_3, \cdot)}{2} = Q(x_0, \cdot), \quad \overline{Q}_{\text{RR}}(-1, \cdot) := \frac{Q(x_1, \cdot) + Q(x_2, \cdot)}{2} = Q(x_1, \cdot).$$

The equalities hold because the paired records have the same input sign. Its binary-input rows are therefore ordinary randomized response. Mixing them with probabilities $(1+\tau)/2$ and $(1-\tau)/2$ gives

$$P_\theta(Z = z) = \frac{1 + z\kappa_\varepsilon\tau}{2}, \quad \partial_\tau P_\theta(Z = z) = \frac{z\kappa_\varepsilon}{2}, \quad z \in \{-1, 1\}.$$

The two mixture probabilities are positive. Its scalar Fisher information is consequently

$$\begin{aligned} \mathcal{I}_{\text{RR}} &:= \sum_{z \in \{-1, 1\}} \frac{\{\partial_\tau P_\theta(Z = z)\}^2}{P_\theta(Z = z)} \\ &= \frac{\kappa_\varepsilon^2}{2(1 + \kappa_\varepsilon\tau)} + \frac{\kappa_\varepsilon^2}{2(1 - \kappa_\varepsilon\tau)} = J_{\text{bin}}. \end{aligned}$$

To relate this scalar information to the full information matrix, define

$$D_{\text{RR}} := \sum_{j=0}^3 Q(x_j, \cdot), \quad f_{\text{RR},j} := \frac{dQ(x_j, \cdot)}{dD_{\text{RR}}}, \quad m_{\text{RR},\theta} := \sum_{j=0}^3 \pi_{\theta,j} f_{\text{RR},j}.$$

The two-coordinate density derivative at fixed assignment probability is

$$\dot{m}_{\text{RR},\theta} := \begin{pmatrix} (f_{\text{RR},1} - f_{\text{RR},0})/2 \\ (f_{\text{RR},3} - f_{\text{RR},2})/2 \end{pmatrix}.$$

Under balance,

$$m_{\text{RR},\theta} = \frac{(1 - \mu_0)(f_{\text{RR},0} + f_{\text{RR},3}) + \mu_0(f_{\text{RR},1} + f_{\text{RR},2})}{2},$$

and

$$v(-1/2)^\top \dot{m}_{\text{RR},\theta} = \frac{f_{\text{RR},0} + f_{\text{RR},3} - f_{\text{RR},1} - f_{\text{RR},2}}{4}.$$

The grouped reference measure and its row densities are

$$\nu_{\text{RR}} := \frac{1}{2}D_{\text{RR}}, \quad \bar{f}_{\text{RR},+} := f_{\text{RR},0} + f_{\text{RR},3}, \quad \bar{f}_{\text{RR},-} := f_{\text{RR},1} + f_{\text{RR},2}.$$

Its mixture density satisfies

$$\overline{m}_{\text{RR},\tau} := \frac{1 + \tau}{2} \bar{f}_{\text{RR},+} + \frac{1 - \tau}{2} \bar{f}_{\text{RR},-} = 2m_{\text{RR},\theta}.$$

Therefore, writing

$$I_{\text{RR}} := I_\theta(Q),$$

we obtain

$$\begin{aligned}\mathcal{I}_{\text{RR}} &= \int \frac{\{(\bar{f}_{\text{RR},+} - \bar{f}_{\text{RR},-})/2\}^2}{2m_{\text{RR},\theta}} d\nu_{\text{RR}} \\ &= \int \frac{\{v(-1/2)^\top \dot{m}_{\text{RR},\theta}\}^2}{m_{\text{RR},\theta}} dD_{\text{RR}} \\ &= v(-1/2)^\top I_{\text{RR}} v(-1/2) = J_{\text{bin}}.\end{aligned}$$

The integrands are defined as zero at zero-density outputs. Indeed, if the sum of the four nonnegative row densities vanishes, all four vanish. If that sum is positive and $f_{\text{RR},0} + f_{\text{RR},3} = 0$, then

$$m_{\text{RR},\theta} = \frac{\mu_0}{2}(f_{\text{RR},1} + f_{\text{RR},2}) > 0.$$

If $f_{\text{RR},0} + f_{\text{RR},3} > 0$, then

$$m_{\text{RR},\theta} \geq \frac{1 - \mu_0}{2}(f_{\text{RR},0} + f_{\text{RR},3}) > 0.$$

Thus the change of reference measure and the divisions above cover all outputs.

The paired-row identities also give, almost everywhere,

$$f_{\text{RR},0} = f_{\text{RR},3}, \quad f_{\text{RR},1} = f_{\text{RR},2}.$$

It follows that, for every $t \in \mathbb{R}$,

$$v(t)^\top \dot{m}_{\text{RR},\theta} = \frac{t(f_{\text{RR},1} - f_{\text{RR},0}) + (t+1)(f_{\text{RR},3} - f_{\text{RR},2})}{2} = v(-1/2)^\top \dot{m}_{\text{RR},\theta}.$$

Integrating the squared directional derivative therefore gives

$$v(t)^\top I_{\text{RR}} v(t) = J_{\text{bin}}, \quad t \in \mathbb{R}.$$

7. We now evaluate the contrast variance using this positive minimizing quadratic value. The matrix I_{RR} is symmetric, and for every $\xi_{\text{dir}} \in \mathbb{R}^2$,

$$\xi_{\text{dir}}^\top I_{\text{RR}} \xi_{\text{dir}} = \int \frac{(\xi_{\text{dir}}^\top \dot{m}_{\text{RR},\theta})^2}{m_{\text{RR},\theta}} dD_{\text{RR}} \geq 0.$$

Thus it is positive semidefinite.

Define the perturbation coefficients by

$$a_{\text{geom}} := \sum_{k,\ell=0}^1 (I_{\text{RR}})_{k\ell}, \quad b_{\text{geom}} := 2\{(I_{\text{RR}}v(-1/2))_0 + (I_{\text{RR}}v(-1/2))_1\}.$$

Symmetry and expansion give, for every $\eta \in \mathbb{R}$,

$$\begin{aligned}0 &\leq v(-1/2 + \eta)^\top I_{\text{RR}} v(-1/2 + \eta) - v(-1/2)^\top I_{\text{RR}} v(-1/2) \\ &= \eta b_{\text{geom}} + \eta^2 a_{\text{geom}}.\end{aligned}$$

If $b_{\text{geom}} \neq 0$, choose

$$\eta_{\text{test}} := -\frac{b_{\text{geom}}}{2(|a_{\text{geom}}| + 1)}.$$

Using $a_{\text{geom}} \leq |a_{\text{geom}}|$, the last expression would satisfy

$$\begin{aligned}\eta_{\text{test}} b_{\text{geom}} + \eta_{\text{test}}^2 a_{\text{geom}} &= \frac{b_{\text{geom}}^2 \{a_{\text{geom}} - 2(|a_{\text{geom}}| + 1)\}}{4(|a_{\text{geom}}| + 1)^2} \\ &\leq -\frac{b_{\text{geom}}^2}{4(|a_{\text{geom}}| + 1)} < 0.\end{aligned}$$

Hence $b_{\text{geom}} = 0$, so the coordinates of $I_{\text{RR}}v(-1/2)$ sum to zero. Since

$$J_{\text{bin}} = v(-1/2)^\top I_{\text{RR}} v(-1/2) = (I_{\text{RR}} v(-1/2))_1,$$

we obtain

$$I_{\text{RR}} v(-1/2) = J_{\text{bin}}(-1, 1)^\top = J_{\text{bin}} c.$$

Define

$$w_{\text{sol}} := \frac{v(-1/2)}{J_{\text{bin}}} \in \mathbb{R}^2.$$

Then

$$I_{\text{RR}} w_{\text{sol}} = c, \quad c^\top w_{\text{sol}} = J_{\text{bin}}^{-1}, \quad c \in \text{range}(I_{\text{RR}}).$$

For any other solution $\tilde{w}_{\text{sol}} \in \mathbb{R}^2$ of $I_{\text{RR}} \tilde{w}_{\text{sol}} = c$, symmetry gives

$$c^\top \tilde{w}_{\text{sol}} = (I_{\text{RR}} w_{\text{sol}})^\top \tilde{w}_{\text{sol}} = w_{\text{sol}}^\top I_{\text{RR}} \tilde{w}_{\text{sol}} = w_{\text{sol}}^\top c = J_{\text{bin}}^{-1}.$$

Thus the solution-value infimum in the contrast-variance convention is

$$c^\top I_{\text{RR}}^+ c = \inf_{\{\tilde{w}_{\text{sol}}: I_{\text{RR}} \tilde{w}_{\text{sol}} = c\}} c^\top \tilde{w}_{\text{sol}} = J_{\text{bin}}^{-1}.$$

8. Apply [Theorem 1](#) to Q , with the constant privacy sequence

$$\varepsilon_n := \varepsilon, \quad n \in \mathbb{N}.$$

Its hypotheses hold because $0 < 1/2 < 1$, the means are interior, the budget is positive, and Q satisfies the stationary privacy inequalities established above. The oracle bound gives

$$V^*(\theta, 1/2, \varepsilon) \leq c^\top I_{\text{RR}}^+ c = J_{\text{bin}}^{-1}.$$

Together with the reciprocal lower bound, this proves

$$V^*(\theta, 1/2, \varepsilon) = J_{\text{bin}}^{-1}.$$

9. Since $\kappa_\varepsilon > 0$ and $1 - \kappa_\varepsilon^2 \tau^2 > 0$, ordinary fraction algebra gives

$$J_{\text{bin}}^{-1} = \frac{1 - \kappa_\varepsilon^2 \tau^2}{\kappa_\varepsilon^2} = \kappa_\varepsilon^{-2} - \tau^2.$$

Moreover,

$$\sinh(\varepsilon/2) > 0, \quad e^\varepsilon = (e^{\varepsilon/2})^2.$$

Expanding the hyperbolic functions and multiplying numerator and denominator by $e^{\varepsilon/2}$ yields

$$\frac{\cosh(\varepsilon/2)}{\sinh(\varepsilon/2)} = \frac{e^{\varepsilon/2} + e^{-\varepsilon/2}}{e^{\varepsilon/2} - e^{-\varepsilon/2}} = \frac{e^\varepsilon + 1}{e^\varepsilon - 1} = \kappa_\varepsilon^{-1}.$$

Consequently,

$$V^*(\theta, 1/2, \varepsilon) = \left(\frac{\cosh(\varepsilon/2)}{\sinh(\varepsilon/2)} \right)^2 - \tau^2.$$

10. The row normalization and event inequalities above establish the claimed stationary Markov privacy property. The constructed information solution establishes estimability, and the two variance identities give

$$c \in \text{range}(I_\theta(Q)), \quad c^\top I_\theta(Q)^+ c = J_{\text{bin}}^{-1} = V^*(\theta, 1/2, \varepsilon).$$

□

Proof of Theorem 9. Throughout, reciprocals at zero are interpreted as zero, empty sums as zero, and the real square root as zero on negative arguments.

1. At the balanced parameter point, the arm means are interior, their sum is one, and the contrast is zero. Thus Theorem 8, with privacy parameter one, gives

$$V^*((1/2, 1/2)^\top, 1/2, 1) = \left(\frac{\cosh(1/2)}{\sinh(1/2)} \right)^2.$$

We retain this identity for the final comparison.

2. The specified public experiment and its private-record representation are retained directly from the hypotheses:

$$\mathcal{L}(X_1, \dots, X_n) = P_\theta^{\otimes n}, \quad n \geq 1,$$

where

$$(P_\theta(\{x_0\}), P_\theta(\{x_1\}), P_\theta(\{x_2\}), P_\theta(\{x_3\})) = (q(1 - \mu_0), q\mu_0, p(1 - \mu_1), p\mu_1).$$

These are the public product experiments in Definition 1.

3. Define the trial contribution and the Laplace scale by

$$A_{\text{trial},i} = \frac{W_i Y_i}{p} - \frac{(1 - W_i) Y_i}{q}, \quad b_{\text{Lap}} = \frac{\Delta_A}{\varepsilon} > 0.$$

Binary assignment and outcome consistency imply, almost surely,

$$A_{\text{trial},i} = \begin{cases} 0, & X_i = x_0, \\ -q^{-1}, & X_i = x_1, \\ 0, & X_i = x_2, \\ p^{-1}, & X_i = x_3. \end{cases}$$

Summing against the four record probabilities therefore gives

$$\mathbb{E}[A_{\text{trial},i}] = -q^{-1}(q\mu_0) + p^{-1}(p\mu_1) = \tau,$$

$$\mathbb{E}[A_{\text{trial},i}^2] = q^{-2}(q\mu_0) + p^{-2}(p\mu_1) = \frac{\mu_0}{q} + \frac{\mu_1}{p}.$$

The trial contribution is bounded, so both moments are finite.

The Laplace density is

$$f_{\text{Lap}}(\ell) = \frac{1}{2b_{\text{Lap}}} \exp\left(-\frac{|\ell|}{b_{\text{Lap}}}\right), \quad \ell \in \mathbb{R}.$$

Its symmetry gives $\mathbb{E}[L_i] = 0$. The gamma integral at shape three gives

$$\int_0^\infty \ell^2 e^{-\ell/b_{\text{Lap}}} d\ell = 2b_{\text{Lap}}^3,$$

and hence

$$\mathbb{E}[L_i^2] = \frac{1}{b_{\text{Lap}}} \int_0^\infty \ell^2 e^{-\ell/b_{\text{Lap}}} d\ell = 2b_{\text{Lap}}^2 = \frac{2\Delta_A^2}{\varepsilon^2}.$$

Since $A_{\text{trial},i}$ is a function of X_i , its independence from L_i yields

$$\mathbb{E}[L_i A_{\text{trial},i}] = \mathbb{E}[L_i] \mathbb{E}[A_{\text{trial},i}] = 0.$$

For the releases of [Definition 14](#), it follows that

$$\mathbb{E}[\tilde{A}_i] = \tau, \quad \mathbb{E}[\tilde{A}_i^2] = \frac{\mu_1}{p} + \frac{\mu_0}{q} + \frac{2\Delta_A^2}{\varepsilon^2}.$$

Define their centered versions by

$$\xi_{\text{OA},i} = \tilde{A}_i - \tau.$$

Expanding the centered square gives

$$\mathbb{E}[\xi_{\text{OA},i}] = 0, \quad \mathbb{E}[\xi_{\text{OA},i}^2] = \mathbb{E}[\tilde{A}_i^2] - 2\tau \mathbb{E}[\tilde{A}_i] + \tau^2 = V_{\text{OA}}(\theta, p, \varepsilon) < \infty.$$

The pairs (X_i, L_i) are independent across subjects. Each pair has the same product law, because the record law and the noise law are common across subjects and its two coordinates are independent. Applying the release function to each pair shows that the centered releases are independent and identically distributed. The iid central limit theorem therefore gives

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_{\text{OA},i} \rightsquigarrow N(0, V_{\text{OA}}(\theta, p, \varepsilon)).$$

For every $n \geq 1$,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_{\text{OA},i} = \sqrt{n}(\hat{\tau}_{\text{OA}} - \tau).$$

At $n = 0$, both normalized expressions are zero under the stated conventions. This proves the asserted convergence in law.

4. Use the staircase notation of [Definition 4](#). Interior assignment and means give nonnegative record probabilities, and every component of b_S is nonnegative. Thus

$$h_S(\theta) = \sum_{j=0}^3 \pi_{\theta,j}(b_S)_j \geq 0,$$

and, for every feasible α and every $t \in \mathbb{R}$,

$$F_\theta(\alpha, t) = \sum_{S \in \mathcal{S}} \alpha_S \frac{(g_S^\top v(t))^2}{h_S(\theta)} \geq 0.$$

Taking the profiled minimum and optimized maximum in [Definition 5](#) gives

$$J^*(\theta, p, \varepsilon) \geq 0, \quad V^*(\theta, p, \varepsilon) = J^*(\theta, p, \varepsilon)^{-1} \geq 0.$$

For every positive integer n , factoring out $\sqrt{n}$ from both square roots and canceling the positive common factor $z_{1-\gamma/2}/\sqrt{n}$ gives

$$\begin{aligned} \frac{z_{1-\gamma/2}\sqrt{V_{\text{OA}}(\theta, p, \varepsilon)/n}}{z_{1-\gamma/2}\sqrt{V^*(\theta, p, \varepsilon)/n}} &= \frac{\sqrt{V_{\text{OA}}(\theta, p, \varepsilon)}}{\sqrt{V^*(\theta, p, \varepsilon)}} \\ &= \sqrt{\frac{V_{\text{OA}}(\theta, p, \varepsilon)}{V^*(\theta, p, \varepsilon)}}. \end{aligned}$$

The identities also respect the zero-reciprocal convention. The sequence is consequently equal to its claimed limit for every $n \geq 1$.

5. Apply the weak law to the centered releases and their squares. The moment identities above yield

$$\frac{1}{n} \sum_{i=1}^n \xi_{\text{OA},i} \xrightarrow{\text{Pr}} 0, \quad \frac{1}{n} \sum_{i=1}^n \xi_{\text{OA},i}^2 \xrightarrow{\text{Pr}} V_{\text{OA}}(\theta, p, \varepsilon).$$

For $n \geq 1$, centering the empirical variance at τ gives the exact identity

$$\begin{aligned} \widehat{V}_{\text{OA}} &= \frac{1}{n} \sum_{i=1}^n (\tilde{A}_i - \widehat{\tau}_{\text{OA}})^2 \\ &= \frac{1}{n} \sum_{i=1}^n \xi_{\text{OA},i}^2 - \left(\frac{1}{n} \sum_{i=1}^n \xi_{\text{OA},i} \right)^2 \\ &= \frac{1}{n} \sum_{i=1}^n \tilde{A}_i^2 - \widehat{\tau}_{\text{OA}}^2. \end{aligned}$$

At $n = 0$, each empirical expression is zero. Continuity of squaring and subtraction proves variance consistency. In particular, for every $\delta > 0$,

$$0 \leq \Pr\left(|\widehat{V}_{\text{OA}} - V_{\text{OA}}| > \delta\right) \leq \Pr\left(|\widehat{V}_{\text{OA}} - V_{\text{OA}}| \geq \delta\right) \longrightarrow 0.$$

6. Choose, for every $n \in \mathbb{N}$,

$$m_n = \lfloor \sqrt{n} \rfloor.$$

For every integer $r_{\text{pilot}} \geq 0$,

$$n \geq r_{\text{pilot}}^2 \implies m_n \geq r_{\text{pilot}},$$

so $m_n \rightarrow \infty$. Moreover, for $n \geq 1$,

$$0 \leq \frac{m_n}{n} \leq \frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}} \longrightarrow 0.$$

For $n \geq 2$,

$$1 \leq m_n \leq \sqrt{n} < n.$$

The remaining values are $m_0 = 0$ and $m_1 = 1$. Thus the schedule satisfies [Assumptions 10](#) and [11](#) over their full stated ranges.

Take the privacy sequence

$$\varepsilon_n = \varepsilon, \quad n \in \mathbb{N}.$$

It satisfies [Assumption 8](#). The fixed measurable strong-saddle selector, the interior parameters, the chosen schedule, and $0 < \gamma < 1$ meet the hypotheses of [Theorem 3](#). Its variance-consistency conclusion supplies, under the transcript law of the selected procedure,

$$\Pr_{\theta}(|\hat{V}^* - V^*(\theta, p, \varepsilon)| > \delta) \longrightarrow 0 \quad (\delta > 0).$$

7. Fix any sequence of couplings $(\nu_n)_{n \geq 0}$ with the specified trial and transcript marginals. Each ν_n is a probability measure because its first marginal is the trial probability law. For measurable events E in the corresponding product space, write

$$\Pr_{\nu_n}(E) = \nu_n(E).$$

Regard $\hat{V}_{\text{OA}}$ and $\hat{V}^*$ as functions of the first and second components, respectively.

We first establish strict positivity of the limiting denominator. In the notation of [Definition 4](#),

$$d = e^\varepsilon - 1 > 0, \quad h_S(\theta) = \sum_{j=0}^3 \pi_{\theta,j}(b_S)_j > 0,$$

because all record probabilities are positive and all components of b_S are at least one. Define the singleton weights, for every $S \in \mathcal{S}$, by

$$\alpha_{\text{RR},S} = \begin{cases} (e^\varepsilon + 3)^{-1}, & |S| = 1, \\ 0, & |S| \neq 1. \end{cases}$$

They are nonnegative, and for every record coordinate j ,

$$\sum_{S \in \mathcal{S}} \alpha_{\text{RR},S}(b_S)_j = \frac{e^\varepsilon + 3}{e^\varepsilon + 3} = 1.$$

Hence $\alpha_{\text{RR}} \in \mathcal{A}_\varepsilon$.

To justify attainment of the profiling minimum, define, for any feasible α ,

$$\lambda_{\text{lin}}(\alpha) = \sum_{S \in \mathcal{S}} \frac{2\alpha_S(g_S^\top v(0))((g_S)_0 + (g_S)_1)}{h_S(\theta)},$$

$$\lambda_{\text{quad}}(\alpha) = \sum_{S \in \mathcal{S}} \frac{\alpha_S((g_S)_0 + (g_S)_1)^2}{h_S(\theta)} \geq 0.$$

Expanding $v(t) = (t, t+1)^\top$ gives, for every real t ,

$$F_\theta(\alpha, t) = F_\theta(\alpha, 0) + t\lambda_{\text{lin}}(\alpha) + t^2\lambda_{\text{quad}}(\alpha).$$

If $\lambda_{\text{quad}}(\alpha) = 0$ and $\lambda_{\text{lin}}(\alpha) \neq 0$, substitution of

$$t = -\frac{F_\theta(\alpha, 0) + 1}{\lambda_{\text{lin}}(\alpha)}$$

would give $F_\theta(\alpha, t) = -1$, contradicting nonnegativity. Thus in this case the linear coefficient is also zero and $t = 0$ is a minimizer. If $\lambda_{\text{quad}}(\alpha) > 0$, define

$$t_{\min}(\alpha) = -\frac{\lambda_{\text{lin}}(\alpha)}{2\lambda_{\text{quad}}(\alpha)}.$$

Then

$$F_\theta(\alpha, t) - F_\theta(\alpha, t_{\min}(\alpha)) = \lambda_{\text{quad}}(\alpha)(t - t_{\min}(\alpha))^2 \geq 0.$$

In both cases a minimizing real coordinate exists. Choose such a coordinate t_{RR} for α_{RR} .

All summands in $F_\theta(\alpha_{\text{RR}}, t_{\text{RR}})$ are nonnegative. The singleton patterns $\{0\}$ and $\{3\}$ therefore give

$$\begin{aligned} 0 &\leq \frac{\alpha_{\text{RR},\{0\}}(dq)^2}{h_{\{0\}}(\theta)} t_{\text{RR}}^2 \leq F_\theta(\alpha_{\text{RR}}, t_{\text{RR}}), \\ 0 &\leq \frac{\alpha_{\text{RR},\{3\}}(dp)^2}{h_{\{3\}}(\theta)} (t_{\text{RR}} + 1)^2 \leq F_\theta(\alpha_{\text{RR}}, t_{\text{RR}}). \end{aligned}$$

Both coefficients are strictly positive. If the objective value were zero, the first inequality would force $t_{\text{RR}} = 0$, while the second would force $t_{\text{RR}} + 1 = 0$. Consequently,

$$0 < F_\theta(\alpha_{\text{RR}}, t_{\text{RR}}) = \min_{t \in \mathbb{R}} F_\theta(\alpha_{\text{RR}}, t) \leq J^*(\theta, p, \varepsilon).$$

The final inequality follows from the optimized maximum in [Definition 5](#). Taking the reciprocal proves

$$V^*(\theta, p, \varepsilon) > 0.$$

8. The formula in [Definition 11](#) expresses $\widehat{V}^*$ as a nonnegative reciprocal times a sum of squares. Thus, for every sample size and every transcript,

$$\widehat{V}^* \geq 0.$$

When $m_n \geq n$, the main-sample sum is empty and its reciprocal normalization is zero, so $\widehat{V}^* = 0$.

For real numerator and denominator inputs, define the total ratio map

$$f_{\text{ratio}}(r_{\text{num}}, r_{\text{den}}) = \begin{cases} \sqrt{r_{\text{num}}/r_{\text{den}}}, & r_{\text{den}} > 0, \\ 0, & r_{\text{den}} \leq 0, \end{cases} \quad (r_{\text{num}}, r_{\text{den}}) \in \mathbb{R}^2.$$

On the half-plane $r_{\text{den}} \geq 0$, this agrees with the displayed square-root ratio: at a zero denominator both expressions are zero. Since $V^* > 0$, the map agrees with the continuous square-root-of-quotient map in a neighborhood of (V_{OA}, V^*) , and is therefore continuous at that pair.

The marginal identities and the union bound give, for every $\eta > 0$,

$$\begin{aligned} & \Pr_{\nu_n} \left(\max\{|\widehat{V}_{\text{OA}} - V_{\text{OA}}|, |\widehat{V}^* - V^*|\} \geq \eta \right) \\ & \leq \Pr \left(|\widehat{V}_{\text{OA}} - V_{\text{OA}}| \geq \eta \right) + \Pr_{\theta} \left(|\widehat{V}^* - V^*| \geq \eta \right) \\ & \leq \Pr \left(|\widehat{V}_{\text{OA}} - V_{\text{OA}}| > \eta/2 \right) + \Pr_{\theta} \left(|\widehat{V}^* - V^*| > \eta/2 \right) \longrightarrow 0. \end{aligned}$$

Hence the pair of variance estimates converges in probability under every prescribed coupling.

For any $\delta > 0$, continuity supplies an $\eta > 0$ such that

$$\max\{|r_{\text{num}} - V_{\text{OA}}|, |r_{\text{den}} - V^*|\} < \eta \implies \left| f_{\text{ratio}}(r_{\text{num}}, r_{\text{den}}) - \sqrt{V_{\text{OA}}/V^*} \right| < \delta.$$

Using the pointwise nonnegativity of $\widehat{V}^*$, we obtain

$$\begin{aligned} & \Pr_{\nu_n} \left(\left| \sqrt{\widehat{V}_{\text{OA}}/\widehat{V}^*} - \sqrt{V_{\text{OA}}/V^*} \right| > \delta \right) \\ & \leq \Pr_{\nu_n} \left(\max\{|\widehat{V}_{\text{OA}} - V_{\text{OA}}|, |\widehat{V}^* - V^*|\} \geq \eta \right) \longrightarrow 0. \end{aligned}$$

This proves the empirical ratio assertion for the exhibited schedule and every sequence of the stated couplings.

9. Finally, at $p = q = 1/2$, $\mu_0 = \mu_1 = 1/2$, and $\varepsilon = 1$,

$$\tau = 0, \quad \Delta_A = 4,$$

so direct substitution gives

$$V_{\text{OA}} = \frac{1/2}{1/2} + \frac{1/2}{1/2} + 2 \cdot 4^2 = 34.$$

To compare this with the balanced value, use

$$\sinh(1/2) > 1/2 > 0, \quad \cosh^2(1/2) - \sinh^2(1/2) = 1.$$

In particular,

$$33 \sinh^2(1/2) > \frac{33}{4} > 1,$$

and therefore

$$\cosh^2(1/2) = 1 + \sinh^2(1/2) < 34 \sinh^2(1/2).$$

Division by the positive quantity $\sinh^2(1/2)$ yields

$$V^* = \left(\frac{\cosh(1/2)}{\sinh(1/2)} \right)^2 < 34 = V_{\text{OA}},$$

as required. □

Proof of Lemma 1. 1. By Assumption 8, the privacy ratio

$$r := e^\varepsilon$$

satisfies $r > 1$. In the four-record order, the probabilities are

$$(\pi_{\theta,1}, \pi_{\theta,2}, \pi_{\theta,3}, \pi_{\theta,4}) = ((1-p)(1-\mu_0), (1-p)\mu_0, p(1-\mu_1), p\mu_1).$$

Thus Assumptions 6 and 7 give

$$\pi_{\theta,j} > 0 \quad (j = 1, \dots, 4), \quad \sum_{j=1}^4 \pi_{\theta,j} = 1.$$

2. Define the row measures, their common dominating measure, and the staircase rays by

$$Q_{\text{row},j} := Q(\cdot \mid x_j), \quad \nu := \sum_{j=1}^4 Q_{\text{row},j}, \quad f_j := \frac{dQ_{\text{row},j}}{d\nu}, \quad b_S(j) := \begin{cases} r, & x_j \in S, \\ 1, & x_j \notin S. \end{cases}$$

Choose nonnegative finite measurable versions of the densities f_j , setting them to zero on any exceptional null set where the Radon–Nikodym derivatives are infinite. The privacy inequalities imply

$$f_j(z) \leq r f_k(z) \quad \text{for } \nu\text{-almost every } z, \text{ for every } j, k.$$

Indeed, integrating over the measurable set where $f_j > r f_k$ would contradict $Q_{\text{row},j}(A) \leq r Q_{\text{row},k}(A)$ if that set had positive ν -measure. There are finitely many pairs, so these inequalities hold simultaneously outside one null set.

For every $z \in Z$, define

$$m_{\text{base}}(z) := \min_{1 \leq j \leq 4} f_j(z), \quad y_j(z) := \begin{cases} 0, & m_{\text{base}}(z) = 0, \\ \frac{f_j(z)/m_{\text{base}}(z) - 1}{r - 1}, & m_{\text{base}}(z) \neq 0. \end{cases}$$

On the common set where the privacy inequalities hold, an index attaining the minimum gives

$$0 \leq m_{\text{base}}(z) \leq f_j(z) \leq r m_{\text{base}}(z).$$

If $m_{\text{base}}(z) = 0$, all four densities vanish. Otherwise, division by the positive minimum and by $r - 1$ gives $0 \leq y_j(z) \leq 1$. In either case,

$$f_j(z) = m_{\text{base}}(z)(1 + (r - 1)y_j(z)).$$

For every subset $T \subseteq \mathcal{X}$, including the empty and full subsets, define

$$\omega_T(z) := \prod_{j: x_j \in T} y_j(z) \prod_{j: x_j \notin T} (1 - y_j(z)).$$

Expanding the products yields

$$\sum_{T \subseteq \mathcal{X}} \omega_T(z) = 1, \quad \sum_{\substack{T \subseteq \mathcal{X} \\ x_j \in T}} \omega_T(z) = y_j(z),$$

and consequently

$$\sum_{T \subseteq \mathcal{X}} \omega_T(z) b_T(j) = 1 + (r-1)y_j(z).$$

To express this sum using proper nonempty subsets, put

$$S_{\dagger} := \{x_1\}, \quad k_{\text{cube}}(z) := \frac{\omega_{\emptyset}(z) + r\omega_{\mathcal{X}}(z)}{r+1},$$

and, for $S \in \mathcal{S}$, define

$$\lambda_S(z) := \max \left\{ 0, m_{\text{base}}(z) \left[\omega_S(z) + k_{\text{cube}}(z) (\mathbf{1}_{\{S=S_{\dagger}\}} + \mathbf{1}_{\{S=\mathcal{X} \setminus S_{\dagger}\}}) \right] \right\}.$$

These functions are measurable and nonnegative on all of Z . On the common set where the privacy inequalities hold, their expressions inside the maximum are nonnegative. Moreover,

$$b_{S_{\dagger}}(j) + b_{\mathcal{X} \setminus S_{\dagger}}(j) = r+1,$$

so the added complementary-pair contribution is precisely $\omega_{\emptyset} + r\omega_{\mathcal{X}}$, the contribution of the empty and full subsets. Therefore

$$f_j(z) = \sum_{S \in \mathcal{S}} \lambda_S(z) b_S(j) \quad \text{for } \nu\text{-almost every } z, \text{ for every } j.$$

The case $m_{\text{base}} = 0$ gives zero on both sides.

3. Define coefficient measures and weights by

$$\Lambda_S(A) := \int_A \lambda_S(z) d\nu(z), \quad \alpha_S := \Lambda_S(Z).$$

Since every ray coordinate is at least one, the preceding decomposition gives

$$0 \leq \lambda_S(z) \leq \lambda_S(z) b_S(1) \leq f_1(z) \quad \nu\text{-almost everywhere.}$$

Thus each coefficient is integrable. Integrating the decomposition gives, for every measurable $A \subseteq Z$,

$$Q_{\text{row},j}(A) = \sum_{S \in \mathcal{S}} b_S(j) \Lambda_S(A).$$

In particular,

$$\alpha_S \geq 0, \quad \sum_{S \in \mathcal{S}} \alpha_S b_S(j) = Q_{\text{row},j}(Z) = 1.$$

Hence $\alpha \in \mathcal{A}_{\varepsilon}$.

Define the postprocessing measures on the whole pattern set by

$$K(S, A) := \begin{cases} \Lambda_S(A)/\alpha_S, & \alpha_S > 0, \\ Q_{\text{row},1}(A), & \alpha_S = 0. \end{cases}$$

Every row is a probability measure, and the finite pattern domain makes K a probability kernel. If $\alpha_S = 0$, the nonnegative measure Λ_S has zero total mass and is the zero measure. Consequently, in both cases,

$$\Lambda_S = \alpha_S K(S, \cdot).$$

It follows that

$$Q_{\text{row},j}(A) = \sum_{S \in \mathcal{S}} \alpha_S b_S(j) K(S, A) = \sum_{S \in \mathcal{S}} Q_\alpha(S | x_j) K(S, A),$$

which proves $Q = K \circ Q_\alpha$. Indexing these measures and weights by the stated fourteen patterns gives the required channel.

4. Both information matrices are symmetric: their defining summands and integrands are symmetric outer products. We next identify their quadratic forms.

With p fixed, define the input score vectors by

$$\sigma_j := \frac{\nabla_\theta \pi_{\theta,j}}{\pi_{\theta,j}} \in \mathbb{R}^2.$$

The positive record probabilities make these vectors well defined. Differentiating the pattern masses, with p and ε fixed, gives

$$h_S(\theta) = \sum_{j=1}^4 \pi_{\theta,j} b_S(j) > 0, \quad g_S = \sum_{j=1}^4 b_S(j) \nabla_\theta \pi_{\theta,j} = \sum_{j=1}^4 \pi_{\theta,j} b_S(j) \sigma_j.$$

For an arbitrary direction $\xi \in \mathbb{R}^2$, define the release masses, score numerators, and release scores by

$$\begin{aligned} \beta_S &:= \sum_{j=1}^4 \pi_{\theta,j} Q_\alpha(S | x_j) = \alpha_S h_S(\theta), \\ n_{S,\xi} &:= \sum_{j=1}^4 \pi_{\theta,j} Q_\alpha(S | x_j) \xi^\top \sigma_j = \alpha_S g_S^\top \xi, \quad \eta_{S,\xi} := \begin{cases} n_{S,\xi} / \beta_S, & \beta_S > 0, \\ 0, & \beta_S = 0. \end{cases} \end{aligned}$$

If $\alpha_S = 0$, both β_S and $n_{S,\xi}$ vanish. If $\alpha_S > 0$, cancellation gives

$$\eta_{S,\xi} = \frac{g_S^\top \xi}{h_S(\theta)}, \quad \frac{n_{S,\xi}^2}{\beta_S} = \alpha_S \frac{(g_S^\top \xi)^2}{h_S(\theta)}.$$

Using zero for the contribution of a zero-mass atom, expansion of the quadratic form therefore yields

$$\sum_{S \in \mathcal{S}} \beta_S \eta_{S,\xi}^2 = \sum_{S \in \mathcal{S}} \alpha_S \frac{(g_S^\top \xi)^2}{h_S(\theta)} = \xi^\top I_\theta(\alpha) \xi.$$

For the original output, define

$$f_{\text{out}}(z) := \sum_{j=1}^4 \pi_{\theta,j} f_j(z), \quad \mathbf{d}_\theta(z) := \sum_{j=1}^4 \nabla_\theta \pi_{\theta,j} f_j(z),$$

$$\zeta_\xi(z) := \begin{cases} \frac{\xi^\top \mathbf{d}_\theta(z)}{f_{\text{out}}(z)}, & f_{\text{out}}(z) > 0, \\ 0, & f_{\text{out}}(z) = 0. \end{cases}$$

Here f_{out} is the output-law density and $\mathbf{d}_\theta$ is its vector of parameter-derivative densities. Since the $\pi_{\theta,j}$ are positive and the f_j are nonnegative, $f_{\text{out}} = 0$ implies that every $f_j = 0$, and hence $\mathbf{d}_\theta = 0$. Thus, with the same zero-denominator convention,

$$\xi^\top I_\theta(Q)\xi = \int_Z \frac{(\xi^\top \mathbf{d}_\theta(z))^2}{f_{\text{out}}(z)} d\nu(z) = \int_Z f_{\text{out}}(z) \zeta_\xi(z)^2 d\nu(z).$$

The first identity follows by expanding the square and exchanging the finite coordinate sums with the integral.

5. We derive the exact information-gap identity. The factorization, weighted first by the input probabilities and then by their directional score numerators, gives the following identities of measures:

$$\begin{aligned} \sum_{S \in \mathcal{S}} \beta_S K(S, \cdot) &= f_{\text{out}} \nu, \\ \sum_{S \in \mathcal{S}} \beta_S \eta_{S,\xi} K(S, \cdot) &= \sum_{j=1}^4 \pi_{\theta,j} \xi^\top \sigma_j Q_{\text{row},j} \\ &= (\xi^\top \mathbf{d}_\theta) \nu. \end{aligned}$$

The second identity uses $\beta_S \eta_{S,\xi} = n_{S,\xi}$, which also holds when $\beta_S = 0$.

The output score is measurable and bounded, because, wherever $f_{\text{out}} > 0$, it is a weighted average of the four directional input scores:

$$\zeta_\xi(z) = \sum_{j=1}^4 \frac{\pi_{\theta,j} f_j(z)}{f_{\text{out}}(z)} \xi^\top \sigma_j, \quad |\zeta_\xi(z)| \leq \max_{1 \leq j \leq 4} |\xi^\top \sigma_j|.$$

Its assigned value zero satisfies the same bound when $f_{\text{out}} = 0$. Hence the following integrals are finite. Probability normalization of each row of K , followed by the two measure identities above, gives

$$\begin{aligned} \sum_{S \in \mathcal{S}} \beta_S \int_Z \eta_{S,\xi}^2 K(S, dz) &= \xi^\top I_\theta(\alpha) \xi, \\ \sum_{S \in \mathcal{S}} \beta_S \eta_{S,\xi} \int_Z \zeta_\xi(z) K(S, dz) &= \int_Z (\xi^\top \mathbf{d}_\theta(z)) \zeta_\xi(z) d\nu(z) \\ &= \xi^\top I_\theta(Q) \xi, \end{aligned}$$

and

$$\sum_{S \in \mathcal{S}} \beta_S \int_Z \zeta_\xi(z)^2 K(S, dz) = \int_Z f_{\text{out}}(z) \zeta_\xi(z)^2 d\nu(z) = \xi^\top I_\theta(Q) \xi.$$

Expanding the square and substituting these three identities proves

$$\xi^\top (I_\theta(\alpha) - I_\theta(Q)) \xi = \sum_{S \in \mathcal{S}} \beta_S \int_Z (\eta_{S,\xi} - \zeta_\xi(z))^2 K(S, dz).$$

6. Each release mass β_S is nonnegative, and each integral of a square is nonnegative. Therefore, for every $\xi \in \mathbb{R}^2$,

$$\xi^\top (I_\theta(\alpha) - I_\theta(Q)) \xi \geq 0.$$

Together with symmetry, this proves

$$I_\theta(\alpha) - I_\theta(Q) \succeq 0.$$

7. Fix $t \in \mathbb{R}$ and suppose that the two quadratic forms agree in the direction $v(t)$. The identities above turn this equality into

$$0 = \sum_{S \in \mathcal{S}} \beta_S \int_Z (\eta_{S,v(t)} - \zeta_{v(t)}(z))^2 K(S, dz).$$

Every summand is nonnegative. For any S with $\alpha_S \neq 0$, nonnegativity of α_S and positivity of $h_S(\theta)$ give

$$\beta_S = \alpha_S h_S(\theta) > 0, \quad \eta_{S,v(t)} = \frac{g_S^\top v(t)}{h_S(\theta)} = \rho_S(\theta, t).$$

Its squared residual consequently has zero integral:

$$\int_Z (\rho_S(\theta, t) - \zeta_{v(t)}(z))^2 K(S, dz) = 0.$$

A nonnegative integrable function with zero integral vanishes almost everywhere, so

$$\zeta_{v(t)}(z) = \rho_S(\theta, t) \quad \text{for } K(S, \cdot)\text{-almost every } z.$$

Now let S, T be two active patterns. If their projected scores were different, define the measurable set

$$E_{S,t} := \{z \in Z : \zeta_{v(t)}(z) \neq \rho_S(\theta, t)\}.$$

The almost-everywhere score identities for S and T would give

$$K(S, E_{S,t}) = 0, \quad K(T, Z \setminus E_{S,t}) = 0.$$

These equalities establish mutual singularity of the two conditional output measures. Thus, when those measures are not mutually singular, their projected scores must satisfy

$$\rho_S(\theta, t) = \rho_T(\theta, t),$$

as claimed. □

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